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Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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ABSTRACT After a load is restored or a line is reclosed, residual dynamics can resemble a new intervention. We study sequential attribution from eight PMUs in an IEEE 39-bus model with nine candidate load buses and nine lines. A bank of switched differential-algebraic predictors compares intervention and recovery histories from estimated past states. Conditional Bayesian evidence integrates load severity, while separate active and cause weights retain the explanation of a restored intervention. In an independent simulation evaluation, 108 interventions are replayed under three network conditions and two noise levels. The physical bank produces 644/648 correct family/source decisions before restoration with 72 unmatched alarms, versus 330/648 with 29 for a fixed causal ExtraTrees adapter. The gain concentrates on loads; line coverage matches the short-window learned comparator. A controlled posthoc study identifies a state-carry benefit under both marginal and profiled severity scoring: retaining the state gives 108/108 timely decisions and one extra alarm, compared with 75/108 and 166 after resetting physical deviations. The results establish recovery-consistent prediction as a useful design choice for sparse-PMU attribution and quantify its coverage–alarm trade-off under network uncertainty.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.

Physical restoration is not electrical recovery. A detector that returns immediately to a flat reference can issue another alarm for the remaining oscillation. Suppressing alarms throughout recovery can instead hide a real intervention. The operating point adds another difficulty: correcting a voltage or current offset need not correct how the network responds to a later input.

Existing work provides strong foundations. DAE state estimators infer unmeasured electrical states [1]; physical outage detectors compare network hypotheses sequentially [2], [3]; and simulated fault trajectories support diagnosis from sparse PMUs [4]. Physical models, transient histories and unknown-onset searches are thus established ingredients. A small state-reconstruction error, however, does not by itself identify the intervention that produced the observations.

We compare physical histories rather than isolated signal features. A history specifies an intervention, source, onset, severity and restoration. DAE-derived predictors start from estimated past states and preserve the state left by previous interventions. Conditional evidence integrates load severity, accounts for instrumental memory, and updates fixed network-model hypotheses. Separate active and cause masses keep a restored intervention available as an explanation of the continuing response.

The paper makes three contributions within this restricted problem:

1) An executable attribution mechanism that makes new load/line interventions compete with retained recovery histories. We specify its physical predictors, evidence updates and decision rules.
2) A comparison that separates detection, correct source

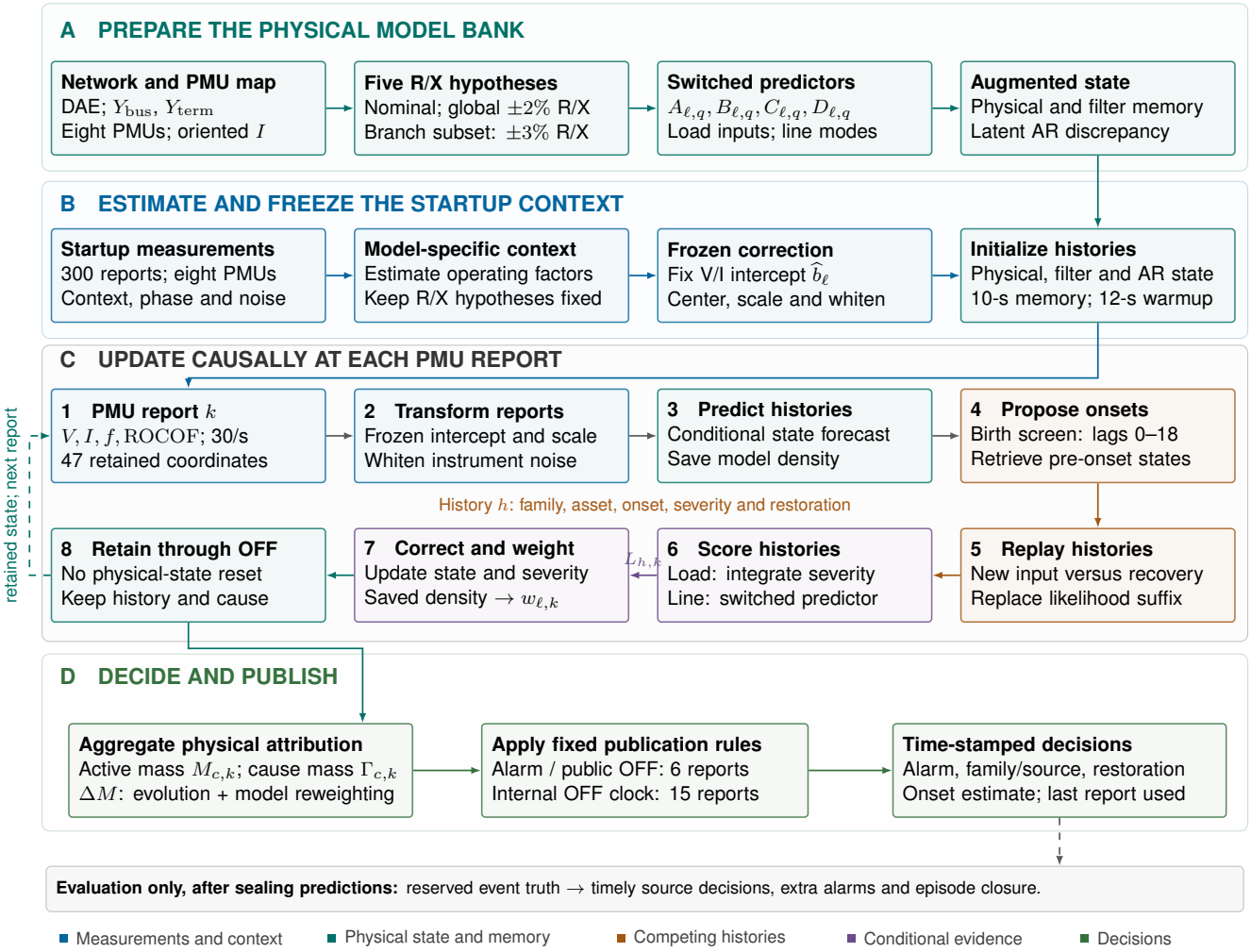


FIGURE 1. Implemented physical-history attribution, organized into model preparation (A), startup context (B), per-report inference (C), and publication (D). The bank predicts before state correction and saves model evidence before proposal selection. Candidate replay replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Reserved event truth is used only after predictions are sealed.

attribution before restoration, extra alarms and episode closure. Nominal and uncertain network models are evaluated alongside two fixed ExtraTrees adapters; Chen's published detector is compared on eligible persistent line openings.

3) A controlled severity-score/state-carry study on identical records. It quantifies the benefit of preserving physical-state deviations at inferred closure under both marginalization and bounded profiling. The common observations and operators connect the decision improvement to a specific state-retention operation.

The independent comparison gives 644/648 timely attributions with 72 extra alarms. The advantage over the fixed learned adapters concerns load coverage, while network mismatch increases alarm burden. This evaluates configured procedures at their original thresholds. It motivates recovery-consistent inference without establishing superiority at equal false-alarm rates. Auxiliary voltage, feature, simultaneous-source and context-signature studies are retained in the supplement with their own cohorts.

II. RELATED WORK

A. STATE ESTIMATION AND PHYSICAL SOURCE INFERENCE

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [1], [5], [6]. Functional observability asks whether a requested state quantity can be recovered [7], [8]. Source inference asks which intervention can explain the trajectory. Descriptor-model diagnosability and nuisance-profiled tests formalize this distinction [9], [10].

Physical hypotheses support outage inference from phasor angles [11], Bayesian inference with uncertain states [12], and sequential detection under stochastic operation [2], [3]. Anguluri *et al.* recover sparse forcing locations, inputs and an unknown initial state using group LASSO for linear dynamics

[13]. These are precedents for source recovery; group LASSO is not an executed comparator in our frozen campaign. Sensor placement also affects source distinguishability [14]–[16].

Recent learned reconstruction also targets sparse sensing. Pomarico *et al.*'s SHRED preprint evaluates dynamic-state reconstruction on IEEE-39 from a limited number of PMUs [17]. Its target is state reconstruction rather than the source/ON/OFF episode contract here. Dey *et al.* detect and localize distribution-network topology changes from phasors using learned statistics, with hardware-in-the-loop validation [18]. Neither sparse sensing nor change localization alone distinguishes the present contribution. These are adjacent tasks, not executed performance baselines in this study.

B. TRAJECTORY-BASED AND SEQUENTIAL DIAGNOSIS

Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window and return candidate lines [4]. Their evaluated target is short-circuit identification before clearing; load changes are an adverse-event test. Our target follows load changes and line openings through restoration and successive interventions. Physical trajectory libraries, remote PMUs and retrospective onset scanning are shared ingredients. Their procedure was not run on our data, so no numerical superiority is inferred.

The Adaptive Matrix CuSum derives sequential identification error/delay results under specified observation laws [19]. Multiple-model filtering already propagates conditional state/mode hypotheses [20]. The multiple-model multiple-hypothesis filter maintains mode-sequence histories and adapts their number through pruning and merging [21]. Retaining histories or adapting bank size is therefore not an original principle of our method. Our pruned history bank uses these ideas, but its approximate covariances, proposal screen and fitted startup context do not inherit their general guarantees. The complete procedure evaluated here makes retained recovery compete with later interventions and reports timely attribution and alarm burden. These comparisons do not isolate recovery memory from the other admission and publication rules.

C. LEARNED DIAGNOSIS AND PHYSICAL UNCERTAINTY

Learned and knowledge-driven representations identify PMU disturbances and assets [22]–[24]. We compare two fixed ExtraTrees adapters from the SGSMA project. Their performance tests transfer of those implementations, not the limits of all supervised learning. A physical catalog and a fixed learned adapter have different access to candidate-source responses. This comparison does not equalize that information or establish a source-excluded generalization result.

Physical stochastic models support latent-state estimation [25], [26], while joint DAE parameter inference addresses uncertain generators and networks [27]. Our stream method uses a finite bank of constant network hypotheses. Trajectory sensitivities and Gaussian evidence have also been combined for Bayesian parameter inference in power-system DAEs [28]. Their use alone does not distinguish an event attribution algorithm. The separate signature test examines a local event–context derivative; it does not identify every plant parameter or replace the stream method with an unrestricted joint estimator.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of -120° and $+120^\circ$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

TABLE 1. Closest inference contracts. Only Chen and the fixed ExtraTrees adapters are executed comparators in the confirmatory stream campaign. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
IMM / M3H	Conditional filters, mode histories and mixture reduction	Hybrid switching-state estimation	Direct history-filter precedent; no common-data baseline here
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

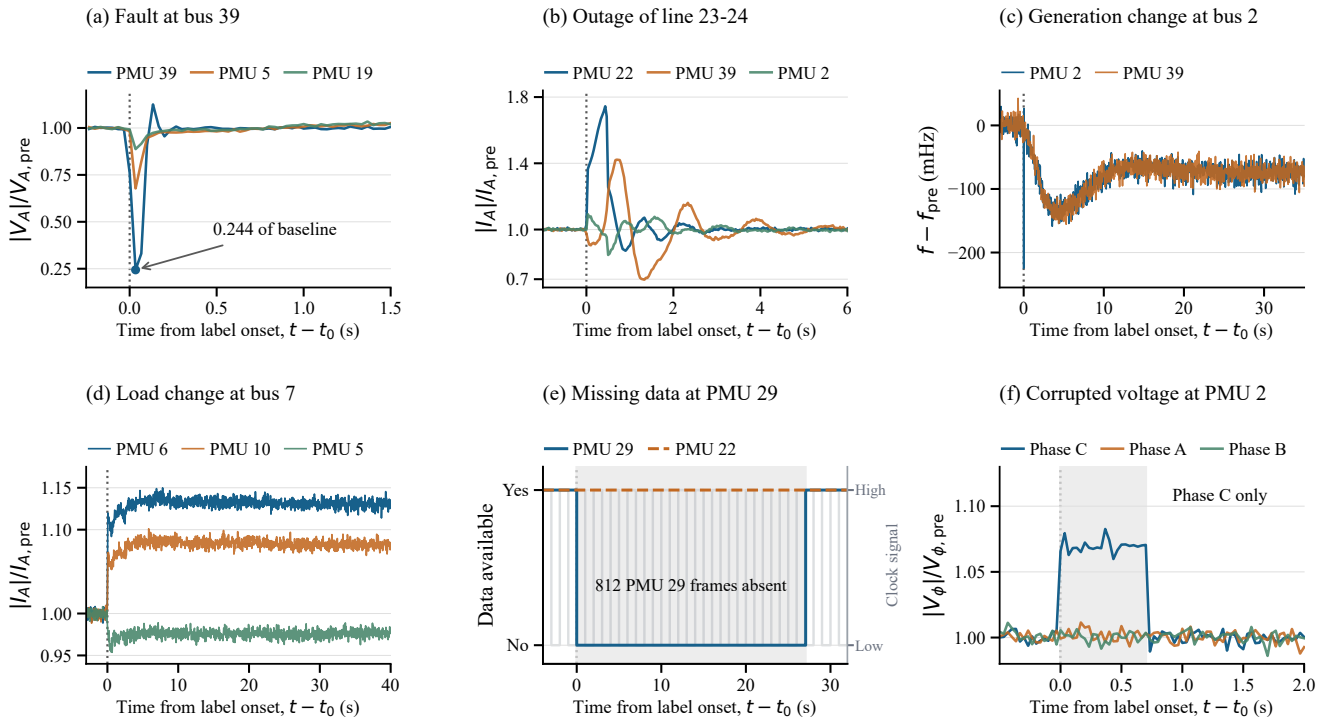


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. H_q selects electrical phasors; η_k retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is applied according to the declared generator contract.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \quad (4)$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

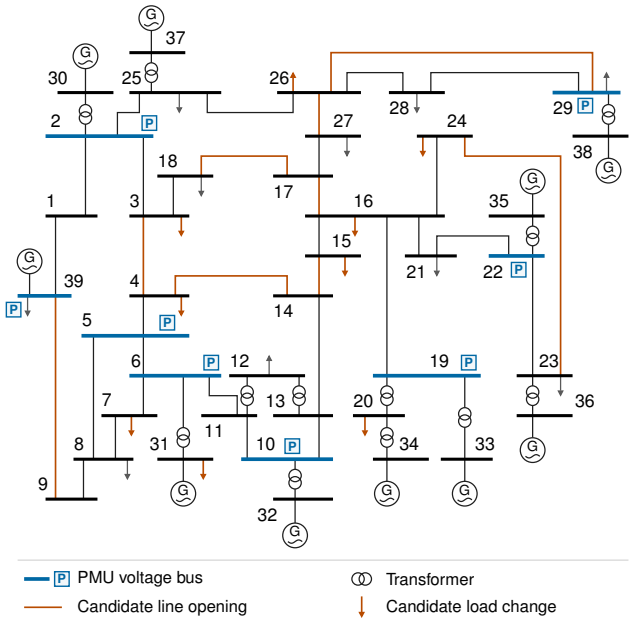


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

TABLE 2. **Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.**

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside the stream test
Unknown	Unresolved diagnosis	Not a new physical family

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insufficient: competing histories must also yield a timely source decision.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

On a regular index-one DAE chart with a fixed angle reference,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= A_q^{red}\delta x + B_q^{red}\delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

They propagate a perturbation at an unmeasured source to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through H_q, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed

colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = {} & \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ & + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

The conditional state mean is represented as $m_h + d_h a$. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral above. The regime covariances are obtained from nominal-operator Riccati equations; The implementation does not propagate an independent exact covariance for every switched history. At episode closure, it retains the mixture mean of the augmented state and records the omitted between-history spread. Thus state continuity and conditional Bayesian amplitude integration are implemented, while a fully Bayesian posterior over states, context and all histories is not.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{14}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t, \kappa)\delta x_h(\kappa), \\ \delta y_h^{net}(t) &\simeq C_0(t)\Phi_0(t, \kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{15}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

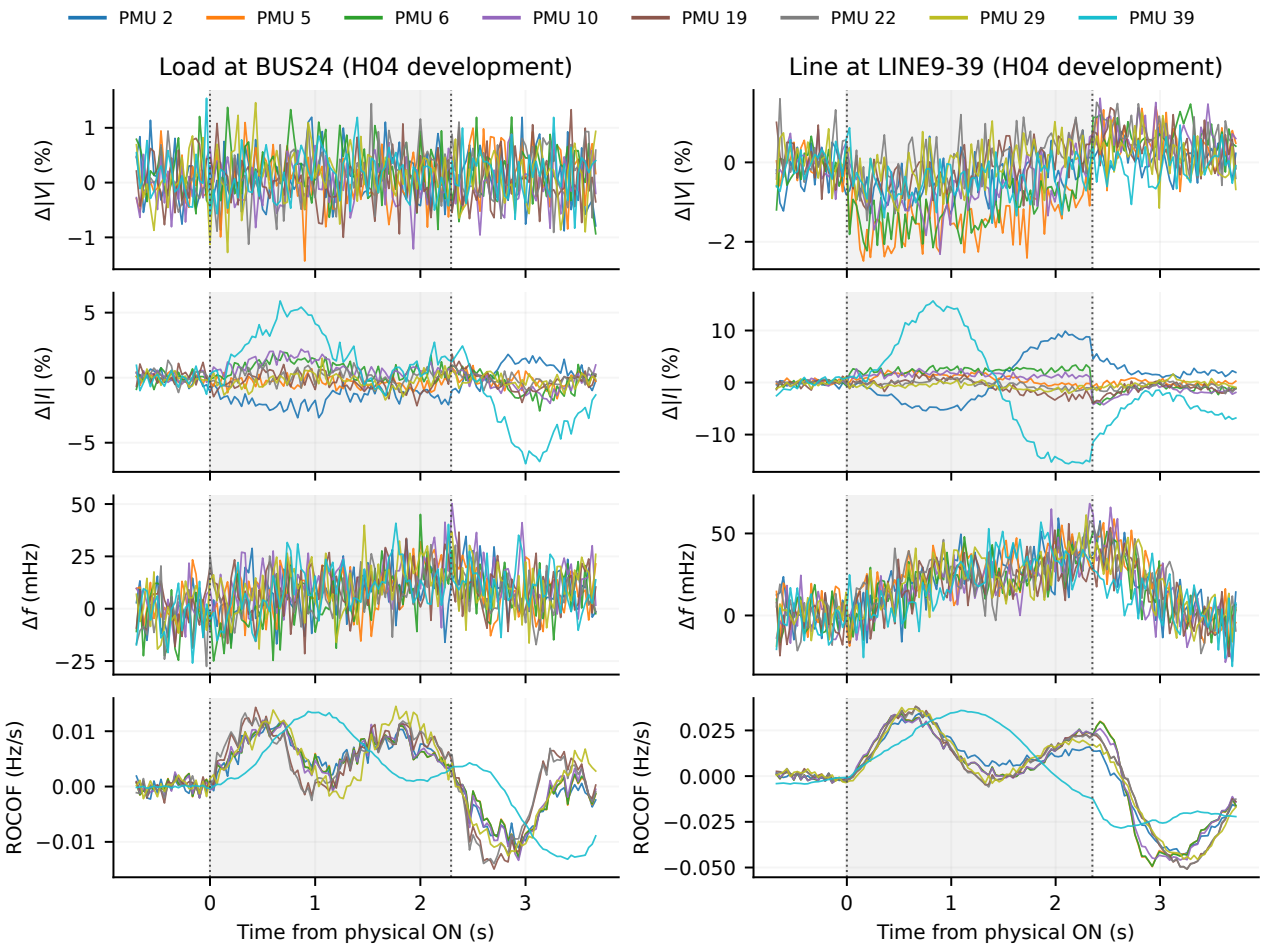


FIGURE 4. Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$M_{c,k} = \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\},$$
$$\Gamma_{c,k} = \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \tag{16}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\Delta M_{c,k} = \sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_{\ell}(w_{\ell,k} - w_{\ell,k-1})M_{\ell,c,k-1}. \tag{17}$$

The terms distinguish within-model evolution from model reweighting. This identity separates within-model evidence from network-model reweighting. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The fixed publication rules require active mass 0.99 for six reports to confirm an alarm, and restoration mass 0.99 for 15 reports to confirm OFF. The archived source/line confirmation and episode-replacement rules determine the public diagnosis. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory

ablation.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The reproducibility archive maps these operations to the frozen `transitions`, `process`, and `predictive_log_density` routines and preserves the complete publication configuration.

G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history h, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \propto L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h') \pi_{\ell,h',k-1}. \tag{18}$$

IMM filtering approximates mode-conditioned state mixtures; the M3H filter retains mode-sequence hypotheses [20], [21]. Equation (18) is that established mechanism. Here the modes are oriented electrical intervention operators, histories include restoration time, and the output separates active intervention from its originating cause through (16). Retrospective births reuse a stored pre-onset state, while closure compresses histories into a retained augmented-state mean. This specifies an application of history filtering rather than a new general recursion. The state-carry experiment tests one operation in this construction; the preliminary single-state control below requires independent, information-matched validation to identify the incremental benefit of explicit recovery hypotheses.

H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, fixed method configuration and successive PMU reports. Reserved event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the frozen centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the frozen short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the archived switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.

The simulator is the IEEE-39 Julia/PowerDynamics research candidate. Each physical parent is integrated continuously, preserving differential states through ON/OFF. Its PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001, including endpoints. At a switch-aligned report the export uses the post-switch algebraic value. Measurement references, load law and stochastic instrument parameters are fixed across events. RAW0001 supplied retrospective calibration context, not independent test data.

The two development campaigns each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

The five-model bank and its nominal version are frozen co-references. Both ExtraTrees artifacts and their causal preprocessing/publication adapters are frozen as well; neither is retrained on the development streams or the independent test. They receive the same 30-fps observations. This is a comparison of those operational implementations on the stated contract, not a rerun of the hackathon leaderboard.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [2]. Its reduced-

TABLE 3. Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.

Family	Candidate physical sources
Load buses	3, 4, 7, 15, 16, 20, 24, 26, 31
Lines	3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29

network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared research-plant adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all simulated instrumental effects. We preserve both matched-law implementation checks and model-mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

The independent campaign specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent results are followed by a development example and the controlled component study.

VI. RESULTS AND CONTROLLED COMPARISONS

A. INDEPENDENT EVALUATION: TIMELY ATTRIBUTION AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched event-stream alarms. Its nominal counterpart makes 643/648 with 92 extras. The frozen causal ExtraTrees reference makes 330/648 with 29; its short-window variant makes 352/648 with 16 extras. These are paired replays of 108 interventions, not 648 independent events. Figure 5 shows why the totals differ: the physical banks recover nearly all loads, while both learned adapters already recover most tested lines. The extra alarm burden of the physical banks rises sharply when true R/X changes fall outside their five-model bank.

Figure 6 separates a fast conditional median from coverage: a method can have short delays on its available diagnoses while leaving many load events without a timely source decision.

A posthoc audit of the original episode matches further separates coverage from classification. Every matched diagnosis emitted before OFF has the correct family and source in all four procedures; their errors in the timely endpoint are missing or late diagnoses. The load advantage therefore concerns timely coverage. It does not demonstrate higher conditional classification accuracy among diagnoses already available before OFF. The supplement shows the complete confusion view and counts unmatched alarms separately.

The paired intervals in Fig. 7 confirm a timely-attribution advantage over this ExtraTrees reference in each tested stratum. They also show a higher rate of unmatched alarms under out-of-bank R/X. Those intervals resample physical parents, preserving correlation across noise replays and network conditions. They do not establish superiority at a common false-alarm operating point. The physical bank's stratum median diagnosis delay is about 0.27 s in matched and in-bank conditions; the ExtraTrees medians range from 0.19 to 0.30 s. There is no general latency advantage.

The separate persistent-line test scores the first stop. Both physical banks and both ExtraTrees adapters identify the right line first on 162/162 openings. The declared Chen2016 adapter does so on 74/162; 66 records stop before the opening and 22 stop afterward on a wrong line. A separate descriptive audit of the same persistent records shows 164 and 179 later unmatched alarms for the two ExtraTrees procedures, versus 0 for the physical bank. Correct first identification therefore does not imply stable maintenance of a sustained event. This audit does not change the first-stop comparison or apply a recurrent score to Chen. Chen also stops on all 72 paired normal records, whereas the continuous methods alarm on none. Chen's normal exposure ends at its first stop (2,522 s); the other methods monitor 42,336 s each. These unequal exposures and the adapter's white-injection assumption prevent a general claim against published quickest-change detection. Full stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

The physical dictionary supplies responses for each catalog source, whereas the learned adapters retain their original

TABLE 4. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Supplementary signature test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Supplementary recovery geometry	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; separate from stream accuracy

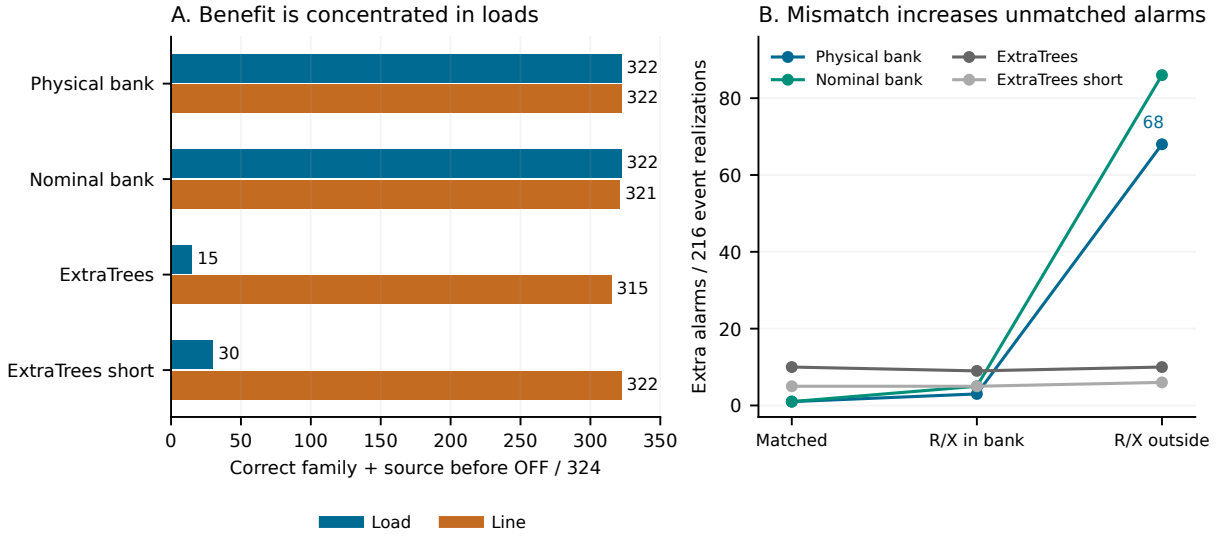


FIGURE 5. Sealed continuous evaluation. (A) Timely correct family and source by event family, with 324 paired realizations per family. (B) Unmatched event-stream alarms across the two noise levels within each R/X condition, with 216 event realizations per condition. The plots display the gain in load attribution and the alarm cost under out-of-bank network error together. Each method retains its own frozen publication rule and false-alarm budget.

TABLE 5. Sealed continuous-stream decisions. Each family has 324 replayed instances from 54 distinct interventions (108 across both families); extras are unmatched alarms on event records. The methods retain unequal alarm operating points.

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

training and feature maps. The comparison therefore measures transfer of those particular implementations, rather than an information-matched test of physics against supervised learning. It also uses different alarm operating points.

TABLE 6. Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

B. COMPUTATIONAL COST

The recorded median history-inference stage exceeds the record duration; its amortized processing rate is below the

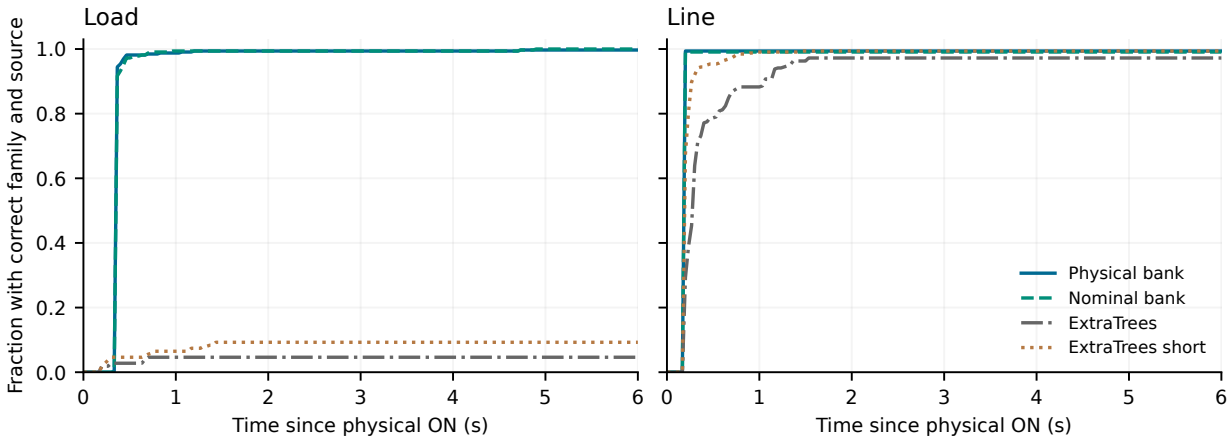


FIGURE 6. **Posthoc descriptive distribution of correct diagnosis time, using all 324 realizations per true family in each denominator. An incorrect or absent diagnosis contributes no cumulative mass. Curves include late correct decisions and are therefore distinct from the before-OFF endpoint; physical OFF varies between events. The correlated replays are not independent observations.**

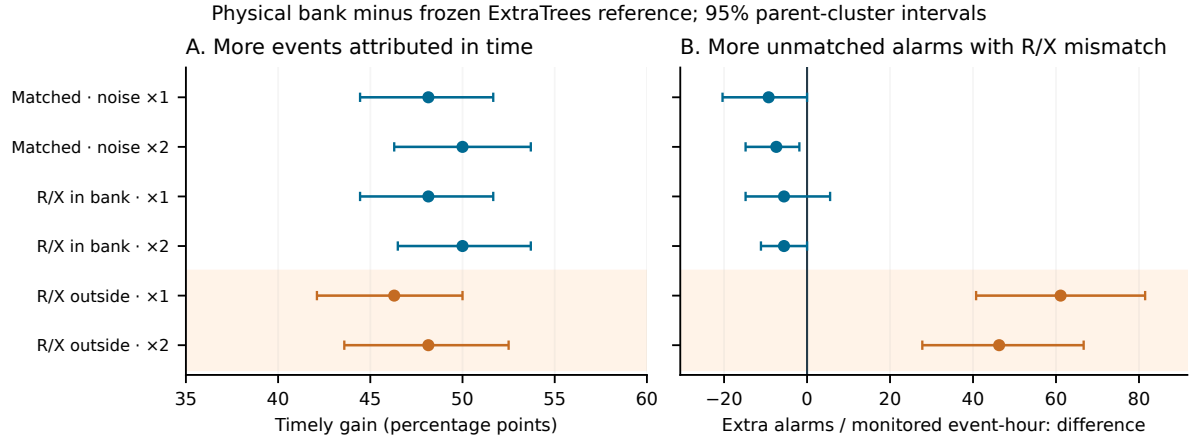


FIGURE 7. **Physical bank minus frozen ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per monitored event-record hour, excluding the 12-s startup from each 120-s record. The original rate differences and interval bounds are rescaled exactly by 120/108; counts and episode matches are unchanged. The orange rows expose the out-of-bank R/X failure. The intervals describe this finite simulated population.**

30-report/s input cadence. The stage ranges show substantial variation. Per-report latency percentiles, peak memory and a matched-hardware speed comparison were not recorded; causal emission timestamps must not be read as wall-clock latency.

C. A CONTINUOUS SEQUENCE AND A DELAYED DECISION

A fixed development trajectory illustrates the timing criterion (Fig. 8). The network continues moving after OFF; a later event must be evaluated against that predicted response. Under the tested out-of-bank network error, the bus-31 source is recovered only after restoration. It is correct eventually and fails timely attribution. The full development counts, active-mass trace, ΔM decomposition and failed admission control remain in the supplement.

D. A CONTINUOUS SINGLE-STATE PHYSICAL CONTROL

An existing open-development control retains one committed augmented physical state through restoration, rather than resetting to equilibrium. It uses the same nominal conditional plant and measurement operators, profiles severity over short alternatives lasting at most 19 reports, and applies the same six-report publication clock to all three procedures. The short GLR achieves 37/54 timely decisions with 37 extras, versus 44/54 and 6 for the nominal history bank, and 51/54 and 3 for the five-model bank. These 54 replays share three development parents; they are separate from the independent test. Different evidence scores, search and severity updates remain, so this is a preliminary continuous-observer comparison, not an isolated causal effect of history retention. Its complete equations and closure outcomes are in the supplement.

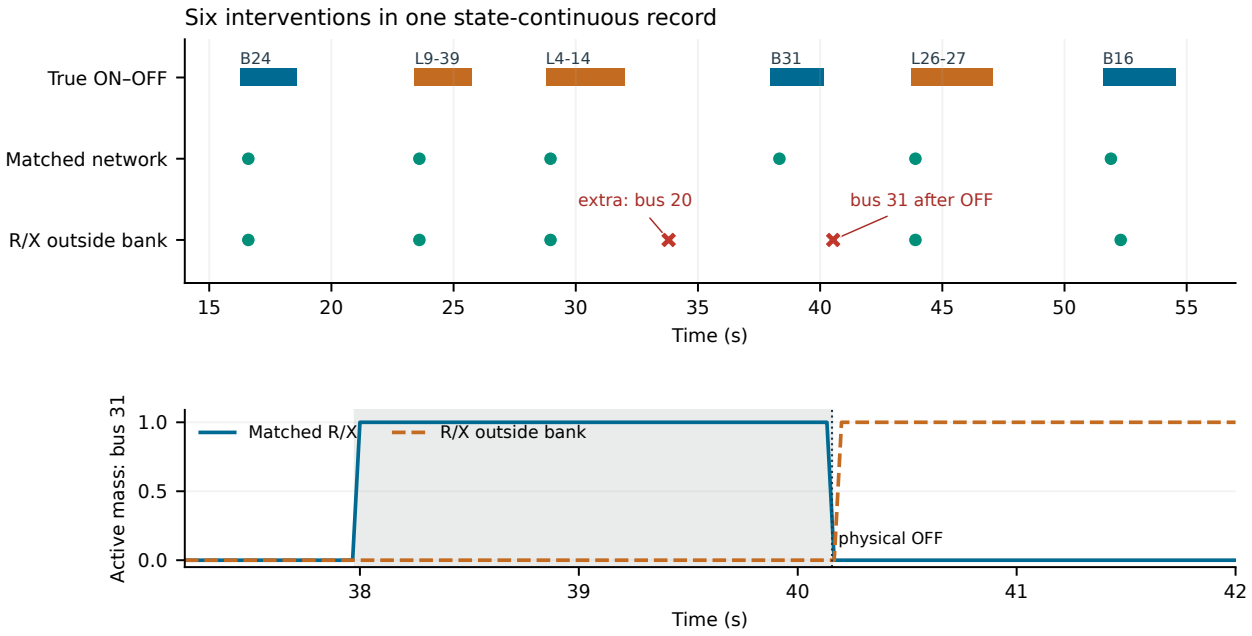


FIGURE 8. **One continuous development trajectory, shown for interpretation rather than selected as an independent test. Top: physical ON–OFF intervals and emitted diagnoses under matched and out-of-bank R/X. Bottom: bus-31 active mass, with the true intervention shaded. A diagnosis after OFF misses the timely endpoint even when it identifies the source. True timing enters only this retrospective display.**

E. CONTROLLED SEVERITY-SCORE AND STATE-CARRY STUDY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{19}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{20}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k/v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

1) Component Outcomes

Table 7 and Fig. 9 report coverage and alarm cost for the four arms. With carry retained, marginal and bounded-profile amplitude scores tie on the reported endpoints. Resetting the first 114 estimated physical-state deviations at inferred closure reduces timely attribution and increases extras under both scores. This reset leaves instrumental state, covariance regime, prior weights, thresholds, clocks and reporting rules intact; simulated states and observations never reset. The profile arm remains a normalized penalized score system and retains conditional Bayesian state estimation. Thus this study supports a physical-state-carry effect at inferred closure in the tested cohort, not the necessity of Bayes or of every form of temporal memory. The supplement reports denominators, normal exposure, persistent episodes, censoring and all paired parent-cluster intervals. These are posthoc within-simulator

TABLE 7. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

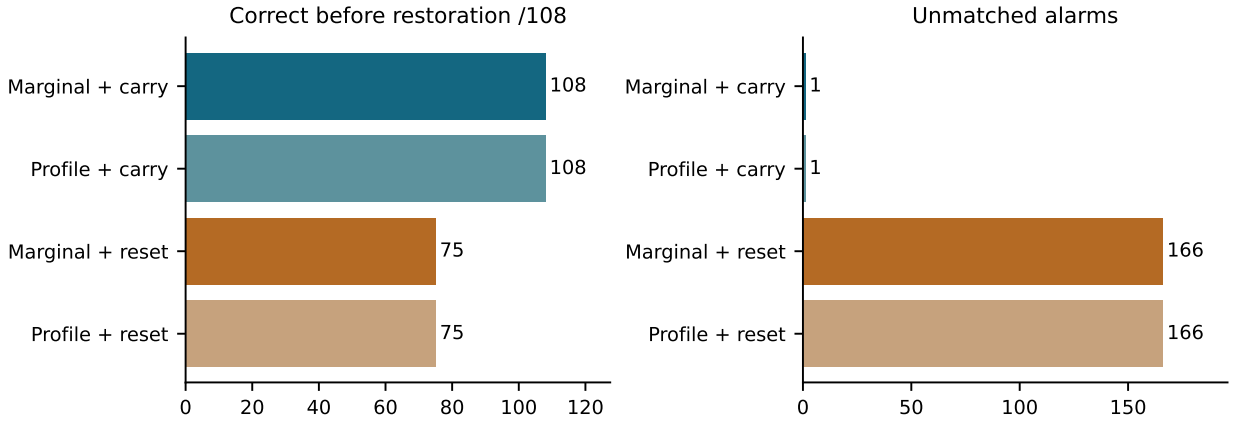


FIGURE 9. Amplitude-scoring and physical-state-carry ablation on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the observed alarm rate may differ between arms.

contrasts at the original thresholds, not a comparison at matched false-alarm rates.

VII. DISCUSSION AND LIMITS

A. PHYSICAL-STATE CONTINUITY AS A DESIGN CHOICE

The strongest controlled result is the state-carry contrast. Physical OFF restores an input or branch, while the machines retain their state. Removing those estimated deviations at inferred closure substantially worsens decisions on the same records. Marginal and bounded-profile severity scores tie on the measured endpoints; both retain conditional state estimation. This supports recovery-consistent prediction, without showing that Bayesian marginalization is necessary for those endpoints.

The history bank gives a concrete way to represent the remaining response and compare it with later interventions. Its complete operational behavior is measured, but the reset control is deliberately severe. A continuous physical observer without explicit recovery-history competition could retain the same consistency. The open-development single-state GLR comparison gives preliminary evidence against one such alternative, with different search and score semantics. Independent validation under matched information and a common calibration contract remains the key unresolved mechanism test.

B. WHAT IMPROVED RELATIVE TO THE LEARNED REFERENCE

The improvement concerns timely coverage of load sources under this known-catalog simulator contract. Line coverage is comparable to the short-window learned adapter. The physical method also issues more extra alarms under network mismatch. The result is therefore a coverage–alarm trade-off at fixed thresholds, rather than dominance at equal alarm rates or a higher official hackathon score. Engineering physical features is useful in the historical learned route; the physical bank additionally computes how each candidate intervention should affect remote PMUs. These routes have different access to source responses and training data.

A fairer architecture comparison would give a physical GLR or M3H control and a learned baseline the same candidate catalog and permitted development records, then select their operating points on validation data before a new test. Neither a posthoc threshold sweep on this test nor a new GNN name would establish that result. The present comparison preserves the original adapters and all their failures.

C. MODEL CORRESPONDENCE AND MEASUREMENT SCOPE

The nonlinear generator and local predictors share the same research plant family. Switched dynamics, nonlinear responses, noise and out-of-bank R/X challenge the inverse model, but do not remove this shared-model advantage. Independent dynamic-model or cross-simulator transfer remains untested. All nine load and nine line candidates were inherited from method development before the independent draw; they are a declared catalog, not a representative sample of all 39 buses or all event families.

The eight-PMU terminal map is a research measurement hypothesis. BUS5 retains the provisional 5–6 assignment,

and BUS39 current angle is excluded. The generator applies temporal and cross-PMU noise structure within each signal family, but does not apply its archived cross-family factor covariance. Frequency/ROCOF noise is added after phase estimation. The experiment therefore does not validate a complete physical PMU noise model, a producer's CT placement or exact reproduction of the historical reference record. It also assumes an event-free startup and available PMUs.

D. APPROXIMATION AND DECISION UNCERTAINTY

History pruning, shared regime covariances, plug-in startup estimates and state-mean compression make the bank approximate. Its masses are attribution weights under those approximations, not validated confidence levels. Discarded mass is recorded by the implementation; a calibrated reliability study and a bank-size sensitivity study are still missing. Response fidelity and source discrimination also differ: sources may remain close after fitting severity and context. The supplementary context-signature and recovery-geometry controls preserve their null source-decision results and do not supply an online identifiability certificate. A useful extension would test whether a causal separation margin predicts errors and delays before using it to justify abstention.

The measured research implementation runs below the input cadence. Decision timestamps, computation delay and queueing must remain separate. These results support further diagnostic development, rather than protection or control deployment.

VIII. CONCLUSION

A restored electrical intervention can remain the cause of the observed response. We formulated load/line attribution through physical histories that start from estimated past states and retain that response through restoration. Active and originating-cause weights separate the current intervention from its continuing physical effect.

The independent simulated comparison gives 644/648 timely family/source decisions and 72 extra alarms for the five-model bank, compared with 330/648 and 29 for the fixed causal ExtraTrees reference. The advantage is concentrated in loads; line coverage is similar to the short-window learned comparator. The controlled study identifies a strong state-carry effect, while marginal and profiled severity scores tie. Together, these results support an executable approach that uses electrical dynamics to attribute interventions from remote PMUs, preserves the response across restoration, and separates timely coverage from alarm burden. Physical-state continuity supplies the consistent reference needed to evaluate subsequent interventions during recovery.

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Digital Object Identifier

Supplementary Material for Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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ABSTRACT This supplement records the detailed literature matrix, electrical conventions, mathematical proofs, Fisher-information analysis, Bayesian estimator construction, frozen experimental contracts, and secondary results supporting the main article. It distinguishes the independently evaluated state component and implemented batch event estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. The latter remains unexecuted. This supplement does not add empirical claims or change any test denominator. Every empirical table and figure is generated from the same hashed evidence snapshot used by the main manuscript; separate deterministic tables audit stated algebraic identities.

INDEX TERMS Bayesian inference, Fisher information, functional observability, phasor measurement units, reproducibility, state estimation, event-source inference.

APPENDIX A DETAILED LITERATURE CAPABILITY MATRIX

Table 1 expands the family-level synthesis in the main paper. "Held source" means that the physical asset is absent from source-labeled training and tuning. The entries describe published tasks and assumptions; they are not a cross-paper performance ranking.

The direct competitors solve different contracts. State and unknown-input estimators define the latent reconstruction problem but not the event taxonomy. Learned localizers rank represented sources but rarely remove the physical asset from every source-labeled fitting operation. Model-based outage methods provide the closest likelihood precedent, while functional observability and placement studies provide the information tests. These distinctions determine the baselines and prevent cross-paper accuracy comparisons with incompatible targets.

APPENDIX B ELECTRICAL MODEL AND EVENT OPERATORS

Within a smooth network mode, the DAE is locally index one after fixing the reference angle. Differentiating the algebraic constraint gives

$$\dot{z} = -g_z^{-1}(g_x f + g_u \dot{u} + g_\theta \dot{\theta}). \tag{1}$$

This reduction is not applied across a topology switch or fault insertion, where algebraic variables may jump. Differential states remain continuous unless a simulator-specific reset is declared.

With $Y_{\mathrm{bus}} = G + \mathrm{j}B$ and $V_i = |V_i|e^{\mathrm{j}\vartheta_i}$, the network injections are

$$P_i = |V_i| \sum_j |V_j|(G_{ij}\cos\vartheta_{ij} + B_{ij}\sin\vartheta_{ij}), \tag{2}$$

$$Q_i = |V_i| \sum_j |V_j|(G_{ij}\sin\vartheta_{ij} - B_{ij}\cos\vartheta_{ij}). \tag{3}$$

For branch $e = (i,j)$ with complex tap t_e, series admittance y_e, and terminal shunts $y_e^{f,sh}$ and $y_e^{t,sh}$,

$$\begin{bmatrix} I_{e,f} \\ I_{e,t} \end{bmatrix} = \begin{bmatrix} (y_e + y_e^{f,sh})/|t_e|^2 & -y_e/t_e^* \\ -y_e/t_e & y_e + y_e^{t,sh} \end{bmatrix} \begin{bmatrix} V_i \\ V_j \end{bmatrix}. \tag{4}$$

The implemented PMU operator selects the actual terminal and orientation. Replacing this current with complete bus-injection current changes the likelihood.

The balanced generator represents one fundamental phasor and exports its phase-A coordinates. B/C are deterministic phase rotations; negative- and zero-sequence disturbances are

TABLE 1. Detailed capability comparison for the closest methodological works.

Work	State target	Event/source target	Physics mechanism	Sparse sensing	Held source	Integrity treatment
Katanic *et al.* [1]	Differential and algebraic DAE states	Unknown injectors	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Warner *et al.* [2]	Simulated transient paths	Short-circuit line candidates	DAE V/I trajectories and moving onset scan	Two PMUs in IEEE-39 case	Not a source-exclusion test	Load forecast/adverse-load tests
Chen / Rovatsos [3], [4]	Network angle increments	Outage identity	QCD; Rovatsos includes transient phases	PMU subset	Not a learned source task	Stochastic injection law
Warner / Fellouris [5]	General sequential observations	Post-change alternative	Adaptive Matrix CuSum	Not a PMU placement study	Finite specified alternatives	Error/delay theory under stated laws
Nugroho *et al.* [6]	Differential and algebraic DAE states	Unknown input and input-sensor failure	Outlier-resistant DAE observer	PMU voltage/current	N/A	Failure tolerance
Qu *et al.* [7]	Machine state	Unknown field voltage and torque	Event-triggered nonlinear state/input estimator	Event-triggered PMUs	N/A	Covariance bounds
Ngo *et al.* [8]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue *et al.* [9]	Distribution-system state	Line outage	Grid-model-informed graph learning	Yes	Not stated	Outlier-resistant SE
Moshtagh *et al.* [10]	Transmission-system state	Topology variation	Topology-aware graph learning	PMU-unobservable systems	Not a source task	PMU failure and bad data
Pandey *et al.* [11]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [12]	None	Fault detection and location	OLS/CNN and PMU placement	Yes	Not stated	No
Li *et al.* [13]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Kummerow *et al.* [14]	None	Known/unknown event type and location	Siamese distance and learned rejection	Limited/full sets	Open-set rejection	Rejection boundary
Hu and Cheng [15]	None	Open-set event detection/classification	Generative window evidence and fusion	Streaming PMUs	Unknown-event category	No source posterior
Rimkus *et al.* [16]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	Imputer only	Corruption via imputation
Zhai *et al.* [17]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmarks	Attack target only
R11 in main article	Retained conditional states	Successive nine-load/nine-line catalog	DAE histories, recovery and fixed R/X mixture	Eight PMU buses	Not tested; no source-label head	Colored instrument; plug-in startup
Separate batch component	Known-context voltage/current	Load supports with $K \leq 2$	Original GH31/W2 likelihood	Eight PMU buses	Different contract from R11	Matched frozen noise

not simulated in this evaluation. Ideal continuous-frequency channels satisfy

$$f_i^{\mathrm{phys}} = f_0 + \frac{1}{2\pi}\frac{d\angle V_i}{dt}, \qquad \mathrm{ROCOF}_i^{\mathrm{phys}} = \frac{1}{2\pi}\frac{d^2\angle V_i}{dt^2}, \tag{5}$$

but an operational likelihood must reproduce the causal filtering of the PMU backend.

The preceding equations define the ideal physical quantities f_i^{phys} and $\mathrm{ROCOF}_i^{\mathrm{phys}}$ implied by the network trajectory. Reported quantities f_i^{meas} and $\mathrm{ROCOF}_i^{\mathrm{meas}}$, however, are outputs of a dynamic frequency-estimation process and can exhibit disturbance- and estimator-dependent transient error and latency. A prior benchmark by the authors compared 16 dynamic frequency estimators spanning six practical groups over 34 disturbance scenarios and quantified these trade-offs under ramps, phase jumps, harmonics, ringdown, and composite IBR disturbances [18]. We therefore distinguish

$$\begin{bmatrix} f_i^{\mathrm{meas}} \\ \mathrm{ROCOF}_i^{\mathrm{meas}} \end{bmatrix} = \begin{bmatrix} f_i^{\mathrm{phys}} \\ \mathrm{ROCOF}_i^{\mathrm{phys}} \end{bmatrix} + \boldsymbol{\varepsilon}_{fr,i}, \qquad \mathrm{Cov}(\boldsymbol{\varepsilon}_{fr,i}) = \boldsymbol{\Omega}_{fr}. \tag{6}$$

Because phasor, frequency, and ROCOF outputs may be estimated from overlapping underlying waveform or phasor information, their errors need not be conditionally independent. An augmented likelihood therefore permits cross-covariance

$$\boldsymbol{\Omega}_{\mathrm{aug}} = \begin{bmatrix} \boldsymbol{\Omega}_{VI} & \boldsymbol{\Omega}_{VI,fr} \\ \boldsymbol{\Omega}_{fr,VI} & \boldsymbol{\Omega}_{fr} \end{bmatrix}; \tag{7}$$

neither the magnitude of $\boldsymbol{\Omega}_{VI,fr}$ nor a block-diagonal approximation is assumed without calibration. Frequency and ROCOF are retained as candidate channels for an augmented sparse-PMU likelihood. Their incremental source-discrimination value, covariance structure, and effect on localization delay are not assumed here and require the dedicated VI, $VI + f$, $VI + \mathrm{ROCOF}$, and $VI + f + \mathrm{ROCOF}$ ablation.

A. F01 MEASUREMENT CONTRACT

The executed channel map is taken from the frozen SIM0001-JuliaPD-F01 research snapshot. The eight voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. The internal voltage phasor is in per unit with unit gain and is exported as phase-to-neutral

TABLE 2. Physical intervention operators represented by the common DAE formulation.

Family	DAE action at source ℓ	Candidate support	Severity coordinates	State continuity
Bus fault	Add a shunt primitive for the clearing interval	admissible buses	complex admittance or resistance/reactance	x continuous; z may jump
Line outage	Remove branch tap, series, charging, and shunt primitives	in-service branches	status and switching time	x continuous; z may jump
Generation change	Modify scheduled injection or a declared governor reference	generator buses	ΔP_G, ΔQ_G, duration	continuous input-driven response
Load change	Modify the active/reactive withdrawal model	load buses	ΔP_L, ΔQ_L, ZIP convention	continuous input-driven response

RMS volts using 345 kV/$\sqrt{3}$; voltage angle uses one common affine phase reference plus the archived relative offsets. The reference interpretation is observational: physical common drift and instrument clock drift are not separately identified. Terminal currents are converted to outward branch convention from the simulator's into-terminal observable, then mapped with the row-specific polarity, gain and phase bias. Current bases use $100\,\mathrm{MVA}/(\sqrt{3}\,V_{\mathrm{terminal}})$; the common voltage base is 199.186 kV.

Frequency is computed as $60 + \dot{\theta}/(360)$ from the reported voltage angle, and ROCOF as $\ddot{\theta}/360$ using a causal 21-report quadratic estimate. The generator fit is robust and magnitude weighted; the predictor uses a fixed unweighted quadratic fit over the same history. Four warmup outputs are unavailable; the confirmatory evaluation uses a 10-s startup context and begins monitoring after 12 s. The 47 retained coordinates exclude the BUS39 current-angle coordinate. Noise multipliers 1 and 2 scale the standard deviation under the frozen stream. The stochastic model archives a cross-signal covariance field, but the executed report operator applies noise by signal family without that cross-family factor. These choices describe a simulated sensor configuration, not a verified eight-PMU CT/PT installation.

APPENDIX C METRICS, CAUSALITY, AND EVALUATION UNITS

For trajectory r, hidden-voltage relative error is

$$\mathrm{HVRE}^{(r)} = \frac{1}{T_r|\mathcal{U}|}\sum_{k=1}^{T_r}\sum_{i\in\mathcal{U}}\frac{|\widehat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|,\epsilon)}. \tag{8}$$

It is a trajectory-and-bus mean reported as a percentage; it is not the standard per-frame synchrophasor TVE conformity statistic. Detection, event-family macro-F1, exact physical Top-1/Top-3, cardinality, candidate-set coverage, and integrity-source Top-1 retain their own populations. A joint outcome is correct only when every applicable component is correct.

A causal estimate at k uses $y_{0:k}$. A fixed-lag smoother, complete-window score, or centered temporal feature is retrospective. For source generalization, the physical asset is absent from source-labeled training, feature fitting, threshold selection, and calibration. Intervals and significance tests use the complete trajectory, event parent, or rebuilt plant as the independent unit; samples within a 30-Hz stream are correlated observations, not replicates.

APPENDIX D PROOFS AND ADDITIONAL DIAGNOSABILITY RESULTS

This appendix develops the mathematical chain used by the paper. It begins with the sampled hybrid event map, separates state-functional recovery from event-source separation, derives the relevant Fisher information after physical nuisance is removed, and then states the finite-horizon and temporal limits. The results are local unless a statement explicitly says otherwise.

A. REGULARITY AND STATISTICAL CONTRACT

Let $u = \mathrm{col}(x, z)$ collect differential and algebraic coordinates and consider the descriptor model

$$E\dot{u} = F(u,p), \qquad y = h(u,p). \tag{9}$$

The analysis uses four assumptions.

1) The reference angle has been fixed and the algebraic Jacobian is nonsingular on the local mode under study, so a regular index-one chart exists between discrete events.
2) The scheduled event time is fixed. In a neighborhood of zero amplitude, the complete callback–solver–sampling map is twice continuously differentiable.
3) Every covariance used as a metric is positive definite on the retained measurement subspace. Candidate-dependent covariances retain their log determinants in likelihood comparisons.
4) The support set is finite and all Bayesian priors are proper. Exact signed-amplitude equivalences use a neighborhood of zero within each support subspace; constrained physical amplitudes instead use their actual difference set.

These assumptions distinguish the local theory from general descriptor left invertibility, state-triggered saltation, and global nonlinear identifiability.

B. SIMULATOR-CONSISTENT EVENT SENSITIVITIES

For a fixed-time, impulse-free event, the differential state is continuous, but the algebraic state must satisfy the post-event constraint. This physical right limit must be distinguished from the values stored inside a simulator callback and from the first sampled frame. The following result makes that distinction explicit.

Theorem 1 (Local consistency-jump expansion): Let $x_0 = x(\tau^-)$ and suppose $g_+(x_0, z_0, 0) = \mathbf{0}$. If g_+ is three

TABLE 3. Eight PMU rows in the frozen F01 measurement map. V is a bus voltage in p.u.; I is an oriented branch-terminal phasor. Gains and phase biases are applied to current channels. BUS5 retains the historical 5–6, terminal-5 hypothesis and its producer CT identity is unverified. BUS39 current phase is partially validated.

PMU	V bus	Edge	Branch	Terminal	Side	Polarity	Gain	Bias (°)	I unit	Provenance
2	2	2	2–3	3	dst	1	0.9581	-1.173	A	F01 archived terminal hypothesis; CT unverified
5	5	9	5–6	5	src	1	1.0361	0.312	A	BUS5 5–6 historical hypothesis; CT unverified
6	6	11	6–7	6	src	1	1.0077	1.495	A	F01 archived terminal hypothesis; CT unverified
10	10	17	10–11	11	dst	-1	1.0041	2.224	A	F01 archived terminal hypothesis; CT unverified
19	19	26	16–19	19	dst	1	1.0384	0.481	A	F01 archived terminal hypothesis; CT unverified
22	22	34	21–22	22	dst	1	1.0189	0.094	A	F01 archived terminal hypothesis; CT unverified
29	29	44	28–29	29	dst	-1	1.0316	-6.170	A	F01 archived terminal hypothesis; CT unverified
39	39	1	1–39	39	dst	-1	0.9298	30.053	A	BUS39 partial phase validation

times continuously differentiable and g_z is nonsingular at $(x_0, z_0, 0)$, there is a unique local consistent branch

$$z_+(a) = z_0 + az_1 + \tfrac{1}{2}a^2 z_2 + O(a^3), \tag{10}$$

where all derivatives below are evaluated at the nominal right-limit mode and

$$z_1 = -g_z^{-1} g_a, \tag{11}$$
$$z_2 = -g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right). \tag{12}$$

The corresponding differential state satisfies $x(\tau^+) = x(\tau^-)$, and its first event-amplitude sensitivity starts with

$$\dot{x}_a(\tau^+) = f_a - f_z g_z^{-1} g_a \equiv B_{r,a}. \tag{13}$$

Its second-order initial forcing is

$$\dot{x}_{aa}(\tau^+) \equiv B_{r,aa} = f_{zz}[z_1, z_1] + 2f_{za}z_1 + f_{aa} - f_z g_z^{-1}\left(g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa}\right). \tag{14}$$

a: Proof.
The implicit-function theorem applied to $g_+(x_0, z_+(a), a) = \mathbf{0}$ gives a unique smooth local branch. Differentiating once gives $g_z z_1 + g_a = \mathbf{0}$, which yields (11). Differentiating a second time gives $g_z z_2 + g_{zz}[z_1, z_1] + 2g_{za}z_1 + g_{aa} = \mathbf{0}$, which yields (12). An impulse-free fixed-time switch does not reset the energy-storage coordinates, so the differential state is continuous. Differentiating $\dot{x} = f_+(x, z_+(a), a)$ at the right limit, with $x_a(\tau^+) = \mathbf{0}$, gives (13). A second differentiation, using $x_{aa}(\tau^+) = \mathbf{0}$ and (12), gives (14). □

The theorem also defines an implementation-independent consistency diagnostic. If a callback changes the parameter but temporarily retains z_0, its algebraic residual is

$$r_{\mathrm{cb}}(a) = g_+(x_0, z_0, a) = ag_a + \tfrac{1}{2}a^2 g_{aa} + O(a^3). \tag{15}$$

Substituting the zeroth-, first-, and second-order approximations to (10) produces constraint residuals of orders $O(a)$, $O(a^2)$, and $O(a^3)$, respectively. Central finite differences of an independently solved algebraic branch can therefore test both derivatives and the expected convergence orders without relying on undocumented callback storage.

For a differentiable PMU map $y = h(x, z, a)$, the direct right-limit signature is

$$y_a(\tau^+) = h_a - h_z g_z^{-1} g_a, \qquad \Delta y_{\mathcal{P}} = aM_{\mathcal{P}} y_a(\tau^+) + O(a^2). \tag{16}$$

This direct algebraic visibility is not the entire event signature. Even when (16) is nearly zero at the retained PMUs, (13) can generate a distinguishable dynamic response over later frames. Instantaneous invisibility therefore does not imply finite-horizon nonidentifiability.

The frozen artifacts begin at the first retained 30-Hz frame and do not record the callback microstate or an independently solved algebraic right limit. Accordingly, the paper treats the right-limit equations above as the physical contract, not as an observed fact about the current ANDES callback implementation. The relevant differentiable object for the existing response dictionary is the sampled map

$$\Psi_{T_s}^{\mathrm{sim}} : (u^-, p(a)) \mapsto u(\tau + T_s; a), \qquad T_s = 1/30\ \mathrm{s}. \tag{17}$$

For event coordinates a_i, define

$$L_i = \left.\frac{\partial \Psi_{T_s}^{\mathrm{sim}}}{\partial a_i}\right|_{a=\mathbf{0}}, \qquad L_{ij}^{(2)} = \left.\frac{\partial^2 \Psi_{T_s}^{\mathrm{sim}}}{\partial a_i \partial a_j}\right|_{a=\mathbf{0}}. \tag{18}$$

For $t \geq \tau + T_s$, the first and second sensitivities obey

$$E\dot{s}_i = F_u s_i + F_p p_i, \tag{19}$$
$$E\dot{s}_{ij} = F_u s_{ij} + r_{ij}, \tag{20}$$

where

$$r_{ij} = F_{uu}[s_i, s_j] + F_{up}[s_i, p_j] + F_{pu}[p_i, s_j] + F_{pp}[p_i, p_j] + F_p p_{ij}. \tag{21}$$

The initial conditions are $s_i(\tau + T_s) = L_i$ and $s_{ij}(\tau + T_s) = L_{ij}^{(2)}$. For the affine event parameterization, $p_{ij} = \mathbf{0}$.

Theorem 2 (Propagation of event-map mismatch): Suppose two second-order models have the same nominal trajectory, first-order sensitivities, parameter derivatives, and forcing in (20), but different second-order event-map initial conditions. Their difference $e_{ij} = s_{ij}^A - s_{ij}^B$ satisfies

$$E\dot{e}_{ij} = F_u e_{ij}, \qquad e_{ij}(t) = \Phi_E(t, t_1) e_{ij}(t_1) \tag{22}$$

on the consistent solution subspace, where $t_1 = \tau + T_s$.

b: Proof.

Subtract (20) for the two models. Every forcing term cancels by assumption, leaving the homogeneous descriptor variational equation. Applying its transition operator gives (22). □

When the retained PMU operator is locally affine, $y = H_y u + d$, the sampled response coefficients are

$$D_i = H_y s_i, \quad Q_i = \tfrac{1}{2} H_y s_{ii}, \quad Q_{ij} = H_y s_{ij}. \tag{23}$$

Thus all quadratic curvature comes from the physical solution and event map, not from PMU post-processing. This is why the dictionary must differentiate the same event semantics used to generate the trajectories.

C. EXACT FUNCTIONAL RECONSTRUCTION

Consider the finite-horizon linear experiment

$$Y = \mathcal{O}x + \varepsilon, \qquad \psi = Lx, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, R), \quad R \succ \mathbf{0}. \tag{24}$$

Theorem 3 (Functional reconstruction equivalence): The following statements are equivalent: (i) $\mathcal{O}x_1 = \mathcal{O}x_2$ implies $Lx_1 = Lx_2$; (ii) $\ker \mathcal{O} \subseteq \ker L$; and (iii) there is a linear map K such that $L = K\mathcal{O}$.

a: Proof.

If (i) holds and $v \in \ker \mathcal{O}$, the observations of v and $\mathbf{0}$ coincide, hence $Lv = \mathbf{0}$ and (ii) follows. If (ii) holds, define $K(\mathcal{O}x) = Lx$ on $\operatorname{range} \mathcal{O}$. The definition is independent of the representative because two representatives differ by a vector in $\ker \mathcal{O} \subseteq \ker L$. Extend K linearly to the observation space to obtain (iii). Finally, (iii) immediately implies (i). □

Let

$$G = \mathcal{O}^{\mathsf{T}} R^{-1} \mathcal{O}. \tag{25}$$

Theorem 4 (Best linear unbiased estimate of an estimable function): If the equivalent conditions of Theorem 3 hold, then

$$\widehat{\psi}_\star = L G^{\dagger} \mathcal{O}^{\mathsf{T}} R^{-1} Y, \tag{26}$$

$$\operatorname{Cov}(\widehat{\psi}_\star) = L G^{\dagger} L^{\mathsf{T}}. \tag{27}$$

It has minimum covariance, in the positive-semidefinite order, among all linear unbiased estimators of Lx [19].

b: Proof.

Because $v^{\mathsf{T}} G v = \|\mathcal{O}v\|^2_{R^{-1}}$, $\ker G = \ker \mathcal{O}$. Estimability gives $L = LG^{\dagger}G$, so taking expectation in (26) yields Lx. Direct covariance propagation gives (27). Let AY be any other linear unbiased estimator and set $T = A - LG^{\dagger}\mathcal{O}^{\mathsf{T}}R^{-1}$. Then $T\mathcal{O} = \mathbf{0}$ and the cross covariance vanishes because $LG^{\dagger}\mathcal{O}^{\mathsf{T}}T^{\mathsf{T}} = \mathbf{0}$. Consequently,

$$\operatorname{Cov}(AY) = \operatorname{Cov}(\widehat{\psi}_\star) + TRT^{\mathsf{T}} \succeq \operatorname{Cov}(\widehat{\psi}_\star). \tag{28}$$

If the kernel inclusion fails, two states induce the same data law but different targets, so no estimator can be unbiased for every state. □

The pseudoinverse has this interpretation only for estimable targets. In the nonlinear DAE, a pointwise kernel test is not sufficient: local factorization requires the condition to hold on a regular constant-rank neighborhood with connected local fibers [20].

D. FISHER INFORMATION WITH PHYSICAL NUISANCE

For a differentiable Gaussian model

$$r \mid \boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{\Sigma}(\boldsymbol{\theta})), \tag{29}$$

the Fisher information contains both mean and covariance contributions. For scalar a,

$$I(a) = \boldsymbol{\mu}_a^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_a + \tfrac{1}{2} \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}_a\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}_a). \tag{30}$$

The second term must be retained when event amplitude changes the covariance. The evaluated load bank freezes $\boldsymbol{\Sigma}$, so only the first term is used.

To expose nuisance geometry, consider

$$r = ad + N\boldsymbol{\nu} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{31}$$

where $\boldsymbol{\nu}$ may contain initial-state, operating-point, parameter, or integrity directions. Let $W^{\mathsf{T}}W = \boldsymbol{\Sigma}^{-1}$, $\overline{d} = Wd$, $\overline{N} = WN$, and

$$P_\perp = I - \overline{N}\overline{N}^{\dagger}. \tag{32}$$

Theorem 5 (Efficient Fisher information after nuisance profiling): For deterministic unknown $\boldsymbol{\nu}$, the efficient information for a is

$$I^{\mathrm{det}}_{a|\nu} = \overline{d}^{\mathsf{T}} P_\perp \overline{d} \tag{33}$$

$$= d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d - d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N(N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}N)^{\dagger}N^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d. \tag{34}$$

It is zero exactly when the event tangent lies in the observable nuisance span.

a: Proof.

After whitening, the score for a is $\overline{d}^{\mathsf{T}}\overline{\varepsilon}$. Nuisance scores span $\{v^{\mathsf{T}}\overline{\varepsilon} : v \in \operatorname{range}\overline{N}\}$. Their Fisher inner product is Euclidean because $\mathbb{E}[\overline{\varepsilon}\overline{\varepsilon}^{\mathsf{T}}] = I$. Removing the nuisance-score projection leaves $(P_\perp \overline{d})^{\mathsf{T}}\overline{\varepsilon}$, whose variance is (33). Expanding the projector by an SVD gives (34). □

If instead $\boldsymbol{\nu} \sim \mathcal{N}(m_\nu, \boldsymbol{\Gamma})$ with a proper prior, marginalization yields

$$r \mid a \sim \mathcal{N}(ad + Nm_\nu, \boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}}), \tag{35}$$

$$I^{\mathrm{Bayes}}_{a|\nu} = d^{\mathsf{T}}(\boldsymbol{\Sigma} + N\boldsymbol{\Gamma}N^{\mathsf{T}})^{-1}d. \tag{36}$$

Woodbury's identity shows that (36) tends to the no-nuisance information as $\boldsymbol{\Gamma} \to \mathbf{0}$ and to (33) as a full-rank nuisance prior becomes diffuse. Thus zero deterministic efficient information is not a universal impossibility statement: a restrictive proper prior can still supply separation.

Corollary 1 (Pre-event information cannot reduce local event information): If pre-event data add nuisance precision $\boldsymbol{\Lambda}_{\mathrm{pre}} \succeq \mathbf{0}$, then

$$I_{a|\nu,\mathrm{pre}} = I_{aa} - I_{a\nu}(I_{\nu\nu} + \boldsymbol{\Lambda}_{\mathrm{pre}})^{-1} I_{\nu a} \tag{37}$$

is nondecreasing in $\boldsymbol{\Lambda}_{\mathrm{pre}}$.

b: Justification.

If $\mathbf{\Lambda}_2 \succeq \mathbf{\Lambda}_1 \succeq \mathbf{0}$, matrix inversion reverses the order. Premultiplication and postmultiplication by the cross-information blocks preserve the positive-semidefinite order, and subtraction reverses it once more. □

For a known source, known positive sign, and fixed covariance, the standardized matched statistic

$$T = \frac{d^\top \mathbf{\Sigma}^{-1} r}{\sqrt{d^\top \mathbf{\Sigma}^{-1} d}} \tag{38}$$

is standard normal under H_0 and has mean $a\sqrt{I}$ under amplitude a. A level-α one-sided test has power $1-\Phi(z_{1-\alpha}-a\sqrt{I})$. Hence the amplitude required for power $1-\beta$ is

$$a_{\alpha,\beta} = \frac{z_{1-\alpha} + z_{1-\beta}}{\sqrt{I}}. \tag{39}$$

This is a conditional known-source benchmark, not the threshold of the 137-support search. Its empirical use in the paper is limited to the exploratory source-wise detectability association.

E. FIRST-ORDER MULTI-EVENT SUPPORT GEOMETRY AND ONSET INFORMATION

Let two first-order source signatures be d_g and d_j under fixed precision $G \succeq \mathbf{0}$. Allow the competing source to refit its amplitude. The local profiled distance satisfies

$$b_\star(a) = a\frac{d_j^\top G d_g}{d_j^\top G d_j} + O(a^2), \tag{40}$$

$$\Delta^2_{g\to j}(a) = a^2 J_{g\to j} + O(a^3), \tag{41}$$

$$J_{g\to j} = d_g^\top G d_g - \frac{(d_g^\top G d_j)^2}{d_j^\top G d_j} \geq 0. \tag{42}$$

Equation (42) is a pairwise local separation quantity, not the Fisher information for amplitudes conditional on the true support.

The pairwise formula extends to arbitrary supports. Let N_c be the unwhitened nuisance Jacobian, let $G_e = [g_1, \ldots, g_M]$ collect the event tangents, and define

$$A = WN_c, \quad F = WG_e, \quad R_N = I - AA^\dagger, \quad \overline{F} = R_N F. \tag{43}$$

Theorem 6 (Profile-likelihood diagnostic quotient): Consider the whitened local experiment

$$y = A\boldsymbol{\nu} + F\boldsymbol{\alpha} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I), \tag{44}$$

with deterministic unrestricted nuisance $\boldsymbol{\nu}$. Let $U_\mathcal{D}$ have orthonormal columns spanning $\mathcal{D} = \mathrm{range}(\overline{F})$, and define

$$z = U_\mathcal{D}^\top R_N y, \qquad d_\mathcal{D} = \dim \mathcal{D} = \mathrm{rank}(\overline{F}). \tag{45}$$

Then every profile-likelihood ratio over $\boldsymbol{\alpha}$ depends on y only through z:

$$\min_{\boldsymbol{\nu}} \|y - A\boldsymbol{\nu} - F\boldsymbol{\alpha}\|_2^2 = \|z - U_\mathcal{D}^\top \overline{F}\boldsymbol{\alpha}\|_2^2 + C(y), \tag{46}$$

where $C(y)$ is independent of $\boldsymbol{\alpha}$. The projector, quotient statistic up to an orthogonal change of coordinates, and $d_\mathcal{D}$ depend only on the nuisance span, not its parameterization.

a: Proof.

Profiling nuisance gives $\|R_N(y - F\boldsymbol{\alpha})\|_2^2$. Decompose this residual orthogonally into $\mathcal{D}$ and $\mathrm{range}(R_N) \cap \mathcal{D}^\perp$. The first component is the squared norm in (46); because $\overline{F}\boldsymbol{\alpha} \in \mathcal{D}$, the second is $C(y) = \|(I - U_\mathcal{D} U_\mathcal{D}^\top) R_N y\|_2^2$ and contains no $\boldsymbol{\alpha}$. Reparameterizing nuisance without changing $\mathrm{range}(A)$ leaves $AA^\dagger$ and therefore every stated object unchanged. □

Thus $d_\mathcal{D}$ is the intrinsic dimension of the *linear profiled* event experiment. It is not a nonlinear global observability count and need not equal the number of PMUs. If candidates use different nuisance spans, or if nuisance has an informative proper prior, the corresponding candidate-specific profile or marginal likelihood must be retained; the common projector above is then insufficient.

Theorem 7 (Signed first-order support-to-support separation): For true support S with fixed severity direction $\boldsymbol{\alpha}_S$ and competing support C with freely fitted local amplitudes, define

$$\gamma^2_{S\to C}(\boldsymbol{\alpha}_S) = \min_{u,\boldsymbol{\beta}_C} \|F_S\boldsymbol{\alpha}_S - Au - F_C\boldsymbol{\beta}_C\|_2^2. \tag{47}$$

Then

$$\gamma^2_{S\to C}(\boldsymbol{\alpha}_S) = \| \left(I - P_{\overline{F}_C}\right) \overline{F}_S \boldsymbol{\alpha}_S\|_2^2, \tag{48}$$

$$= \boldsymbol{\alpha}_S^\top K_{S|C} \boldsymbol{\alpha}_S, \tag{49}$$

$$K_{S|C} = \overline{F}_S^\top \left(I - P_{\overline{F}_C}\right) \overline{F}_S \succeq \mathbf{0}. \tag{50}$$

Here $P_{\overline{F}_C} = \overline{F}_C \overline{F}_C^\dagger$, so rank deficiency in either nuisance or event coordinates does not change the projected trajectory.

b: Proof.

For fixed $\boldsymbol{\beta}_C$, minimizing (47) over u removes the component in $\mathrm{range}\, A$ and leaves $\|\overline{F}_S\boldsymbol{\alpha}_S - \overline{F}_C\boldsymbol{\beta}_C\|_2^2$. A second least-squares projection gives (48). Symmetry and idempotence of the orthogonal projector give (49) and its positive-semidefinite order. □

The theorem exposes two distinct weak-event mechanisms. For nonzero residualized single-source signatures $\overline{f}_i$ and $\overline{f}_j$,

$$\gamma^2_{i\to j}(a_i) = a_i^2 \|\overline{f}_i\|_2^2 (1 - \rho_{ij}^2), \qquad \rho_{ij} = \frac{\overline{f}_i^\top \overline{f}_j}{\|\overline{f}_i\|_2 \|\overline{f}_j\|_2}. \tag{51}$$

Small $\|\overline{f}_i\|$ is event–nuisance invisibility; large $|\rho_{ij}|$ is event–event aliasing after nuisance removal.

For general supports, write $H = S \cap C$, $B = S \setminus C$, and $D = C \setminus S$. The shared amplitudes can be refitted, so

$$\gamma^2_{S\to C} = \| \left(I - P_{[A,F_H,F_D]}\right) F_B \boldsymbol{\alpha}_B\|_2^2. \tag{52}$$

Thus the shared true component cancels exactly at first order for signed amplitudes or an interior parameter point. In particular, for $S = \{i,j\}$ and $C = \{i,k\}$ the relaxed margin is independent of the amplitude at i. Enlarging a competitor can only reduce the margin because its explanatory subspace grows.

The exclusive-support quotient removes the zero directions introduced by a literal support intersection. Define

$$\begin{aligned} Z_H &= [A, F_H], \quad R_H = I - Z_H Z_H^{\dagger}, \\ \widetilde{F}_B &= R_H F_B, \quad \widetilde{F}_D = R_H F_D. \end{aligned} \tag{53}$$

Then

$$\begin{aligned} \gamma^2_{S\to C}(\boldsymbol{\alpha}_B) &= \min_{\boldsymbol{\beta}_D} \|\widetilde{F}_B\boldsymbol{\alpha}_B - \widetilde{F}_D\boldsymbol{\beta}_D\|_2^2 \\ &= \boldsymbol{\alpha}_B^{\mathsf{T}} K_{B|D;H}\boldsymbol{\alpha}_B, \\ K_{B|D;H} &= \widetilde{F}_B^{\mathsf{T}}(I - P_{\widetilde{F}_D})\widetilde{F}_B \succeq \mathbf{0}. \end{aligned} \tag{54}$$

The first equality follows by minimizing first over the common span $\operatorname{range} Z_H$ and then over $\boldsymbol{\beta}_D$. Moreover, $K_{B|D;H} \succ \mathbf{0}$ if and only if

$$\ker \widetilde{F}_B = \{\mathbf{0}\}, \qquad \operatorname{range} \widetilde{F}_B \cap \operatorname{range} \widetilde{F}_D = \{\mathbf{0}\}. \tag{55}$$

Indeed, a zero quadratic form means that $\widetilde{F}_B\boldsymbol{\alpha}_B$ lies in the competing range; the two conditions force $\boldsymbol{\alpha}_B = \mathbf{0}$, and the converse is immediate. Under the coefficient normalization $\|\boldsymbol{\alpha}_B\|_2 = 1$, Rayleigh–Ritz gives the worst relaxed direction and margin as

$$\min_{\|\boldsymbol{\alpha}_B\|_2=1} \gamma^2_{S\to C} = \lambda_{\min}(K_{B|D;H}). \tag{56}$$

This value depends on the physical scaling of the amplitude coordinates.

For a scale-free comparison, let Q_B, Q_D be orthonormal bases of the two conditional ranges. If $\theta_{\min}$ is their smallest principal angle,

$$\lambda_{\min}\left[Q_B^{\mathsf{T}}(I - Q_D Q_D^{\mathsf{T}})Q_B\right] = \sin^2\theta_{\min}. \tag{57}$$

The angle must be computed after removing nuisance and shared directions; otherwise every shared event creates a trivial zero angle. If the residual ambient dimension is $q_H = m - \operatorname{rank} Z_H$, then $r_B + r_D > q_H$ forces a nonzero intersection between the two conditional *spans*. This is a span-relaxed dimensional obstruction, not yet a one-sided physical ambiguity.

If $S \subset C$ and active amplitudes may equal zero, the competitor contains the true response and $\gamma_{S\to C} = 0$. If zero is excluded but belongs to the amplitude-domain closure, the infimum remains zero. Pure best-fit geometry therefore cannot select cardinality among nested support models. The marginal likelihood in Supplementary Appendix E supplies the required amplitude-volume and support-prior terms.

F. PHYSICAL TANGENT-CONE SEPARATION

Let $\mathcal{A}_C$ be the physical amplitude domain of a competing event model at the nominal point, and let $T_C = T_{\mathcal{A}_C}(\mathbf{0})$ denote its Bouligand tangent cone. Signed load changes give $T_C = \mathbb{R}^{|C|}$; one-sided trip, shedding, outage-continuation, and fault-severity coordinates give nonnegative rays. Missing PMU data are a discrete measurement-mode change and are not forced into this continuous construction.

Theorem 8 (Constrained first-order separation): Suppose a competing trajectory family has a local C^2 effective chart ϕ with $\phi(\mathbf{0}) = y_0$, full-column-rank derivative J, and local parameter cone $K = \mathbb{R}^p \times T_C$, where T_C is closed and polyhedral. Assume no distant branch approaches y_0. For $y_\star(\epsilon) = y_0 + \epsilon b + o(\epsilon)$,

$$d_C^2(\epsilon) = \epsilon^2 \operatorname{dist}^2(b, JK) + o(\epsilon^2). \tag{58}$$

a: Proof.

For every $v \in K$, conicity gives $\phi(\epsilon v) = y_0 + \epsilon Jv + o(\epsilon)$, which yields the upper bound after taking the infimum. Conversely, an $o(\epsilon)$-optimal local coordinate u_ϵ satisfies $\|\phi(u_\epsilon) - y_0\| = O(\epsilon)$. Full column rank and differentiability imply $\|u_\epsilon\| = O(\epsilon)$. Every convergent subsequence of u_ϵ/ϵ has a limit in the closed cone K, and division of the residual by ϵ gives the matching lower bound. The isolated-branch assumption prevents a nonlocal chart from attaining a smaller leading distance. □

With A the whitened unconstrained nuisance tangent and F_C the event tangent, nuisance profiling is still exact:

$$\gamma^2_{\text{phys}} = \min_{\boldsymbol{\beta}_C \in T_C} \|R_N(b - F_C\boldsymbol{\beta}_C)\|_2^2 = \operatorname{dist}^2(R_N b, R_N F_C T_C). \tag{59}$$

The observable projection is unique even when the coefficient vector is not. Let $\mathcal{C} = R_N F_C T_C$ and $\mathcal{L} = \operatorname{span}\mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{L}$,

$$\gamma_{\text{phys}} \geq \gamma_{\text{span}}, \quad \gamma_{\text{phys}} = \gamma_{\text{span}} \iff \boldsymbol{\Pi}_{\mathcal{L}}(R_N b) \in \mathcal{C}. \tag{60}$$

Thus a positive span margin certifies physical separation, whereas a zero span margin does not prove one-sided ambiguity.

For $T_C = \mathbb{R}_+^{|C|}$, the optimum is a nonnegative least-squares projection. With residual $r = R_N(b - F_C\boldsymbol{\beta}_C)$, the KKT conditions are

$$\boldsymbol{\beta}_C \geq \mathbf{0}, \quad (R_N F_C)^{\mathsf{T}} r \leq \mathbf{0}, \quad \beta_j (R_N f_j)^{\mathsf{T}} r = 0. \tag{61}$$

For one true signature x and one admissible competing ray d,

$$\gamma^2_{\text{ray}} = \|x\|_2^2 - \frac{[d^{\mathsf{T}}x]_+^2}{\|d\|_2^2}, \qquad [v]_+ = \max(0, v). \tag{62}$$

An antiparallel signature can therefore have zero span margin and a strictly positive physical margin.

For a shared one-sided support H, the exact difference coordinate satisfies $\boldsymbol{\delta}_H \in T_H - \boldsymbol{\alpha}_H$, which need not be a subspace. Hence the physical margin can depend on the shared true amplitude; the unconditional cancellation statement applies only when the relaxed solution is feasible. Competitor enrichment still cannot increase the margin whenever the old feasible set is contained in the new one. Likewise, a nested superset still has zero margin when it can match the shared amplitudes and set every added amplitude to zero.

For nontrivial closed cones $\mathcal{C}_S, \mathcal{C}_C$, the directed scale-free separation

$$\delta(\mathcal{C}_S \to \mathcal{C}_C) = \min_{\substack{x \in \mathcal{C}_S \\ \|x\|_2 = 1}} \operatorname{dist}(x, \mathcal{C}_C) \tag{63}$$

TABLE 4. Deterministic checks of conditional and one-sided geometry. These toy systems are not IEEE 39-bus evidence.

Check	Recomputed outcome
Conditional quotient	full–quotient error 0.0e+00 (pass).
Worst coefficient direction	eigenvalue–grid error 1.9e-08 (pass).
Conditional angle	naive 0.0e+00 deg; conditional 18.689 deg (pass).
Ray versus span	squared margins 0.0 versus 1.0 (pass).
One-sided shared source	squared-margin sequence 5.0, 2.0, 1.0, 1.0 (pass).
Cone enrichment	2000 trials; max violation 3.6e-15 (pass).
Nested cone	squared margin 2.4e-31 (pass).

vanishes exactly when the cones share a nonzero vector. It reduces to $\sin\theta_{\min}$ for subspaces, but need not be symmetric. Consequently, the classical signed-sparse condition $\operatorname{spark}(\overline{F}) > 2K$ remains sufficient under one-sided constraints, while its failure need not imply a physically feasible ambiguity.

For a multi-source support S, the Fisher information conditional on that known support is

$$I_S(a) = J_S(a)^{\mathsf{T}} G J_S(a), \quad J_S = \frac{\partial \boldsymbol{\mu}_S}{\partial a_S}. \tag{64}$$

Positive definiteness of I_S permits local amplitude estimation conditional on S; it does not exclude another support with the same mean.

Unknown onset has a separate degeneracy. For $\boldsymbol{\mu}(a,\tau) = ad(\tau)$ and fixed covariance,

$$I(a,\tau) = \begin{bmatrix} d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d & ad^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\dot{d} \\ a\dot{d}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d & a^2\dot{d}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\dot{d} \end{bmatrix}. \tag{65}$$

Profiling amplitude by a Schur complement gives

$$I_{\tau|a} = a^2\left[\dot{d}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\dot{d} - \frac{(d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\dot{d})^2}{d^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}d}\right]. \tag{66}$$

At $a = 0$, onset is unidentifiable. If $\dot{d}$ is proportional to d, delay and gain are locally confounded even for nonzero amplitude. The evaluated bank avoids this unresolved problem by fixing onset.

G. CANDIDATE INDISTINGUISHABILITY AND SPARSE INJECTIVITY

For candidate h and fixed integrity support S, define the affine mean set

$$\mathcal{M}_h(S) = s_{h,H} + \operatorname{range}[\mathcal{O}_{h,H}, \mathcal{E}_{S,H}]. \tag{67}$$

The profiled margin in the main paper is half the squared distance between two such sets under R_H^{-1}.

Theorem 9 (Local candidate indistinguishability): If $D_{ij,S}(H) = 0$, the candidate manifolds intersect. At any intersection, the two candidates induce the same Gaussian law; no decision rule can distinguish them uniformly over the declared nuisance set.

a: Proof.

Zero distance between closed affine sets means that there exist nuisance vectors for which their means coincide. The covariance is common by the stated contract, so the two conditional probability measures are identical. For any decision region, the probabilities of choosing i and j under this common law sum to one; at least one conditional error is therefore at least one half. □

For two fixed simple candidates with squared Mahalanobis distance $\Delta^2 = 2D_{ij,S}$ and equal priors, projection onto their mean difference reduces the optimal test to one dimension, giving

$$P_e^\star = \Phi(-\Delta/2) = \Phi(-\sqrt{D_{ij,S}/2}). \tag{68}$$

This formula does not apply directly to amplitude-marginalized support mixtures.

For the local persistent-event model $Y_T = \mathcal{H}_T a + \varepsilon_T$, define the physically admissible difference set

$$\mathcal{D}_K = \{a - b : a, b \text{ admissible and } K\text{-sparse}\}. \tag{69}$$

Theorem 10 (Finite-horizon sparse event uniqueness): Every admissible K-sparse event is uniquely determined in the noise-free local model if and only if

$$\ker \mathcal{H}_T \cap \mathcal{D}_K = \{\mathbf{0}\}. \tag{70}$$

For a signed neighborhood in each support subspace, $\mathcal{D}_K = \mathcal{A}_{2K}$ locally, recovering the condition in the main paper.

b: Proof.

If two admissible events produce the same output, their nonzero difference belongs to both sets in (70). Conversely, any nonzero vector in the intersection is, by definition of $\mathcal{D}_K$, the difference of two admissible K-sparse events that produce the same output. □

Unknown initial-state nuisance can be removed locally by whitening and projection:

$$\widetilde{\mathcal{H}}_T = \left(I - P_{\operatorname{range}(W_T\mathcal{O}_T)}\right) W_T \mathcal{H}_T, \qquad W_T^{\mathsf{T}} W_T = R_T^{-1}. \tag{71}$$

Theorem 10 then applies with $\widetilde{\mathcal{H}}_T$. Define the restricted margin

$$\gamma_{2K}(T) = \min_{1\le|U|\le 2K} \sigma_{\min}(\widetilde{\mathcal{H}}_{T,U}). \tag{72}$$

If two K-sparse explanations both fit data within whitened residual ϵ, then

$$\|a - b\|_2 \le \frac{2\epsilon}{\gamma_{2K}(T)}. \tag{73}$$

Indeed, the triangle inequality bounds $\|\widetilde{\mathcal{H}}_T(a-b)\|_2$ by 2ϵ, while the definition of γ_{2K} gives the reverse lower bound. For the evaluated $K \le 2$ bank, the relevant union size is four.

H. INFORMATION CANNOT DECREASE WHEN VALID MEASUREMENTS ARE ADDED

For a true trajectory at severity a and a competing physical family with parameter $\boldsymbol{\eta} \in \Theta_C$, define the exact profiled prefix distance

$$d_T^2(a) = \inf_{\boldsymbol{\eta}\in\Theta_C} r_T(a,\boldsymbol{\eta})^{\mathsf{T}} \boldsymbol{\Sigma}_T^{-1} r_T(a,\boldsymbol{\eta}), \tag{74}$$

where every $\boldsymbol{\Sigma}_T$ is a principal prefix of one positive-definite process covariance and the physical competitor domain does not change with T. Partition the difference between candidate means into old and added blocks, $\boldsymbol{\delta} = \mathrm{col}(\boldsymbol{\delta}_1, \boldsymbol{\delta}_2)$, with joint covariance $\boldsymbol{\Sigma} \succ \mathbf{0}$. Block Gaussian elimination gives

$$\boldsymbol{\delta}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{\delta} = \boldsymbol{\delta}_1^{\mathsf{T}}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\delta}_1 + \widetilde{\boldsymbol{\delta}}_2^{\mathsf{T}}\boldsymbol{\Sigma}_{2|1}^{-1}\widetilde{\boldsymbol{\delta}}_2, \tag{75}$$

$$\widetilde{\boldsymbol{\delta}}_2 = \boldsymbol{\delta}_2 - \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\delta}_1, \tag{76}$$

$$\boldsymbol{\Sigma}_{2|1} = \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}. \tag{77}$$

Theorem 11 (Prefix monotonicity and temporal rescue): Under the prefix and fixed-domain contract above,

$$d_{T+1}^2(a) \geq d_T^2(a). \tag{78}$$

If both minima are attained, equality holds if and only if an element of $\operatorname{Argmin} d_T^2$ also has zero conditional innovation residual in the added block. Suppose further that, locally,

$$d_T^2(a) = c_T a^{2k_T} + o(a^{2k_T}), \qquad c_T > 0. \tag{79}$$

Then $k_{T+1} \leq k_T$; if the order remains k, then $c_{T+1} \geq c_T$.

a: Proof.

The conditional covariance in (75) is positive definite, so its second quadratic form is nonnegative for every fixed $\boldsymbol{\eta}$. Taking the infimum over the same Θ_C proves (78). If the minima are attained, equality of the infima forces an extended-horizon minimizer to minimize the prefix term and to make the innovation term zero; the converse is immediate. If $k_{T+1} > k_T$, the ratio $d_{T+1}^2(a)/d_T^2(a)$ tends to zero as $a \to 0$, contradicting (78). When both orders equal k, division by a^{2k} and passage to the limit yield $c_{T+1} \geq c_T$. □

The theorem permits a longer prefix to reveal a first-order direction that was absorbed at an earlier horizon, or to reveal a higher-order obstruction that was previously invisible. It does not say that a concrete estimator improves: changing its normalization, nuisance set, covariance model, or optimization breaks the prefix contract. In particular, the archived RAWSIM39 calibration renormalized each event–signal–horizon block to unit weight and is not a prefix likelihood for this theorem.

If whitened model errors under two supports satisfy $\|We_S\| \leq \epsilon_S$ and $\|We_R\| \leq \epsilon_R$, the reverse triangle inequality gives the robust separation bound

$$d_{\mathrm{true}} \geq [d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+, \qquad \mathcal{J}_{\mathrm{true}} \geq \tfrac{1}{2}[d_{\mathrm{nom}} - \epsilon_S - \epsilon_R]_+^2. \tag{80}$$

This is the required bridge for future simulator-shift claims; it is not evaluated by the current nominal load bank.

I. TEMPORAL INFORMATION UNDER THE FROZEN AR(1) CONTRACT

For true support S, competing support R, true amplitude a, and competitor amplitude b, let $\boldsymbol{\delta}_t(b) = \boldsymbol{\mu}_{S,t}(a) - \boldsymbol{\mu}_{R,t}(b)$. The frozen temporal covariance is

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}, \qquad [R_T]_{ij} = \rho^{|i-j|}, \quad |\rho| < 1. \tag{81}$$

For fixed b, its squared Mahalanobis separation has the innovation form

$$q_T(b) = \|\boldsymbol{\delta}_1(b)\|_{\boldsymbol{\Omega}^{-1}}^2 + \frac{1}{1-\rho^2}\sum_{t=2}^{T}\|\boldsymbol{\delta}_t(b) - \rho\boldsymbol{\delta}_{t-1}(b)\|_{\boldsymbol{\Omega}^{-1}}^2. \tag{82}$$

Define the profiled simple-model separation

$$\mathcal{J}_{S\to R}(T) = \tfrac{1}{2}\inf_{b\in\mathcal{A}_R} q_T(b). \tag{83}$$

It equals the minimum KL divergence over fixed-amplitude simple Gaussian competitors. It is not the KL divergence or Bayes factor of the amplitude-mixture support model.

Theorem 12 (Persistent AR(1) information rate): If the amplitude domain is compact and fixed and $\boldsymbol{\delta}_t(b)$ converges uniformly to $\boldsymbol{\delta}_\infty(b)$, then

$$\lim_{T\to\infty}\frac{2\mathcal{J}_{S\to R}(T)}{T} = \frac{1-\rho}{1+\rho}\inf_{b\in\mathcal{A}_R}\|\boldsymbol{\delta}_\infty(b)\|_{\boldsymbol{\Omega}^{-1}}^2. \tag{84}$$

a: Proof.

Under uniform convergence, $\boldsymbol{\delta}_t - \rho\boldsymbol{\delta}_{t-1}$ converges uniformly to $(1-\rho)\boldsymbol{\delta}_\infty$. Divide (82) by T; the first-frame term vanishes and the Cesàro mean converges uniformly to $(1-\rho)^2(1-\rho^2)^{-1}\|\boldsymbol{\delta}_\infty\|_{\boldsymbol{\Omega}^{-1}}^2$. Compactness and uniform convergence permit interchange of infimum and limit, giving (84). □

Monotonicity of $\mathcal{J}_{S\to R}(T)$ follows independently from Theorem 11; the AR(1) calculation supplies the explicit limiting rate.

If the profiled equilibrium distance is positive, separation grows as $\Theta(T)$. If an admissible competitor matches equilibrium and its innovation differences are square summable, evaluating (83) at that competitor gives a finite upper bound; monotonicity then implies convergence to a finite ceiling. Exact trajectory overlap gives zero separation for every horizon.

For a true pair $\{i,j\}$, a competing pair $\{i,k\}$ can refit the shared source. With whitened equilibrium columns $\bar{h}_i, \bar{h}_j, \bar{h}_k$,

$$a_j^2\|(I - P_{i,k})\bar{h}_j\|_2^2 \leq a_j^2\|(I - P_i)\bar{h}_j\|_2^2. \tag{85}$$

The inequality follows from $\mathrm{span}(\bar{h}_i) \subseteq \mathrm{span}(\bar{h}_i, \bar{h}_k)$. It formalizes why identifying which weak second source occurred cannot be easier than rejecting the shared single-source explanation.

J. HIGHER-ORDER CONTACT AND LOCAL DETECTION SCALE

First-order Fisher information can vanish while nonlinear separation remains. In whitened coordinates, let the no-event nuisance manifold and event curve be

$$F_0(b) = Nb + \tfrac{1}{2}H_0[b,b] + O(\|b\|^3), \tag{86}$$
$$c(a) = ad + a^2 q + O(a^3). \tag{87}$$

Theorem 13 (Relative first- and second-order separation): Assume the nuisance manifold has constant rank and use a regular chart after removing redundant coordinates. Define the profiled first-order coefficient

$$\gamma_1^2 = \|P_\perp d\|_2^2, \qquad P_\perp = I - (WJ_c)(WJ_c)^\dagger, \tag{88}$$

where J_c is the nuisance tangent before whitening and d is the already whitened event tangent in (87). If $\gamma_1 > 0$, the local squared distance to the nuisance manifold is

$$\Delta^2(a) = a^2\gamma_1^2 + O(a^3). \tag{89}$$

If $\gamma_1 = 0$, so the whitened event tangent lies in the nuisance tangent space, choose the regular-chart vector v satisfying $Nv = d$ and define

$$\boldsymbol{\kappa} = P_\perp\left(q - \tfrac{1}{2}H_0[v,v]\right). \tag{90}$$

Then

$$\Delta^2(a) = a^4\gamma_2^2 + O(a^5), \qquad \gamma_2^2 = \|\boldsymbol{\kappa}\|_2^2. \tag{91}$$

a: Proof.

The local minimizer satisfies $DF_0(b)^\mathsf{T}[F_0(b) - c(a)] = \mathbf{0}$. At the origin, its derivative with respect to the regular chart coordinate is $N^\mathsf{T}N \succ \mathbf{0}$, so the implicit-function theorem gives a smooth minimizing branch. Write $b_\star(a) = av + a^2 w + O(a^3)$. At first order, $N^\mathsf{T}(Nv - d) = \mathbf{0}$, yielding residual $aP_\perp d + O(a^2)$ and (89). If $d = Nv$, the first-order residual vanishes. At second order, minimization projects $q - \frac{1}{2}H_0[v,v]$ onto $\operatorname{range} N$; the remaining normal component is (90). Squaring the residual gives (91). The projector is unchanged by any full-rank reparameterization of the same nuisance tangent space, so γ_1 is a property of the local manifolds and covariance metric rather than of a particular nuisance coordinate basis. □

The same constant-rank qualification makes γ_2 intrinsic. Redundant original parameters may give several representatives of the same observable tangent, but their normal acceleration equals the second fundamental form of the same manifold branch. If $\gamma_2 = 0$, the conclusion is only $\Delta^2(a) = o(a^4)$ under the corresponding little-o expansion; an a^6 law or exact overlap requires a separate third-order analysis.

K. OPERATIONAL ERROR AND SUPPORT-RECOVERY LIMITS

For two simple hypotheses with a shared covariance, $P_i = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, define

$$d^2 = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0). \tag{92}$$

The equal-prior log likelihood ratio is Gaussian with mean $\mp d^2/2$ and variance d^2 under P_0 and P_1, respectively. Hence

$$P_e^\star = \Phi(-d/2), \quad D_{\mathrm{KL}}(P_1\|P_0) = d^2/2, \quad C_{\mathrm{Ch}} = d^2/8. \tag{93}$$

The last identity gives $P_e^\star \le \frac{1}{2}\exp(-d^2/8)$. If a common affine nuisance is $A\boldsymbol{\eta}$, orthogonal projection after whitening removes it exactly and replaces d by the quotient margin. Thus $P_e^\star = \Phi(-\gamma/2)$ is exact for that quotient Gaussian experiment. It is not an exact formula for two nonlinear amplitude mixtures with different priors or dimensions.

Suppose the first nonzero contact order at horizon T is k and

$$d_T^2(a) = T\mathcal{I}_k|a|^{2k}[1 + o(1)], \qquad \mathcal{I}_k > 0. \tag{94}$$

For target simple-Gaussian error $\alpha < 1/2$, inversion of (93) yields

$$|a|_{\min}(\alpha, T) \sim \left[\frac{2\Phi^{-1}(1-\alpha)}{\sqrt{T\mathcal{I}_k}}\right]^{1/k}. \tag{95}$$

Equivalently, if $a_T = cT^{-\beta}$, the optimal binary error tends to one half for $\beta > 1/(2k)$, to a nontrivial constant at equality, and to zero for $\beta < 1/(2k)$. This gives the $T^{-1/2}$ and $T^{-1/4}$ boundaries for first- and second-order separation.

Now let $\Theta = \bigcup_{S\in\mathfrak{S}}\Theta_S$ and define the minimax exact-support risk

$$R^\star = \inf_{\widehat{S}}\sup_{\theta\in\Theta}\Pr_\theta\{\widehat{S} \ne S(\theta)\}. \tag{96}$$

Theorem 14 (Binary and minimum-severity limits): For every pair $\theta_i \in \Theta_{S_i}$ with $S_0 \ne S_1$,

$$R^\star \ge \frac{1}{2}[1 - \mathrm{TV}(P_{\theta_0}, P_{\theta_1})] \ge \frac{1}{2}\left[1 - \sqrt{\tfrac{1}{2}D_{\mathrm{KL}}(P_{\theta_0}\|P_{\theta_1})}\right]_+. \tag{97}$$

If $S^+ = S \cup \{j\}$ and $P_{S^+,a_j} \to P_S$ in total variation as $a_j \downarrow 0$, then uniform exact-support recovery over all $a_j > 0$ has minimax risk at least one half.

a: Proof.

Restriction of any support rule to a two-point subset cannot reduce its worst-case risk; the optimal equal-prior binary risk is $[1 - \mathrm{TV}(P,Q)]/2$. Pinsker's inequality gives the second bound. For the nested pair, the binary risk tends to one half as the active amplitude tends to zero, so the supremum over positive amplitudes is at least that limit. □

The theorem requires either a declared $|a_j| \ge a_{\min} > 0$ contract or a decision that may abstain near absence. It also distinguishes a source-ranking error from an impossible exact-cardinality decision.

For M equiprobable supports, Fano's inequality gives

$$P_e \geq 1 - \frac{I(V;Y) + \log 2}{\log M}. \tag{98}$$

With $\overline{P} = M^{-1}\sum_j P_j$, convexity of KL in its second argument implies

$$\begin{aligned} I(V;Y) &= \frac{1}{M}\sum_i D_{\mathrm{KL}}(P_i\|\overline{P}) \\ &\leq \frac{1}{M^2}\sum_{i,j} D_{\mathrm{KL}}(P_i\|P_j) = \overline{D}_{\mathrm{KL}}. \end{aligned} \tag{99}$$

Therefore a necessary condition for $P_e \leq \alpha$ is

$$\overline{D}_{\mathrm{KL}} \geq (1-\alpha)\log M - \log 2. \tag{100}$$

If $M = \binom{N}{K}$ and $\overline{D}_{\mathrm{KL}}(T) \leq T\overline{\mathcal{I}}_k a^{2k}/2$, then necessarily

$$T \geq \frac{2[(1-\alpha)\log\binom{N}{K} - \log 2]}{\overline{\mathcal{I}}_k a^{2k}}. \tag{101}$$

This is an impossibility certificate, not a sufficient sample-size guarantee [21], [22]. For sparse $K \ll N$, it has the familiar scaling $T \gtrsim K\log(N/K)/(\overline{\mathcal{I}}_k a^{2k})$, now expressed through the physical Hybrid-DAE information rate.

L. DIAGNOSABILITY-PRESERVING DISCREPANCY COVARIANCE

A residual correction can improve fit while making physical events harder to distinguish. The following calculation makes that tradeoff explicit. Work after whitening and removal of common nuisance. Let $Q_U \in \mathbb{R}^{m\times r}$ be an orthonormal basis for the discriminant directions that must be protected, and let $Q_D \in \mathbb{R}^{m\times q}$ be an orthonormal discrepancy basis. The former should be frozen from critical candidate pairs; the latter must be estimated from mismatch-only records so that unlabeled events are not absorbed as model error.

Theorem 15 (Discrepancy retention and finite intensity): Let $P_D = Q_D Q_D^{\mathsf{T}}$ and model the whitened residual with covariance

$$C_s = I + sP_D, \qquad s = \tau^2 \geq 0. \tag{102}$$

Then $C_s^{-1} = I - \alpha P_D$, where $\alpha = s/(1+s)$, and the worst information fraction retained over the protected subspace is

$$\rho(s, Q_D) = 1 - \alpha\,\sigma_{\max}^2(Q_U^{\mathsf{T}}Q_D). \tag{103}$$

Set $c^2 = \sigma_{\max}^2(Q_U^{\mathsf{T}}Q_D)$ and $\delta = 1 - \rho_{\min}$. The constraint $\rho \geq \rho_{\min}$ imposes no finite upper bound on s when $c^2 \leq \delta$; otherwise it requires

$$s \leq s_{\max} = \frac{\delta}{c^2 - \delta}. \tag{104}$$

If the true residual covariance is $I + C_m$ and $E_D = \mathrm{tr}(Q_D^{\mathsf{T}} C_m Q_D)$, the expected Gaussian negative twice-log-likelihood relative to $s = 0$, up to constants, is

$$\Delta\mathcal{L}(s) = q\log(1+s) - \frac{s}{1+s}(q + E_D). \tag{105}$$

For fixed Q_D, its constrained minimizer is finite:

$$s^\star = \min\{E_D/q, s_{\max}\}, \tag{106}$$

with $s_{\max} = +\infty$ in the first geometric case.

TABLE 5. Deterministic checks for the discrepancy-retention identities. These are algebraic tests, not power-system experiments.

Check	Error	Tolerance	Status
Low-rank covariance inverse	4.53e-16	1.0e-12	PASS
Principal-angle retention	2.22e-16	1.0e-12	PASS
Retention variance cap	0.00e+00	1.0e-12	PASS
Finite full-likelihood intensity	5.50e-09	1.0e-07	PASS
Safe projected PCA	1.78e-15	1.0e-10	PASS
Protected-distance floor	1.11e-16	1.0e-12	PASS

a: Proof.

The inverse follows from $P_D^2 = P_D$. For $x = Q_U a$, the information ratio is the Rayleigh quotient of $I - \alpha Q_U^{\mathsf{T}} P_D Q_U$; its smallest eigenvalue is (103) because the singular values of $Q_U^{\mathsf{T}}Q_D$ are the principal-angle cosines. Solving $\alpha c^2 \leq \delta$ gives (104). Finally, $\log|C_s| = q\log(1+s)$ and taking the expectation of $y^{\mathsf{T}}C_s^{-1}y$ under $\mathrm{Cov}(y) = I + C_m$ gives (105). Its derivative is $(qs - E_D)/(1+s)^2$, which gives (106). □

Under strict no-loss protection, $\rho_{\min} = 1$ and $s > 0$ require $\mathrm{range}(Q_D) \subseteq \mathrm{range}(Q_U)^{\perp}$. For fixed rank, the basis that captures the most mismatch energy is therefore the top-q eigenspace of $P_{U^\perp} C_m P_{U^\perp}$, provided that the orthogonal complement has dimension at least q. This is a safe projected-PCA baseline, not a learned-estimator result.

Maximizing only αE_D omits both the covariance determinant and the whitened measurement-noise contribution. It therefore drives s toward an infinite-variance limit whenever geometry permits. That utility is a valid description of reduced quadratic mismatch penalty, but it is not a normalized Bayesian tuning objective. The general joint orientation problem for $q > 1$ is an offline nonconvex design; spectral, semidefinite, or moment relaxations belong in a separate optimization study unless they are executed on the frozen event and mismatch dictionaries.

M. APPLIED GATE TAXONOMY FROM AN ARCHIVED ANDES CHECKPOINT

The frozen RAWSIM39 V1.2 calibration ledger provides an applied boundary check on the theory. It contains complete first-order screening and rank-selection decisions for 18 plant/event directions, but not the Jacobian and higher-order arrays needed to evaluate γ_2 or γ_3. Table 6 therefore classifies only what the recorded gates establish. In particular, failure of the curvature gate is not relabeled as second-order identifiability, and a small orthogonal fraction is not relabeled as exact redundancy.

N. THEORY-TO-EVIDENCE BOUNDARY

Equation (89) supplies a direct falsification test: for independently generated nonlinear trajectories with decreasing $|a|$, the profiled squared distance divided by a^2 should converge to γ_1^2. That convergence study has not been run for the present ANDES bank; the retrospective distance association and the gate-only RAWSIM39 taxonomy are not substitutes for it.

If n independent Gaussian replications have local distance $\Delta(a) = |a|^r\|k_r\| + O(|a|^{r+1})$, their KL divergence to the

TABLE 6. Conservative taxonomy of the frozen RAWSIM39 V1.2 calibration checkpoint. The gate sequence was 18 candidates, 10 individually eligible ($\kappa = 722.91$), eight rank-selected ($\kappa = 7.83$), and seven released after one 6.15σ local-domain rejection.

Category	Count	Directions	Evidence boundary
Regular local	7	plant_inertia_scale, plant_governor_droop_scale, generation_delta_p_pu, generation_delta_q_pu, generation_amplitude, generation_peak_s, load_delta_p_fraction	Released only under the local restricted posterior
Weak first-order	6	plant_damping, plant_governor_lag_scale, plant_turbine_time_scale, plant_avr_gain_scale, plant_avr_time_scale, plant_subtransient_x_scale	Low information; order and temporal rescue unresolved
Curvature unresolved	2	fault_duration_s, line_duration_s	Linear surrogate failed; not evidence of order 2
Near-redundant first-order	2	generation_tau_s, generation_residual	Small new orthogonal fraction; exact redundancy unproved
Local-domain failure	1	fault_reactance_pu	MAP left the declared local prior domain

TABLE 7. Recomputed checks of higher-order evidence and information limits. Controlled systems are not IEEE 39-bus performance evidence.

Check	Recomputed outcome
Evidence classes	slopes 0.9994, 0.4951, and 0.0003 for regular, order-2, and redundant additions (pass).
Prefix monotonicity	500 correlated-noise trials; max violation 0.0e+00 (pass).
Temporal rescue	first-order $\gamma_1^2(2) = 0.452724$; second-order relative error 1.0e-04 (pass).
Gaussian error map	quadrature error 5.6e-17 (pass).
Fano lower bound	$M = 8$ bounds 0.5993/0.3974; no Monte Carlo violation (pass).

TABLE 8. Deterministic checks of the local separation identities. These toy systems are not IEEE 39-bus evidence.

Check	Prediction and recomputed outcome
Regular first-order DAE	slope 2 and $\gamma_1^2 = 0.4527244641$; 2.000000; relative error 3.8e-12 (pass).
Shared-tangent second order	slope 4 and $\gamma_2^2 = 2.89$; 4.000000; coefficient 2.890000 (pass).
Rank-deficient nuisance	unique projected fit and $\gamma_1^2 = 0.5$; rank 2/3; max variation 1.2e-16 (pass).
Signed shared-source cancellation	margin independent of shared amplitude; squared-margin range 0.0e+00 (pass).
Competitor enrichment	$\gamma(S \to C_2) \leq \gamma(S \to C_1)$; 0.0196 versus 0.5096 (pass).
Nested support	geometric margin 0; 6.1e-35 (pass).
Principal angle	unit margin $\sin\theta$; absolute error 0.0e+00 (pass).
Nuisance enlargement	profiled margin cannot increase; sequence 2.0, 1.0, 0.0 (pass).
Diagnostic quotient	profiled costs factor through $d_{\mathcal{D}}$ coordinates; rank 4/6; max error 1.1e-15 (pass).
Nuisance reparameterization	unchanged nuisance span gives the same projector; spectral error 9.5e-16 (pass).

closest null is $n\Delta^2(a)/2$. Pinsker's inequality bounds the power of any level-α test by

$$\text{Power}_a \leq \alpha + \min\{1, \sqrt{n}\Delta(a)/2\}. \tag{107}$$

Thus $n|a_n|^{2r} \to 0$ prevents power from separating from size. The reference scales are $n^{-1/2}$ for first-order contact and $n^{-1/4}$ for second-order contact; singular-information asymptotics are established more broadly [23]. Correlated frames in one PMU window are not independent replications and must be handled through their joint covariance.

All audited load directions in the present bank have nonzero first-order separation. The second-order result is therefore a boundary theorem, not the empirical explanation for weak-pair failure.

TABLE 9. Open checks for the simulator event semantics.

Check	Evidence required for closure
Right limit	Solver-scale $\|g_+(x(\tau+\epsilon), z(\tau+\epsilon), a)\|$ after re-equilibration; the callback residual may remain $O(a)$.
Algebraic derivatives	Agreement of analytic z_1, z_2 with central differences on a decreasing symmetric amplitude grid.
Taylor order	Constraint and state-approximation slopes $1, 2, 3$, or amplitude-halving ratios $2, 4, 8$.
Event families	One declared semantics and tolerance set for load, generation, and continuous branch-admittance perturbations.
Dynamic first order	Convergence of $\Delta y/a$ to $B_{r,a}$ over the declared early-time window.
Dynamic second order	Analytic second variation within a preregistered tolerance of nonlinear trajectories; the current discrepancy is unresolved.

TABLE 10. Status of the mathematical results relative to the frozen evidence.

Result	Present evidence boundary
Consistency jump	DAE contract only; callback/right-limit logging and derivative convergence remain open.
Functional equivalence and BLUE	Exact in the local linear model; nonlinear TEST accuracy remains empirical.
Fisher, support, and cone geometry	Signed and one-sided toy identities pass; IEEE 39-bus convergence and physical cone margins remain untested.
Sparse injectivity and γ_{2K}	T120 diagnostics concern the frozen load bank, not general descriptor left-invertibility.
Temporal order and rate	Prefix and controlled correlated-noise identities pass; the frozen long-horizon load study remains diagnostic and lacks independent operating-point validation.
Relative curvature	The controlled parabola passes; curvature-dominated weak-event behavior was not observed in the frozen atlas.
Error and minimax limits	Gaussian quotient, minimum-severity, and Fano formulas pass controlled checks; no IEEE 39 pairwise oracle-error atlas has been frozen.
RAWSIM39 gate taxonomy	The 18-direction ledger supports five conservative gate categories; derivative arrays needed to assign higher-order classes are unavailable.
Discrepancy retention	Exact for the whitened low-rank covariance model and checked algebraically; the protected dictionary, mismatch covariance, learned correction, and shifted IEEE 39-bus test remain unexecuted.

APPENDIX E ESTIMATOR AND LIKELIHOOD DETAILS

This appendix distinguishes a state-reconstruction component from two event estimator contracts. Sections E-A–E-C specify the state recursion and the implemented batch physical-manifold Bayes estimator $\mathcal{E}_{\mathrm{B}}$. Section E-F specifies the proposed joint Hybrid-DAE estimator $\mathcal{E}_{\mathrm{J}}$. Filtering, smoothing, quadrature, and optimization are numerical operations within those contracts, not additional estimator proposals.

A. BAYESIAN RECURSION AND THE STATE HIERARCHY

For a Markov state model, Bayesian filtering alternates prediction and correction [24]. More broadly, this construction treats the PMU task as a Bayesian inverse problem in which physical candidate models and latent states are compared through predictive evidence [25]:

$$p(x_k \mid y_{0:k-1}) = \int p(x_k \mid x_{k-1}) p(x_{k-1} \mid y_{0:k-1})\, dx_{k-1}, \tag{108}$$

$$p(x_k \mid y_{0:k}) = \frac{p(y_k \mid x_k) p(x_k \mid y_{0:k-1})}{\int p(y_k \mid x_k) p(x_k \mid y_{0:k-1})\, dx_k}. \tag{109}$$

The denominator in (109) is the predictive evidence. Omitting it is harmless for a state mean under one fixed model, but not when physical candidates are compared.

B0 returns the equilibrium. B1 conditions the equilibrium prior on one PMU frame:

$$K_{\mathrm{LMMSE}} = P_0 C^{\mathsf{T}} (C P_0 C^{\mathsf{T}} + R)^{-1}, \tag{110}$$

$$\widehat{\delta x}_k^{\,\mathrm{B1}} = K_{\mathrm{LMMSE}} (y_k - y_0). \tag{111}$$

B2 retains state and covariance between frames. For the linear Gaussian model, substituting Gaussian densities into (108)–(109) yields

$$m_k^- = A_d m_{k-1}^+, \qquad P_k^- = A_d P_{k-1}^+ A_d^{\mathsf{T}} + Q, \tag{112}$$

$$S_k = C P_k^- C^{\mathsf{T}} + R, \qquad K_k = P_k^- C^{\mathsf{T}} S_k^{-1}, \tag{113}$$

$$m_k^+ = m_k^- + K_k (y_k - C m_k^-), \quad P_k^+ = (I - K_k C) P_k^-. \tag{114}$$

Theorem 16 (Linear-Gaussian reduction): Equations (112)–(114) are the exact posterior mean and covariance generated by the Bayesian recursion for the stated linear Gaussian model.

a: Proof.

An affine transformation of the Gaussian posterior at $k-1$, followed by addition of independent Gaussian process noise, gives (112). The pair (x_k, y_k) is jointly Gaussian with cross covariance $P_k^- C^{\mathsf{T}}$ and measurement covariance S_k. The standard conditional-Gaussian formula gives (113)–(114). Equivalently, multiplying the predictive density by the likelihood and completing the square gives posterior precision $(P_k^-)^{-1} + C^{\mathsf{T}} R^{-1} C$; Woodbury's identity produces the same covariance and mean. □

The implementation may use the Joseph covariance form

$$P_k^+ = (I - K_k C) P_k^- (I - K_k C)^{\mathsf{T}} + K_k R K_k^{\mathsf{T}} \tag{115}$$

to preserve symmetry and positive semidefiniteness numerically.

B3 uses the Rauch–Tung–Striebel backward recursion [26]:

$$J_t = P_t^+ A_d^{\mathsf{T}} (P_{t+1}^-)^{-1}, \tag{116}$$

$$m_{t|k} = m_t^+ + J_t (m_{t+1|k} - m_{t+1}^-), \tag{117}$$

$$P_{t|k} = P_t^+ + J_t (P_{t+1|k} - P_{t+1}^-) J_t^{\mathsf{T}}. \tag{118}$$

It estimates a past state using later measurements. The negative B3 result therefore diagnoses model mismatch; it does not imply that correct future information generally lacks value.

The nominal uncertainty correction rescales the rectangular voltage covariance as

$$\mathbf{\Sigma}_{i,k}^{\mathrm{U2}} = T \mathbf{\Sigma}_{i,k} T^{\mathsf{T}}, \qquad T = \mathrm{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}), \tag{119}$$

with temperatures fitted on independent calibration trajectories. The observed-only adequacy statistic is a five-frame trailing mean of

$$a_k = (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu)^{\mathsf{T}} \mathbf{\Sigma}_\nu^{-1} (\boldsymbol{\nu}_k - \boldsymbol{\mu}_\nu). \tag{120}$$

Neither operation changes the physical state estimate.

B. CONSTRAINED PHYSICAL RECENTERING

The static recentering estimator treats active/reactive load changes and active generator changes as nuisance coordinates d. It solves

$$(\widehat{V}, \widehat{d}) = \arg\min_{V,d} \|e_y(V)\|_2^2 + \gamma^2 \|g_{\mathrm{AC}}(V,d)\|_2^2 + \lambda q_d^{-2} \|d\|_2^2. \tag{121}$$

Development data fix $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$. Because the AC constraint appears through a finite penalty rather than a normalized stochastic density, the implemented objective is a regularized physical estimator, not a fully normalized posterior.

For an exactly constrained local MAP, let $r(\boldsymbol{\chi})$ be the stacked residual, $c(\boldsymbol{\chi}) = \mathbf{0}$ the equality constraints, and $\mathbf{\Omega}$ the residual precision. A Gauss–Newton step solves

$$\begin{bmatrix} H & G_c^{\mathsf{T}} \\ G_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{\chi} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla J \\ -c \end{bmatrix}, \quad H = J_r^{\mathsf{T}} \mathbf{\Omega} J_r + H_{\mathrm{prior}}. \tag{122}$$

Theorem 17 (Feasible local covariance): If G_c has full row rank and Z spans $\ker G_c$, then $Z^{\mathsf{T}} H Z \succ \mathbf{0}$ makes the KKT matrix nonsingular. The local constrained Gaussian covariance is

$$P_\chi = Z (Z^{\mathsf{T}} H Z)^{-1} Z^{\mathsf{T}}. \tag{123}$$

a: Proof.

Any vector in the KKT nullspace satisfies $G_c v = \mathbf{0}$, hence $v = Zq$. Premultiplying the first block equation by v^{T} gives $q^{\mathsf{T}} Z^{\mathsf{T}} H Z q = 0$, so $q = \mathbf{0}$ and $v = \mathbf{0}$. Full row rank then forces the multiplier to zero. Parameterizing feasible perturbations by $\delta\boldsymbol{\chi} = Zq$ gives Gaussian precision $Z^{\mathsf{T}} H Z$ in q; its linear pushforward is (123). □

This covariance is local and unimodal. The reported joint NEES-like failure shows that it is not calibrated for the full shifted-plant experiment.

C. IMPLEMENTED BATCH PHYSICAL-MANIFOLD BAYES $\mathcal{E}_{\mathrm{B}}$

Let r be the centered T-frame PMU window. For support S with amplitudes a_S, the evaluated mean is

$$\boldsymbol{\mu}_S(a_S) = \sum_{i \in S} (a_i D_i + a_i^2 Q_i) + \sum_{\substack{i,j \in S \\ i<j}} a_i a_j Q_{ij}. \tag{124}$$

The W2 likelihood is

$$q_S(a_S) = \|r - \boldsymbol{\mu}_S(a_S)\|^2_{\boldsymbol{\Sigma}_T^{-1}}, \tag{125}$$

$$p(r \mid S, a_S) = \frac{\exp[-q_S(a_S)/2]}{(2\pi)^{n_y T/2}|\boldsymbol{\Sigma}_T|^{1/2}}, \tag{126}$$

$$\boldsymbol{\Sigma}_T = R_T(\rho) \otimes \boldsymbol{\Omega}. \tag{127}$$

The covariance is fitted only on normal calibration records and remains fixed across supports. The selected priors are

$$p(K = 0, 1, 2) = (0.20, 0.50, 0.30), \tag{128}$$

$$p(S \mid K) = \binom{16}{K}^{-1}, \tag{129}$$

$$a_i \mid i \in S \sim \mathcal{N}(0, \sigma_a^2), \qquad \sigma_a = 0.05. \tag{130}$$

For reference, the linear single-source model has a closed form.

Proposition 1 (Conjugate amplitude posterior and Bayes factor): Let $r = ad + \varepsilon$, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, and $a \sim \mathcal{N}(0, \sigma_a^2)$. Define $I = d^\mathsf{T}\boldsymbol{\Sigma}^{-1}d$ and $s = d^\mathsf{T}\boldsymbol{\Sigma}^{-1}r$. Then

$$v = (I + \sigma_a^{-2})^{-1}, \qquad m = vs, \tag{131}$$

$$a \mid r, S \sim \mathcal{N}(m, v), \tag{132}$$

$$\log \frac{p(r \mid S)}{p(r \mid H_0)} = \tfrac{1}{2}\frac{s^2}{I + \sigma_a^{-2}} - \tfrac{1}{2}\log(1 + \sigma_a^2 I). \tag{133}$$

a: Proof.

Terms in the joint log density that depend on a are $-\frac{1}{2}[(I + \sigma_a^{-2})a^2 - 2sa]$. Completing the square gives mean m and variance v. Integrating the resulting Gaussian contributes $\sqrt{v/\sigma_a^2}\exp(vs^2/2)$ relative to H_0; taking its logarithm gives (133). The logarithmic term is the amplitude-volume or Occam penalty. □

Corollary 2 (One-sided amplitude evidence): Under Proposition 1, replace the signed Gaussian slab by the proper half-normal prior $p(a) = 2\mathcal{N}(a; 0, \sigma_a^2)\mathbb{I}(a \geq 0)$. If $BF_\pm$ denotes the signed Bayes factor in (133), then

$$BF_+ = 2\Phi(z)BF_\pm, \qquad z = \frac{\sigma_a s}{\sqrt{1 + \sigma_a^2 I}}. \tag{134}$$

Conditional on the active model, the severity posterior is $\mathcal{N}(m, v)$ truncated to $[0, \infty)$.

b: Proof.

The fixed-amplitude likelihood ratio is $\exp(as - Ia^2/2)$. Multiplication by the half-normal density and completion of the square give twice the signed Gaussian integral restricted to $a \geq 0$. The unrestricted posterior has standardized probability $\Phi(m/\sqrt{v}) = \Phi(z)$ on that half-line, which yields (134). □

The factor $2\Phi(z)$ preserves direction: a large score with the prohibited sign is evidence against the one-sided event even though the signed factor depends on s^2. Proper slab scales are essential because Bayes factors across different support dimensions are undefined up to an arbitrary constant under an improper amplitude prior. Under the one-sided null $a = 0$, the likelihood ratio statistic satisfies

$$2\Delta\ell = [Z]_+^2, \qquad Z \sim \mathcal{N}(0, 1), \tag{135}$$

and therefore has the boundary mixture $\frac{1}{2}\delta_0 + \frac{1}{2}\chi_1^2$, rather than the regular χ_1^2 law.

The quadratic multi-source mean is nonconjugate. Its evidence is

$$Z_S = p(r \mid S) = \int p(r \mid S, a_S) \prod_{i \in S} p(a_i)\, da_S. \tag{136}$$

With Q-node Gauss–Hermite abscissas x_q and weights w_q, the Gaussian-prior transformation $a_i = \sqrt{2}\sigma_a x_{q_i}$ gives

$$Z_S \approx \pi^{-|S|/2} \sum_{q_1=1}^{Q} \cdots \sum_{q_{|S|}=1}^{Q} \left(\prod_{j=1}^{|S|} w_{q_j} \right) p(r \mid S, \sqrt{2}\sigma_a x_q). \tag{137}$$

The implementation uses $Q = 31$ and evaluates (137) in the log domain using log-sum-exp. Independent adaptive Gauss–Kronrod integration checks representative two-dimensional cases. These checks support numerical integration under the frozen model; they do not validate the physical model itself.

D. POSTERIOR OUTPUTS AND THE CARDINALITY–LOCALIZATION DISTINCTION

The normalized support posterior is

$$p(S \mid r) = \frac{Z_S p(S)}{\sum_{R \in \mathcal{H}} Z_R p(R)}. \tag{138}$$

It induces cardinality, source-inclusion, and amplitude distributions:

$$p(K = k \mid r) = \sum_{|S|=k} p(S \mid r), \tag{139}$$

$$p(i \in S \mid r) = \sum_{S:\, i \in S} p(S \mid r), \tag{140}$$

$$p(a_i \mid r) = p(i \notin S \mid r)\delta_0 + \sum_{S:\, i \in S} p(S \mid r) p(a_i \mid S, r). \tag{141}$$

Equation (141) is a model-averaged spike-and-slab output: source absence contributes a point mass at zero.

The experimentally observed gap between cardinality and support has an exact algebraic decomposition. Let $N_k = \binom{16}{k}$, $\ell_S = \log Z_S$, and $\ell_k^\star = \max_{|S|=k} \ell_S$. Define the effective multiplicity

$$m_{\mathrm{eff},k} = \sum_{|S|=k} \exp(\ell_S - \ell_k^\star). \tag{142}$$

Proposition 2 (Exact effective-multiplicity identity): Under a uniform prior over supports of cardinality k,

$$\log p(r \mid K = k) = \ell_k^\star - \log N_k + \log m_{\mathrm{eff},k}, \tag{143}$$

$$\max_{|S|=k} p(S \mid K = k, r) = m_{\mathrm{eff},k}^{-1}. \tag{144}$$

a: Proof.

Factor $\exp(\ell_k^\star)$ from $N_k^{-1}\sum_{|S|=k}\exp(\ell_S)$ and take logarithms to obtain (143). Conditional normalization within cardinality k divides the largest evidence by the same sum, giving (144). □

Many similar pair supports can therefore increase evidence for $K = 2$ while preventing any one pair from dominating. Entropy separates the same two questions exactly because K is a deterministic function of S:

$$H(S \mid r) = H(K \mid r) + \sum_k p(K = k \mid r) H(S \mid K = k, r). \tag{145}$$

The first term measures cardinality uncertainty; the second measures localization ambiguity within each cardinality.

For set-valued output, sort supports by posterior mass and define the smallest credible set

$$\mathcal{C}_q(r) = \arg\min_{\mathcal{C}} |\mathcal{C}| \quad \text{subject to} \quad \sum_{S\in\mathcal{C}} p(S \mid r) \geq q. \tag{146}$$

Candidate-set size and coverage are more informative than an arbitrary Top-m when the posterior is multimodal. The present paper has not yet calibrated an abstention threshold or credible-set coverage prospectively.

E. STEP-BY-STEP EVALUATED BAYESIAN ALGORITHM

Table 11 records the exact order of the load-support computation. Steps marked "evaluated" produced the 137-hypothesis results; optional outputs are mathematically defined but are not promoted to validated claims.

The direct quadrature cost per record is

$$1 + GQ + \binom{G}{2} Q^2 \tag{147}$$

likelihood-node evaluations for $K \leq 2$. With $G = 16$ and $Q = 31$, this is 115,817 nodes before vectorization or reuse. The count explains why the support bank, integration order, and horizon must be frozen together when runtime is compared.

An incremental implementation of $\mathcal{E}_{\mathrm{B}}$ is computational, not a new estimator. If the exact innovations factorization includes the same initial AR(1) density, then at every frozen prefix it must reproduce the batch quadratic form, evidence, and posterior to numerical tolerance. This is a regression test, not evidence for the joint model below.

F. PROPOSED JOINT HYBRID-DAE BAYES $\mathcal{E}_{\mathrm{J}}$

The proposed extension would infer state trajectory X, algebraic variables Z, event family c, onset τ, support S, amplitudes a_S, operating nuisance $\boldsymbol{\eta}$, and integrity corruption b. Its posterior factorization is

$$\begin{aligned} p(X, Z, \boldsymbol{\eta}, b, c, \tau, S, a_S \mid Y) &\propto p(\boldsymbol{\eta}) p(c) p(\tau \mid c) p(S \mid c) \\ &\times p(a_S \mid S, c) p(\boldsymbol{\chi}_0 \mid \boldsymbol{\eta}) \\ &\times \prod_{t=0}^{T-1} p(\boldsymbol{\chi}_{t+1} \mid \boldsymbol{\chi}_t, \boldsymbol{\eta}, c, \tau, S, a_S) \\ &\times \prod_{t=0}^{T} p(y_t \mid \boldsymbol{\chi}_t, b_t, c, \tau, S, a_S) p(b_t), \end{aligned} \tag{148}$$

subject to the candidate DAE constraints. Here $\boldsymbol{\chi}_t$ denotes a consistent state/nuisance chart. A constrained negative twice-log density, excluding the discrete candidate priors, is

$$\begin{aligned} \mathcal{J}_S^\circ = \|\boldsymbol{\chi}_0 - \overline{\boldsymbol{\chi}}_0\|^2_{P_0^{-1}} + \sum_{t=0}^{T} \|r_{S,t}^{\mathrm{pmu}}\|^2_{R_t^{-1}} \\ + \sum_{t=0}^{T-1} \|r_{S,t}^{\mathrm{dyn}}\|^2_{Q_t^{-1}} + 2\lambda_b \sum_{t=0}^{T} \|b_t\|_1. \end{aligned} \tag{149}$$

Candidate evidence must be evaluated in a common feasible measure. The boundary correction is exposed by an explicit concentration parameter rather than by informally assuming a concentrated posterior.

Let $\mathcal{R}_0$ fix operating point, onset, and load family and replace the candidate prediction with the same quadratic manifold, W2 covariance, support prior, and amplitude prior used by $\mathcal{E}_{\mathrm{B}}$. Then the defining regression condition is

$$p_{\mathrm{J}}(S, a_S \mid Y, \mathcal{R}_0) = p_{\mathrm{B}}(S, a_S \mid r, \boldsymbol{\eta}_0, \tau_0, c_{load}). \tag{150}$$

The equality is not claimed for an unrestricted nonlinear DAE likelihood; it holds only when the complete statistical contract is shared.

Theorem 18 (Local cone–Laplace evidence): Let $L_{S,\epsilon}$ be the log likelihood plus the log of a proper prior in identifiable local coordinates of dimension d_S. Suppose the rescaled feasible domains converge locally to a closed polyhedral cone K_S and

$$L_{S,\epsilon}(\boldsymbol{\xi}_0 + \epsilon u) - L_{S,\epsilon}(\boldsymbol{\xi}_0) = h_S^{\mathsf{T}} u - \tfrac{1}{2} u^{\mathsf{T}} H_S u + r_\epsilon(u), \tag{151}$$

where $H_S \succ \mathbf{0}$ and $r_\epsilon \to 0$ uniformly on compact sets. Assume an integrable dominating envelope and negligible evidence from distant modes. Then

$$\begin{aligned} Z_{S,\epsilon} &= e^{L_{S,\epsilon}(\boldsymbol{\xi}_0)} \epsilon^{d_S} (2\pi)^{d_S/2} |H_S|^{-1/2} \\ &\quad \times \exp\left(\tfrac{1}{2} h_S^{\mathsf{T}} H_S^{-1} h_S\right) \Pr(U_S \in K_S)[1 + o(1)], \\ U_S &\sim \mathcal{N}(H_S^{-1} h_S, H_S^{-1}). \end{aligned} \tag{152}$$

a: Proof.

Set $\boldsymbol{\xi} = \boldsymbol{\xi}_0 + \epsilon u$, which contributes the Jacobian ϵ^{d_S}. The compact-uniform expansion gives pointwise convergence of the rescaled integrand to $\exp(h_S^{\mathsf{T}} u - u^{\mathsf{T}} H_S u/2)$. The assumed

TABLE 11. Evaluated 137-hypothesis Bayesian load-support algorithm.

Step	Operation	Computation and frozen choice	Status
0	Declare contract	Fix 16 candidate load buses, $K \leq 2$, known onset, 30-frame window, eight PMUs, nominal topology, and no corrupted channels	Evaluated
1	Validate plant	Check equilibrium, terminal-current orientation, sampling, nonlinear solve acceptance, and first-order response before building the bank	Evaluated
2	Build physical dictionary	Generate centered D_i, self-curvature Q_i, and cross-interaction Q_{ij} from simulator-consistent interventions	Evaluated
3	Fit noise on CAL	Estimate channel covariance and AR(1) dependence from normal calibration records; freeze W2 before event TEST	Evaluated
4	Freeze priors and integrator	Set $p(K)$, uniform support priors, Gaussian amplitude slab, GH31 nodes/weights, and independent GK checks	Evaluated
5	Form the residual	Align the known onset, subtract the frozen normal mean, retain the fixed PMU/channel order, and apply the declared temporal covariance	Evaluated
6	Enumerate supports	Construct H_0, 16 single-source supports, and 120 two-source supports; no learned source label prunes the bank	Evaluated
7	Integrate each support	Evaluate (136) with (137); retain quadrature-weighted amplitude moments	Evaluated
8	Normalize once	Add the support prior in log space, subtract the global log-sum-exp, and obtain (138)	Evaluated
9	Read decisions in order	Report event presence, $p(K \mid r)$, conditional support ranking, source inclusion, and conditional amplitude intervals separately	Evaluated
10	Quantify ambiguity	Compute entropy, effective multiplicity, or $\mathcal{C}_q$ without relabeling them as validated abstention guarantees	Defined; partial evidence
11	Record provenance	Store the plant identifier, support, amplitude, noise seed, likelihood version, and frozen artifact hash for every independent record	Evaluated

envelope permits dominated convergence, while the distant-mode condition localizes the original integral. Completing the square in the limiting quadratic integral gives the Gaussian normalizer and the cone probability in (152). □

For an interior model $K_S = \mathbb{R}^{d_S}$ and the probability factor is one. For one or more boundary amplitudes it is a Gaussian orthant probability, not generally 2^{-q}. Half-normal prior normalizers supply their own factors in addition to this cone probability. Chart Jacobians, prior normalizers, and candidate-dependent covariance determinants belong in $L_{S,\epsilon}$ or the measure term. Comparing only optimized residuals drops both volume and boundary terms and does not produce a posterior.

For a small-noise Gaussian Hybrid-DAE trajectory map $\boldsymbol{\mu}_S(\boldsymbol{\xi}_0 + \epsilon u) = \boldsymbol{\mu}_0 + \epsilon J_S u + O(\epsilon^2)$,

$$h_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} z, \qquad H_S = J_S^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} J_S, \tag{153}$$

so positive curvature is equivalent to first-order identifiability in the reduced coordinates. This is a theoretical local specialization, not a validation of the unevaluated joint estimator. If H_S is singular, merely replacing its inverse by a pseudoinverse is invalid: redundant coordinates must first be quotiented out, while genuine higher-order identification changes the evidence scale. For example,

$$\int_0^\infty e^{-na^4/4}\, da = \frac{\Gamma(1/4)}{4^{3/4}} n^{-1/4}, \tag{154}$$

not the $n^{-1/2}$ rate of regular quadratic Laplace theory.

Proposition 3 (Exact reduction along redundant fibers): Suppose a parameter chart $\boldsymbol{\theta} \leftrightarrow (\boldsymbol{\xi}, \boldsymbol{\zeta})$ is one-to-one on the prior support and the observable map satisfies $\boldsymbol{\mu}(\boldsymbol{\theta}) = \overline{\boldsymbol{\mu}}(\boldsymbol{\xi})$ there. If $\widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})$ is the transformed proper prior, then the evidence equals the reduced integral with induced prior

$$\overline{\pi}(\boldsymbol{\xi}) = \int \widetilde{\pi}(\boldsymbol{\xi}, \boldsymbol{\zeta})\, d\boldsymbol{\zeta}. \tag{155}$$

Thus an exactly redundant coordinate contributes no vanishing small-noise power; its effect is the pushforward prior measure, not a pseudodeterminant.

b: Proof.

The likelihood is independent of $\boldsymbol{\zeta}$. Fubini's theorem integrates the proper transformed prior along each fiber and gives $Z = \int p(y \mid \boldsymbol{\xi}) \overline{\pi}(\boldsymbol{\xi}) d\boldsymbol{\xi}$ exactly. □

For a merely local chart, the same reduction is asymptotic only after a localization condition excludes evidence from distant branches.

Theorem 19 (Mixed regular and higher-order evidence): In whitened coordinates, suppose the first observable expansion after redundancy removal is

$$\boldsymbol{\mu}(\boldsymbol{\xi}, a) = \boldsymbol{\mu}_0 + A\boldsymbol{\xi} + \frac{c_k}{k!} a^k + o(\|\boldsymbol{\xi}\| + |a|^k), \tag{156}$$

where A has rank r, the proper prior density is continuous and positive at the origin, and

$$c_{k,\perp} = (I - P_A) c_k \neq \mathbf{0}. \tag{157}$$

Assume a dominating envelope and negligible evidence away from this branch. For the Gaussian-normalizer-free evidence kernel with noise scale ϵ,

$$M_\epsilon \sim C \epsilon^{r+1/k}, \tag{158}$$

$$C = \pi(\mathbf{0}, 0)(2\pi)^{r/2} |A^{\mathsf{T}} A|^{-1/2} \tag{159}$$

$$\times \frac{\Gamma(1/(2k))}{k} \left[\frac{2(k!)^2}{\|c_{k,\perp}\|_2^2} \right]^{1/(2k)}. \tag{160}$$

c: Proof.

Use $\boldsymbol{\xi} = \epsilon u$ and $a = \epsilon^{1/k} t$. The Jacobian is $\epsilon^{r+1/k}$ and the limiting kernel is $\exp[-\|Au + c_k t^k / k!\|^2 / 2]$. The component of c_k in $\operatorname{range} A$ disappears after a translation of u. Gaussian integration over the remaining regular coordinates yields $(2\pi)^{r/2} |A^{\mathsf{T}} A|^{-1/2}$; integration of the normal component over $t \in \mathbb{R}$ gives the gamma-function factor. Dominated convergence supplies the prior density at the origin and the stated limit. □

For one additional amplitude, the theorem separates three cardinality cases. A regular direction supplies an $O(\epsilon)$ Occam factor, a direction first visible at order k supplies $O(\epsilon^{1/k})$,

TABLE 12. Deterministic checks of boundary-aware evidence identities. These toy systems are not evaluated estimator results.

Check	Recomputed outcome
Signed/one-sided Bayes factors	max quadrature error 2.7e-12 (pass).
Broad nuisance limit	relative information error 2.1e-09 (pass).
Boundary likelihood ratio	zero mass 0.4972; positive mean 0.9987 (pass).
Nonlinear cone–Laplace	relative-error slope 0.9497 (pass).
Singular quartic evidence	scaling slope -0.2500 (pass).

and an exactly redundant extension supplies an $O(1)$ factor determined by its induced prior. For $\epsilon \asymp n^{-1/2}$ these become $n^{-1/2}$, $n^{-1/(2k)}$, and n^0. These are local evidence rates; the RAWSIM39 gate ledger does not contain the higher-order derivatives required to assign them to its rejected parameters.

For a sliding window, eliminating old Gaussian variables creates an arrival prior. Write its canonical log kernel as $-\frac{1}{2}z^{\mathsf{T}}\Lambda z + h^{\mathsf{T}}z + c$. If the information matrix and vector are partitioned into variables a to eliminate and retained variables b, exact integration yields

$$\Lambda_{\text{new}} = \Lambda_{bb} - \Lambda_{ba}\Lambda_{aa}^{-1}\Lambda_{ab}, \tag{161}$$

$$h_{\text{new}} = h_b - \Lambda_{ba}\Lambda_{aa}^{-1}h_a, \tag{162}$$

$$c_{\text{new}} = c + \tfrac{1}{2}h_a^{\mathsf{T}}\Lambda_{aa}^{-1}h_a + \tfrac{n_a}{2}\log(2\pi) - \tfrac{1}{2}\log\det\Lambda_{aa}. \tag{163}$$

The first two lines preserve the next MAP problem; the last line is also required to preserve evidence between candidates.

Table 13 gives the intended sequence. It is a specification for the next experiment, not a claim about the results reported in the main paper.

For a posterior state $\pi_t(S) = p(S \mid Y_{1:t})$, an idealized unit-cost support decision with per-frame delay cost c satisfies the Bellman equation

$$V(\pi) = \min\left\{1 - \max_S \pi(S),\ c + \mathbb{E}[V(\pi_{t+1}) \mid \pi_t = \pi]\right\}. \tag{164}$$

The one-step expected information gain is

$$I(S; y_{t+1} \mid Y_{1:t}) = H(S \mid Y_{1:t}) - \mathbb{E}[H(S \mid Y_{1:t+1})]. \tag{165}$$

Equations (164)–(165) motivate adaptive horizon selection. They are not an executed stopping rule for the current fixed-onset, fixed-window bank.

Finally, a hidden-voltage or derived-output posterior must average across support uncertainty. With $w_S = p(S \mid Y)$, conditional output mean m_S, and covariance P_S,

$$\overline{m} = \sum_S w_S m_S, \tag{166}$$

$$\mathrm{Cov}(\psi \mid Y) = \sum_S w_S P_S + \sum_S w_S (m_S - \overline{m})(m_S - \overline{m})^{\mathsf{T}}. \tag{167}$$

The second term is between-support ambiguity. Nonlinear output moments may be computed from posterior samples or a local delta method; phasors are averaged in rectangular coordinates before magnitude and angle are formed.

G. ESTIMATOR CLAIM BOUNDARY

The state recursion is an independently evaluated component. $\mathcal{E}_{\mathrm{B}}$ computes normalized Bayesian evidence under the frozen quadratic W2/L2 model and is the implemented reference event estimator. $\mathcal{E}_{\mathrm{J}}$, its reduction test, learned discrepancy, unknown-onset/family inference, changing-PMU operation, virtual outputs, and sequential stopping remain specified but unexecuted. The physical recentering is a regularized constrained estimator with separately assessed uncertainty; it is not relabeled as either event estimator.

APPENDIX F COMPLETE FROZEN EXPERIMENTAL CONTRACTS

The PowerDynamics validation ladder checks static translation, PMU terminal-current orientation, equilibrium residual, local modes, nonlinear acceptance, and first-order response before estimator scoring. All 980 main mismatch cases and 175 refinement cases pass the declared acceptance gates. The physical-recentering TEST includes 20 equilibria at each of three shift scales; a solve is accepted only when the optimizer succeeds, outputs are finite, and the maximum AC residual is at most 5×10^{-5}. Rejections remain in the denominator.

For GLOBAL-137, the physical dictionary, covariance, priors, onset, and quadrature are frozen before new trajectories are generated. Three disjoint noise realizations accompany each event path, and 100 additional H_0 records represent no event. The identifiers have zero overlap with 1,816 historical records. The T120 bank preserves exact prefixes at 30, 60, 90, and 120 frames; independent operating-point validation remains pending.

The ANDES ordinary split uses three replicas for training, one for development, and one for testing at each source configuration. The source-holdout audit removes the physical asset from every source-labeled operation and supplies the true event family at evaluation. These contracts answer interpolation and source generalization separately.

TABLE 13. Candidate-conditioned joint estimator design and current implementation boundary.

Step	Operation	Required calculation	Present status
1	Pre-event update	Use only pre-event measurements to update state and operating nuisance; retain cross covariance and normalized arrival evidence	State filter evaluated; joint nuisance pending
2	Integrity mask	Mark missing channels exactly and introduce sparse corruption variables for available but suspect channels	Ontology defined; common inference pending
3	Candidate generation	Enumerate physical family, support, onset, and severity domain without using source identity as a learned label	Load family evaluated; other families pending
4	Candidate prediction	Apply the candidate DAE/event map and the current PMU membership to generate a physical trajectory	Load bank evaluated
5	Candidate-conditioned solve	Fit state and nuisance on the feasible DAE manifold, conditional on the physical candidate and amplitude quadrature node	Not executed jointly
6	Learned discrepancy	If used, predict only candidate-residual discrepancy and covariance; do not replace the physical hypothesis or consume target source labels	Not implemented
7	Evidence normalization	Retain GH31 amplitude marginalization when $\lvert S\rvert \leq 2$; integrate or approximate the remaining coordinates with all determinants, priors, and Schur normalizers	GH31 load evidence evaluated; joint evidence pending
8	Posterior outputs	Return state-functional mixture, cardinality, support, integrity state, severity, and candidate-set uncertainty	Separate state/load outputs only
9	Sequential decision	Continue observing when expected information gain justifies delay; use distinct thresholds for detection, cardinality, and exact source	Not evaluated
10	Reconfigure and verify	After accepting a physical change, rebuild the operating center and check subsequent innovations before declaring a persistent model update	Static recentering only

TABLE 14. Complete frozen-contract inventory. The main paper groups these contracts by research question.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families and seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Regularized physical recentering	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Load tangent audit	16 load buses; centered $\pm 0.5\%$ nonlinear simulations	Candidate-bus intervention
Bayesian load inference V1	Eight amplitudes; 2,560 event / 320 no-event TEST records	Source-amplitude-noise record
Bayesian load inference V2	1,000 normal CAL; 500 normal + 640 event DEV; weak and finite TEST blocks	Source-amplitude-noise record
Prospective 137-hypothesis inference	Fresh $H_0 + 16$ singles $+120$ pairs; 3,488 paths; 10,564 decisions	Physical trajectory or complete record, as declared
Weak-regime mechanism audit	Frozen GH31 prefixes at 5–30 frames; 2,400 paths	Retrospective source-pair trajectory
Long-horizon diagnostic	Exact prefixes through 120 frames; 32 nominal pair paths; 137 hypotheses	Diagnostic trajectory, not population recovery
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied	101 event parents, each evaluated across three seeds

APPENDIX G SECONDARY RESULTS

The main paper retains the figures needed for the scientific argument. This appendix records secondary diagnostics and exact values that would otherwise interrupt that argument. The two compact tables first preserve the exact mismatch and weak-signal values; the full-width panels then show the corresponding temporal, likelihood, and supervised-baseline diagnostics.

TABLE 15. Median B2 hidden-voltage HVRE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	HVRE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

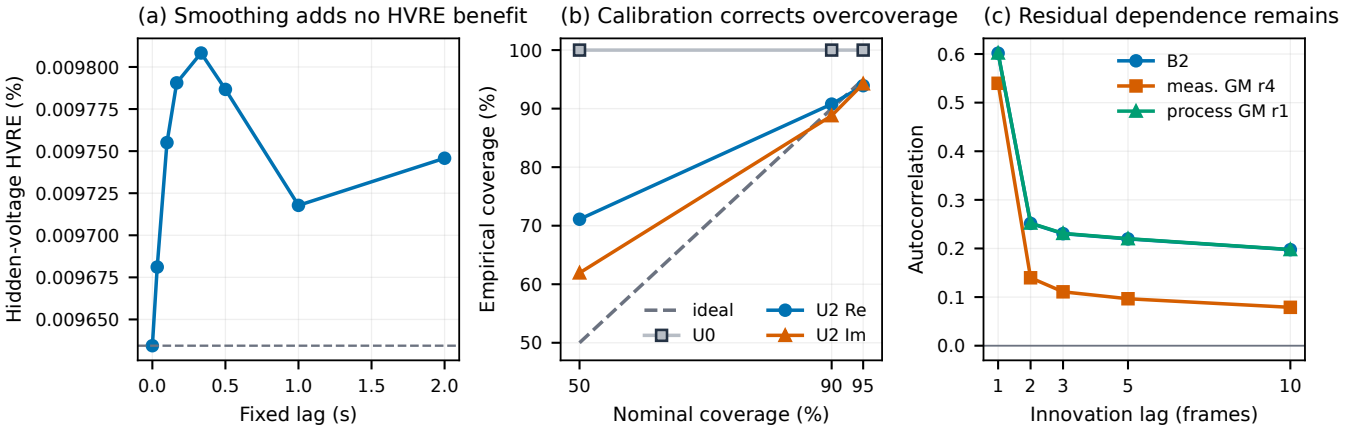


FIGURE 1. **Secondary temporal and uncertainty diagnostics for B2/B3 and calibration variants. These panels diagnose model adequacy; they do not upgrade nominal marginal coverage to a complete uncertainty claim.**

TABLE 16. **Sign-pooled weak load-event TEST results for the frozen W2/L2 model. Each row contains 1,600 event cases (16 source buses, two signs, 50 noise replicas); the false-positive rate uses 800 no-event records.**

$\lvert a \rvert$ (%)	AUROC	FPR (%)	FNR (%)	Top-1 (%)	Top-3 (%)
0.010	0.733	0.25	83.1	13.0	37.1
0.020	0.906	0.25	45.0	31.8	69.0
0.050	0.995	0.25	4.5	73.4	95.9
0.100	1.000	0.25	0.2	93.2	99.4
0.200	1.000	0.25	0.0	98.9	100.0
0.400	1.000	0.25	0.0	100.0	100.0

TABLE 17. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	HVRE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

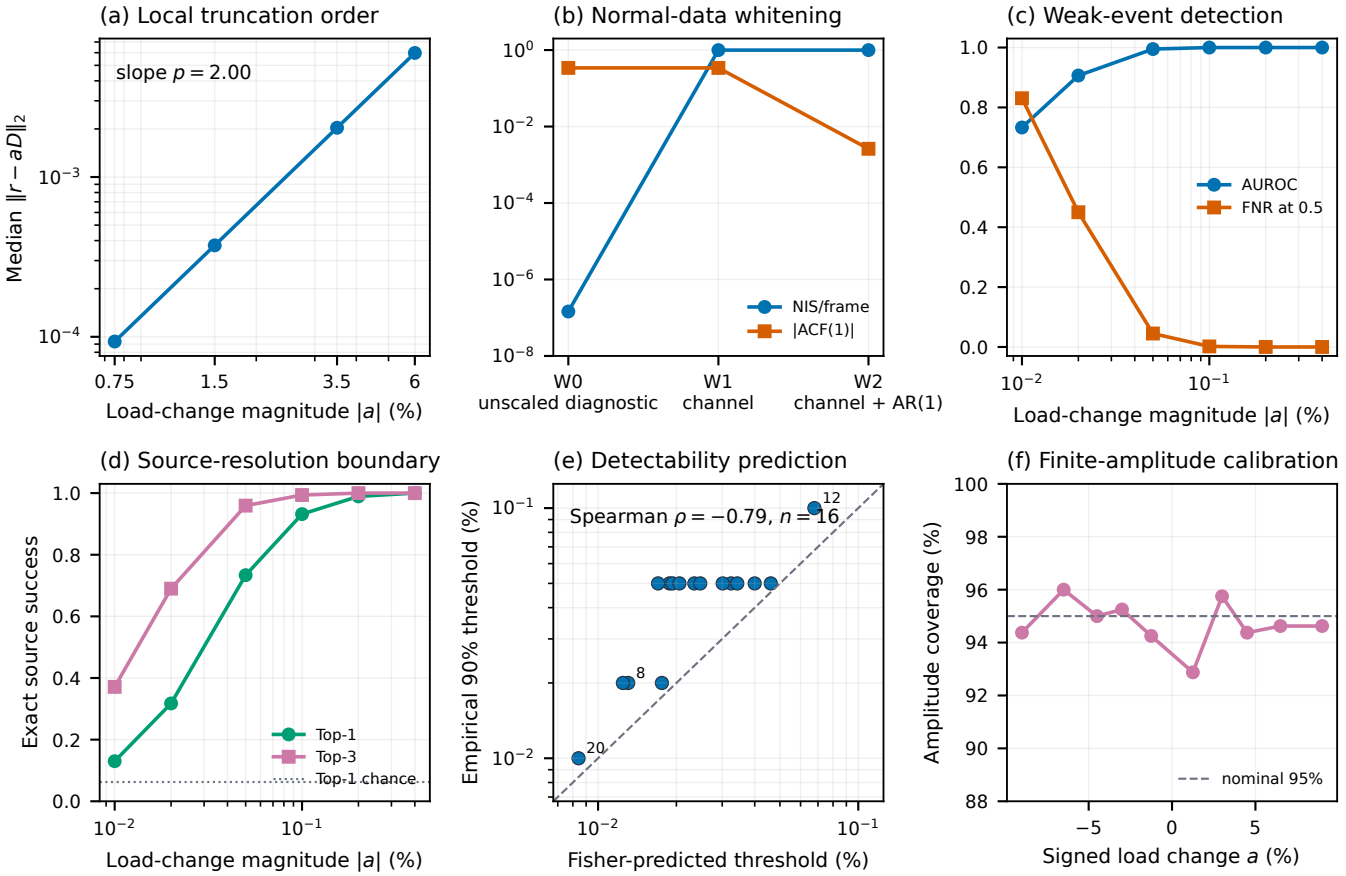


FIGURE 2. Single-source development evidence for the frozen load likelihood. This archived component study is distinct from the causal and signature experiments in the main article.

TABLE 18. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

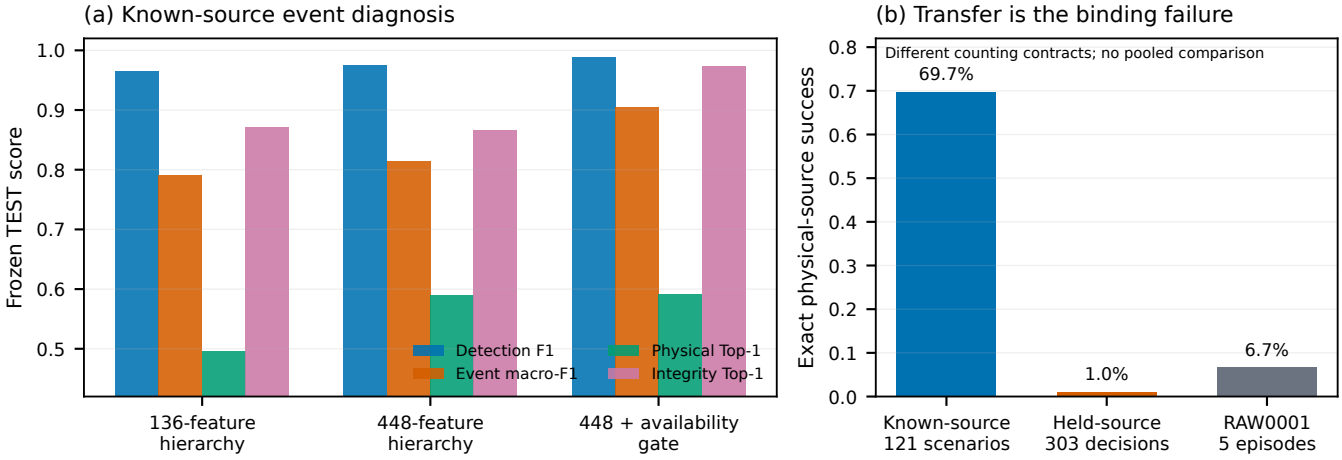


FIGURE 3. **Secondary evidence for the supervised ANDES baseline, including known-source performance and the source-holdout failure.**

APPENDIX H EXPLORATORY HIERARCHICAL SINGLE-EVENT PILOT

This appendix records the September 24, 2026 estimator handoff as a *provisional* numerical branch of the joint design $\mathcal{E}_{\mathrm{J}}$. Its source documents are the 30-page mathematical derivation book and 24-page technical integration dossier identified in the repository integration ledger. Their statements are technical source material, not independent instructions or substitutes for campaign outputs. The dossiers label many checks "artifact-verified" in packages supplied elsewhere; those packages were not supplied with this manuscript checkout. The latest strict-final comparison is explicitly tagged terminal-reported in both documents. All values below are therefore transcribed as *dossier-reported* diagnostics and excluded from the article's frozen primary tables.

A. STATE, MEASUREMENTS, AND EVENT OPERATORS

The pilot's declared model has 114 differential states, 78 algebraic rectangular-voltage coordinates, and 32 measured rectangular channels: voltage at eight PMUs and terminal current at eight monitored branches. Sampling is at $1/30$ s. Its default matrices are $P_0 = 10^{-2} I_{114}$, $Q = 10^{-6} I_{114}$, and $R = 10^{-6} I_{32}$. These numbers identify the pilot likelihood, not the separately frozen covariance W2 of $\mathcal{E}_{\mathrm{B}}$. Three-phase phasors, frequency, ROCOF, and data-present fields in the historical competition files do not silently become inputs to this 32-channel physical likelihood.

For a fixed candidate H and severity ϑ_H, the semi-explicit system is $\dot{x} = f_H(x, z; \vartheta_H)$, $\mathbf{0} = g_H(x, z; \vartheta_H)$, and $y = h_H(x, z) + v$. At a regular point with invertible $G_z = \partial g_H / \partial z$, the implicit-function theorem gives

$$Z_x = -G_z^{-1} G_x, \quad Z_\vartheta = -G_z^{-1} G_\vartheta, \quad A_c = F_x - F_z G_z^{-1} G_x. \tag{168}$$

The corresponding reduced event pathway is $B_\vartheta = F_\vartheta + F_z Z_\vartheta$ and the direct measured pathway is $D_\vartheta = H_\vartheta + H_z Z_\vartheta$. A generator-reference change can satisfy $D_\vartheta = \mathbf{0}$ while $B_\vartheta \neq \mathbf{0}$: it is invisible to a raw algebraic screen but visible after dynamic propagation. The event operator and its duration must be replayed consistently in truth generation and candidate fitting. The pilot dossier reports a five-frame fault pulse, persistent load steps unless recovery is declared, and finite generator and line events with recovery. Some earlier inference runs used persistent approximations for the latter two; those rankings have a known model-contract discrepancy.

The current single-event catalog has 39 fault buses, 19 load buses, nine controlled generators, 46 in-service branches, and a separate H_0 comparison. Thus it has 113 physical candidates and one null candidate; the screening rank denominators may be smaller when a candidate is numerically unscorable. Fault severity uses a log-resistance coordinate, load severity two active/reactive changes, generator severity a reference-power change, and an outage has a discrete branch and switching profile. Missing or bad PMU channels belong to the integrity variable of $\mathcal{E}_{\mathrm{J}}$ and were not jointly inferred in this pilot.

B. JOINT GAUSS–NEWTON STEP AND ITS INTERPRETATION

For one candidate, write the conditional MAP residual as $r_H(X, \vartheta)$ after whitening and appending proper prior residuals. Let $\Delta u = \mathrm{col}(\Delta X, \Delta\vartheta)$ and $g = \mathrm{col}(g_X, g_\vartheta)$. At an iterate, its local quadratic is

$$q(\Delta u) = g^{\mathsf{T}} \Delta u + \tfrac{1}{2} \Delta u^{\mathsf{T}} H \Delta u, \qquad H = \begin{bmatrix} H_{XX} & H_{X\vartheta} \\ H_{\vartheta X} & H_{\vartheta\vartheta} \end{bmatrix}. \tag{169}$$

Eliminating ΔX gives $q_{\mathrm{red}}(\Delta\vartheta) = \mathrm{const} + b_H^{\mathsf{T}} \Delta\vartheta + \tfrac{1}{2} \Delta\vartheta^{\mathsf{T}} S_H \Delta\vartheta$, where $S_H = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ and $b_H = g_\vartheta - H_{\vartheta X} H_{XX}^{-1} g_X$. Hence its unconstrained local decrease is $-\tfrac{1}{2} b_H^{\mathsf{T}} S_H^{-1} b_H$ when $S_H \succ 0$. This algebra needs no conditional convergence of X: it is the full-block Gauss–Newton step expressed with a smaller parameter solve. By contrast, if $\psi(\vartheta) = \min_X J_H(X, \vartheta)$, the identity $\nabla^2 \psi = H_{\vartheta\vartheta} - H_{\vartheta X} H_{XX}^{-1} H_{X\vartheta}$ is a local *profile-Hessian* statement only at a regular conditional optimum, and for the true Hessian rather than an arbitrary Gauss–Newton approximation. An unsettled state trajectory can change the sign of the reduced gradient without invalidating Schur elimination.

The numerical realization factors the temporally sparse state block and obtains its action on g_X and the few columns of $H_{X\vartheta}$ by solves; it does not form H_{XX}^{-1}. One Armijo step length updates both state and severity and evaluates the actual nonlinear DAE objective. The pilot selected Rodas4P for transition evaluation after the reported cross-integrator audit. That choice is a numerical guard, not a new inference theorem. Levenberg–Marquardt remains an independent finalizer candidate for the unresolved fault/line fits [27], [28].

C. DETECTION, PHYSICAL SCREENING, AND RACING

Let $\boldsymbol{\nu}_k = y_k - \hat{y}_{k|k-1}^{H_0}$ be the observed innovation. An alarm tests the adequacy of H_0 on measured channels; it is not itself the physical onset. The pilot refines onset over a bounded interval and fits the common initial state $\widehat{X}_{\mathrm{pre}}$ on data before that onset. Every candidate starts from this same reference. This prevents a strong event from contaminating its own null-model baseline.

For a smooth candidate nested at H_0, the efficient local score is $T_H = b_H^{\mathsf{T}} S_H^{-1} b_H$ under positive definite S_H. If a severity is physically one-sided, maximizing the local quadratic over its tangent cone replaces the unconstrained inverse step and changes the null law; the ordinary signed score is then only a proposal proxy. For a fault or topology switch, let s_H be its short-horizon physical response from $\widehat{X}_{\mathrm{pre}}$ and r the corresponding null residual. With $W^{\mathsf{T}} W = R_H^{-1}$ and nuisance projector P_N, define $\widetilde{s}_H = (I - P_N) W s_H$ and $\widetilde{r} = (I - P_N) W r$. A signed matched score is $(\widetilde{r}^{\mathsf{T}} \widetilde{s}_H)^2 / \|\widetilde{s}_H\|^2$, with the allowed severity domain imposed before interpreting it as a physical fit. The two screening scores need not share a calibrated scale across families.

The frozen pilot consequently ranks within each family, retains a family quota, and allocates increasing nonlinear

TABLE 19. Single-event pilot ledger transcribed from the two author-supplied September 24 dossiers. All rows require the corresponding campaign archives for an independent paper audit.

Check or campaign	Dossier-reported result and precise interpretation	Source tier
Linear Gaussian oracle E0	RTS and batch MAP means agree near 10^{-16}; covariance/log-likelihood checks near 10^{-14} after an omitted evidence constant was repaired. This guards the numerical algebra only.	Dossier AV
Three-node and five-bus toys	E1: five development and 30 evaluation seeds, hidden algebraic two-sigma coverage 0.927 versus nominal 0.954. E2: first-order classification 0.2875 over 10 hypotheses × eight repetitions; not a nonlinear full-catalog result.	Dossier AV
Transition/derivative audit	Rodas4P/Rodas5P: 23 shared successful candidates with maximum endpoint difference 4.9×10^{-7}; Rodas4P evaluated three of three selected Rodas5P failures. Finite-difference coverage was 25/25; one random direction retained a moderate derivative discrepancy.	Dossier AV
Joint update and severity	Full-block and Schur steps agree in 30/30 synthetic block systems to numerical precision. Four fault-resistance initializations of 0.01, 0.03, 0.05, and 0.08 ended in 0.02006–0.02133 around truth 0.021; only the first two passed the stated strict stationarity gate.	Dossier AV
Small hidden-event bank	A hidden LOAD@16 and LINE(23,24) ranked first against selected decoys after joint severity and line-index corrections. These were not full-catalog tests.	Dossier AV
H_0 and onset pilots	Fifty-one disjoint normal calibration windows yielded no alarms at two reported thresholds on that same set; this is neither independent alarm testing nor false alarms per hour. Blind onset alternatives differed by 26–27 frames in two cases.	Dossier AV
Stage-A development	Four cases; 451/456 attempted physical scores were valid (98.9%); family-stratified shortlist retained 4/4 truths.	Dossier AV
Blind mini-holdout	FAULT@20, LOAD@28, GEN@38, and LINE(4,14): within-family truth ranks 3/38, 1/19, 1/9, and 1/46. The shortlist and subsequent racing retained 4/4 truths; GEN survived at the six-candidate cutoff. Four quiet windows were a proposal-level control only.	Dossier AV
Strict final comparison	The later terminal summary reports 0/4 exact winners after 24 final fits. Fault and line include unconverged truth/competitors; load and generator have better-scoring converged neighbors. This is a failure audit, not an identifiable physical error rate.	TR; final archive absent

budgets $B1, B2, B3, B5, B10$ before strict finalization. A candidate with a failed score is recorded as unscorable rather than inserted into a tie by catalog order. The dossier reports that ranking was nonmonotone with iteration budget and that the generator truth survived the blind racing cutoff at rank six of six. Thus survival is a search diagnostic; it says nothing by itself about a final converged winner or computational savings. In the reported racing run all 21 candidates were still evaluated to ten iterations, so a wall-clock advantage has not been demonstrated.

The operational order of this *pilot* is fixed: (1) process only the declared 32-channel PMU sequence and evaluate the H_0 innovation; (2) record alarm time, refine physical onset candidates, and fit a common pre-event state; (3) score every admissible physical family/location with its own event operator and explicit numerical status; (4) retain the frozen within-family quotas; (5) apply the same staged nonlinear budgets to all survivors while checkpointing objective and stationarity; (6) finalize every remaining candidate to the same strict gate; and (7) compare only validly converged modes with consistent priors, normalizers, and boundary treatment. Outputs before step (6) are candidate sets and diagnostics, not final source decisions. This route is retrospective whenever onset refinement or smoothing uses samples after the decision time; it has no measured end-to-end latency guarantee.

Several discarded results in the dossier were caused by implementation changes rather than by power-system information limits. A generator screen based only on the algebraic Jacobian missed its differential pathway; a previous joint update computed but did not apply the severity step; and an outage candidate was once switched at frame zero instead of the inferred onset. A full-window H_0 fit absorbed an event into the null reference, while independent caps on state and severity steps invalidated a quadratic proposal score. Unscored fault candidates inherited catalog order through a tied sentinel value. One detection-delay calculation used the window start rather than its end, and some failed integrator returns were encoded as nonfinite values despite finite unfinished states. The corrected pilot treats each as a separate numerical or semantic gate. Its four-case truth survival must not be merged with superseded rankings from those earlier variants.

D. EVIDENCE AND THE FINAL COMPARISON GATE

For converged, regular, identifiable interior modes only, the local Laplace approximation is

$$\log p(Y \mid H) \simeq -J_H(\widehat{u}_H) + \tfrac{d_H}{2}\log(2\pi) - \tfrac{1}{2}\log|\nabla^2 J_H(\widehat{u}_H)|, \tag{170}$$

where u_H uses a common feasible measure and J_H includes all model-dependent prior and likelihood normalizers. The normalizer, prior scale, curvature determinant, and any boundary or quotient correction must be retained when comparing families of different dimension. Section E-F derives the boundary-aware alternative. A candidate that stops at $B10$ or violates the strict stationarity/solver gate cannot be made comparable merely by evaluating (170). The pilot has not demonstrated exact normalization against the load-bank restriction $\mathcal{R}_0$.

E. DOSSIER-REPORTED NUMERICAL LEDGER

Table 19 preserves the numerical chronology without treating small pilot denominators as population rates. "Dossier AV" means the PDF claims an external campaign package supports the entry; it does not mean this repository has reopened that package. "TR" is the dossier's own terminal-report tier.

The case-specific terminal report places converged LOAD@27 and LOAD@26 ahead of true LOAD@28. It quotes approximate objectives -29986.9, -29985.9, and -29980.7, respectively, with evidence values around 4609.98, 4610.22, and 4607.17. Those values are deliberately absent from the main manuscript: their sign convention and normalizing terms must be checked against the final archive before quantitative model comparison. No exact generator

margin was supplied. The fault and line comparisons are not physically interpretable until both truth and nearest competitors pass a common strict convergence gate.

F. REQUIRED CLOSURE BEFORE A MULTI-FAMILY CLAIM

The next frozen audit must first replay each event operator, measurement conversion, onset, duration, and initial state under the same truth and inference model. It must then use an independent finalizer for the fault/line candidates, retaining every failed solve in the denominator. For converged load/generator mislocalizations, successive oracles should fix the true pre-event state, timing, severity, and trajectory, then remove noise and add a counterfactual nearby PMU. Only a persistent small whitened truth-to-competitor margin after those controls supports an information-limited explanation. The final-survivor archive must supply per-candidate status, objective, prior and likelihood constants, Hessian measure, and evidence normalization. A larger source-disjoint holdout, operational normal exposure, and the $\mathcal{R}_0$ reduction test are still needed. The current pilot cannot be used to upgrade the separate $\mathcal{E}_{\mathrm{B}}$ load-support result into a validated multi-family joint estimator.

APPENDIX I WHAT PHYSICAL STRUCTURE AND LEARNED FEATURES RECOVER

The continuous estimator builds on three distinct questions: what can be reconstructed at unmeasured buses, what response distinguishes candidate sources, and how evidence should be assigned when several explanations fit. The following component experiments answer parts of these questions. They use different cohorts and are not ablations of the final continuous estimator. Figure 4 places their measured effects side by side without pooling their scores.

A. RECOVERING UNMEASURED VOLTAGES IS NOT YET EVENT ATTRIBUTION

In 100 nominal test trajectories, the mean hidden-voltage relative error falls from 0.3402% for the equilibrium reference to 0.00990% for snapshot weighted least squares and 0.00963% for a causal Kalman reconstructor. Most of this improvement comes from conditioning the physical network on the measured phasors; temporal filtering adds a smaller improvement. This supports reconstruction under that model and sensor placement, not unique recovery of every source. The independently evaluated Kalman reconstructor is not silently substituted for the conditional observer in the main article.

B. PHYSICAL FEATURES REMAIN A USEFUL LEARNED ROUTE

The historical feature study raises physical-source Top-1 from 49.6% with 136 features to 54.8% with graph-temporal features added, and 59.0% with the full 448-feature set. Voltage level, relative angle, frequency, time scale and spatial coherence carry complementary information; Fig. 4(b) reports an executed feature comparison rather than an inferred explanation of tree importance. Electrical features make a classifier physics-informed in its representation. They do not enforce Kirchhoff balance on its predicted label or compute the PMU response of a candidate intervention.

For phase-to-neutral RMS phasors and a declared current orientation,

$$S_{ij} = P_{ij} + \mathrm{j}Q_{ij} = \sum_{\phi\in\{a,b,c\}} V_{i\phi} I^{*}_{ij,\phi}. \tag{171}$$

This expression requires voltage and current at the same terminal. In the frozen map, the bus-2 voltage is paired with a current measured at bus 3, and the bus-10 voltage with a current at bus 11. Their raw channel products are therefore power proxies, not exact terminal powers. Even a co-located terminal power is not the bus's net injection unless all relevant terminals are accounted for. Frequency and ROCOF complement the phasors but share the instrument's angle processing; the likelihood must account for that dependence. Fundamental-frequency phasors also do not determine harmonic distortion. Thus adding nominally physical features does not by itself supply missing harmonic or nodal-injection measurements.

C. MIXED-SOURCE RESPONSE COEFFICIENTS

For simultaneous load changes, a second-order response approximation includes products of the input amplitudes. In the archived two-load experiment,

$$s_{q,n}(a) = \sum_i D_{qni} a_i + \sum_i Q_{qni} a_i^2 + \sum_{i<j} Q_{qnij} a_i a_j, \tag{172}$$

where q indexes PMU coordinates, n elapsed reports and i,j load sources. Centered nonlinear simulations estimate the coefficients; the Q terms include their Taylor scaling. The mixed term represents the interaction of two simultaneous load changes. Across 32 archived two-source responses, median whitened error is 0.205 for D, 0.469 after self-curvature, and 0.00307 after mixed terms. This is evidence for response fidelity in that local experiment. It is not a measured classification gain from inserting this second-order tensor into the continuous load/line bank, which uses switched local predictors instead.

The coefficients in (172) form a tensor over measured coordinate, elapsed time and source indices. The useful term is the physical cross-source interaction Q_{qnij}; the array's shape alone does not provide an accuracy or computational advantage. In the continuous estimator, M is instead a vector of aggregated source/family weights at each report, and ΔM is its temporal difference, as defined in the main article. That implementation does not require a tensor decomposition or a tensor-learning library.

D. WHY INTEGRATE SEVERITY RATHER THAN SELECT THE BEST FIT?

An extra source with zero amplitude can reproduce a smaller source set. Best-fit residual alone therefore cannot decide

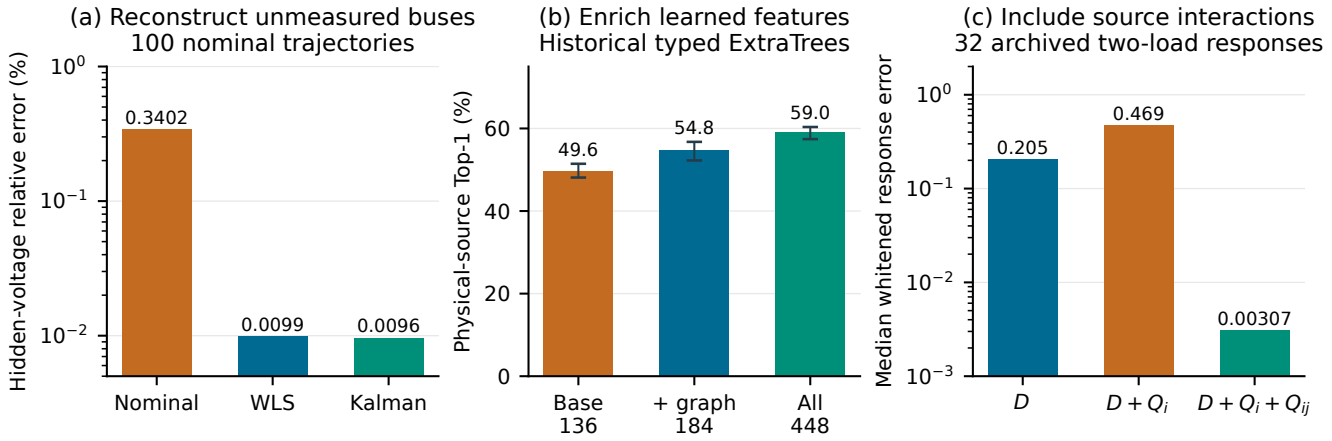


FIGURE 4. **Three separate component experiments. (a) Mean relative error of the 31 unmeasured bus voltages under a matched nominal model; this is not a PMU compliance TVE test. (b) Mean physical-source Top-1 over three model seeds, each scored on 7,915 labeled physical samples, in the historical typed-ExtraTrees feature study; whiskers show the seed range, not a confidence interval. These 136–448-feature models are neither the final hackathon submission nor the causal comparators. (c) Median whitened prediction error at the archived 120-report horizon for two-load responses: mixed-source terms matter, while adding only self-curvature worsens this metric. Each panel has its own units and cohort.**

TABLE 20. **Archived joint cardinality-and-support decisions. Onset, load family and operating context are supplied. These are noise replays of shared physical trajectories, not independent continuous detections.**

Amplitude regime	Replays	Joint support Top-1
Finite / finite	1,440	100.00%
Moderate / moderate	1,440	98.82%
Weak / strong	1,440	66.25%
Weak / weak	5,760	24.38%

cardinality. In the archived load-support estimator, the model-selection quantity is

$$Z_S = \int p(r \mid s_S(a), \Sigma) p(a \mid S)\, da, \qquad p(S \mid r) \propto Z_S p(S). \tag{173}$$

Integration accounts for the prior volume of amplitudes consistent with the data, rather than rewarding only the best-fitting amplitude. This is standard Bayesian model comparison applied to an electrical response model, not a new Bayes rule. The archived implementation evaluates the null, 16 single-source and 120 pair supports with 31-point Gauss–Hermite quadrature per amplitude. Table 20 shows both its successes and its loss of resolution for weak sources.

An audit found a one-source evidence normalization offset in this archive: the implemented cardinality prior is proportional to $(0.20, 0.50/(2\pi), 0.30)$ rather than the declared $(0.20, 0.50, 0.30)$. The table preserves the executed results under that effective prior; it does not validate calibrated posterior probabilities or a corrected replay. The continuous estimator uses separate, independently checked local evidence code. Consequently these component results motivate the physical/Bayesian construction, but do not establish that adding Bayes or tensor storage alone causes its final accuracy advantage.

APPENDIX J EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - a s_k$. The affine conditional state is $m_k + a d_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{174}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{175}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{176}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a} U - \tfrac{1}{2} \widehat{a}^2 G - C_{ref}. \tag{177}$$

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

TABLE 21. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c)P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The H04 comparison contains no such post-startup missing observations.

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

APPENDIX K OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of R11.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h},$$
$$E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{178}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the frozen common angle drift, the three predicted histories are

$$\mu_i^{\mathrm{N}}(a) = y_0 + aD_i^0 + a^2 Q_i^0,$$
$$\mu_i^{\mathrm{R}}(a) = \mu_i^{\mathrm{N}}(a) + \widehat{b},$$
$$\mu_i^{\mathrm{E}}(a) = \mu_i^{\mathrm{R}}(a) + a\widehat{\eta}E_i. \tag{179}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The frozen F01 plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three frozen instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its archived instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

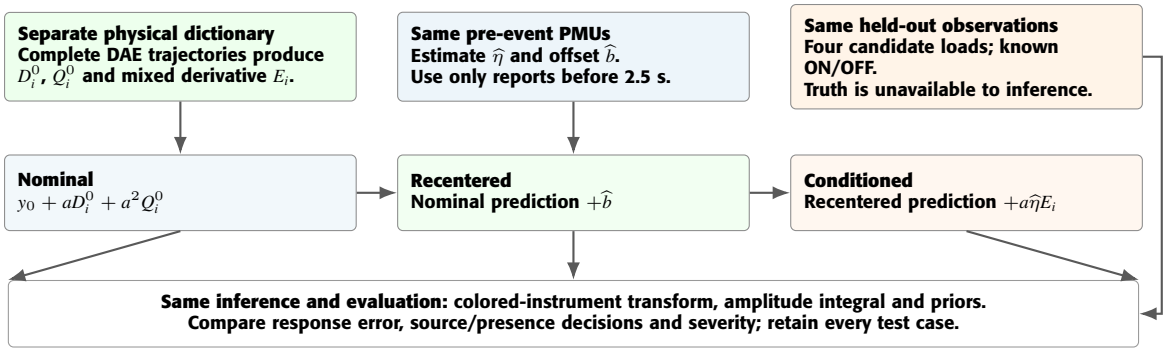


FIGURE 5. Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 6 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the frozen impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the frozen artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

APPENDIX L RECOVERY GEOMETRY AND SOURCE SEPARATION

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{180}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{181}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [29], [30].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The supplementary archive retains the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the

TABLE 22. Controlled signature test, separate from causal detection. Response error is the median whitened incremental error at the generating source/severity over 80 physical cases. Joint correctness includes presence and source on 240 noisy event records; source-only ranking conditions on a load. Severity error is median absolute relative error at the generating source. Normal selections use 15 fixed windows, not normal-hours exposure.

Model	Response error (%)	Joint correct	Source-only correct	Normal event selections	Severity error (%)
Nominal	1.25	114/240	114/240	12/15	3.45
Recentered	1.25	240/240	240/240	0/15	1.2
Conditioned	0.0561	240/240	240/240	0/15	0.329

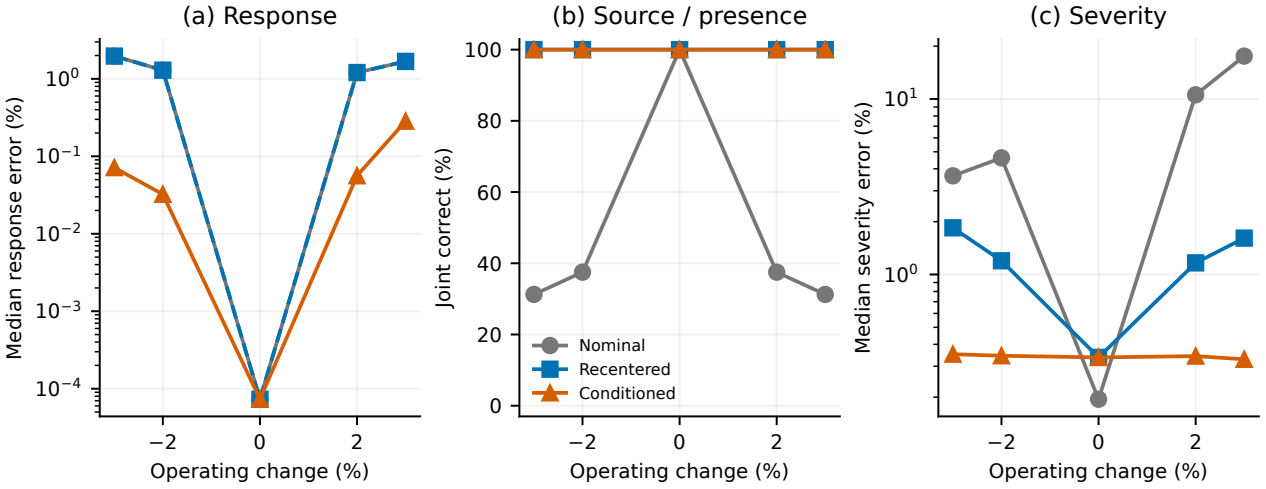


FIGURE 6. Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.

causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete F01 trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied. Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate F01 measurement operator is followed by its frozen innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 8). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an R11 improvement nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

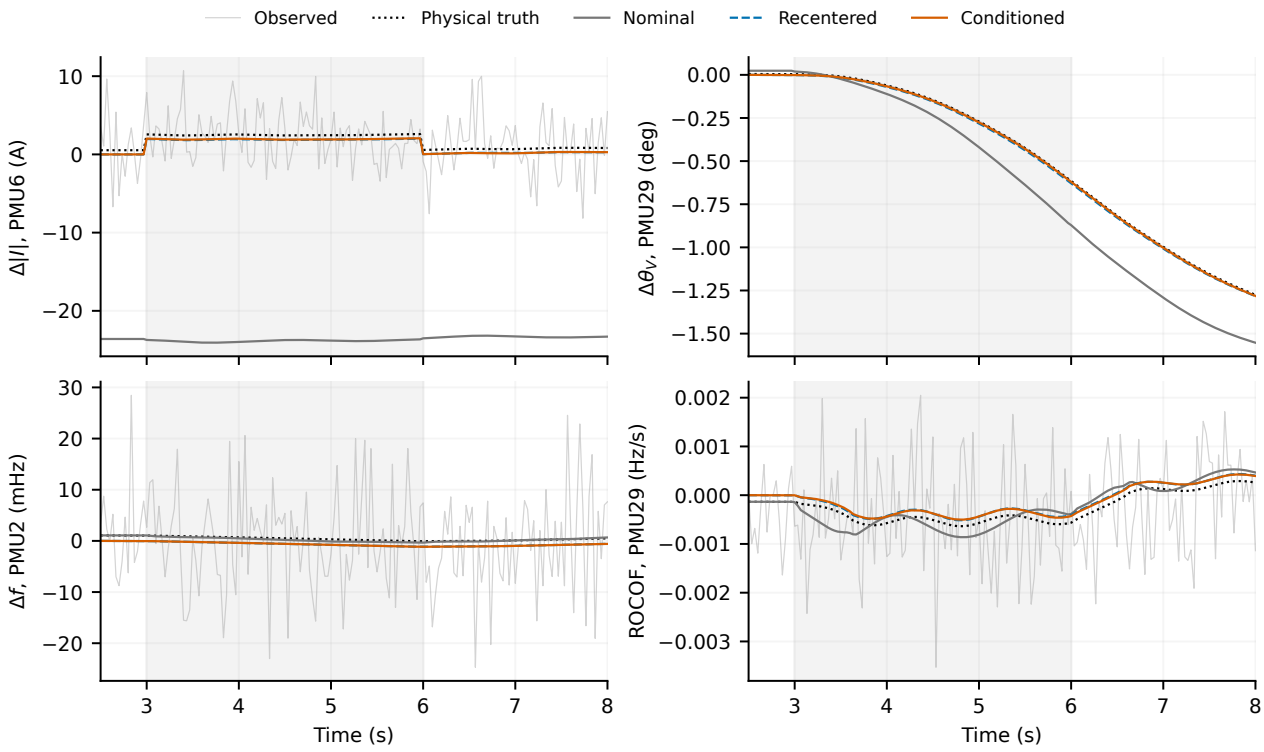


FIGURE 7. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

TABLE 23. **What the causal component actually resolves in H04 development. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.**

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

APPENDIX M DEVELOPMENT STREAMS AND ADMISSION CONTROL

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In H04, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 23). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the frozen adapters; it neither rules out supervised load detection nor reproduces the historical multiclass hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 10 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and the mass-increment identity in the main article separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative experiment leaves R11 and nominal frozen for C01.

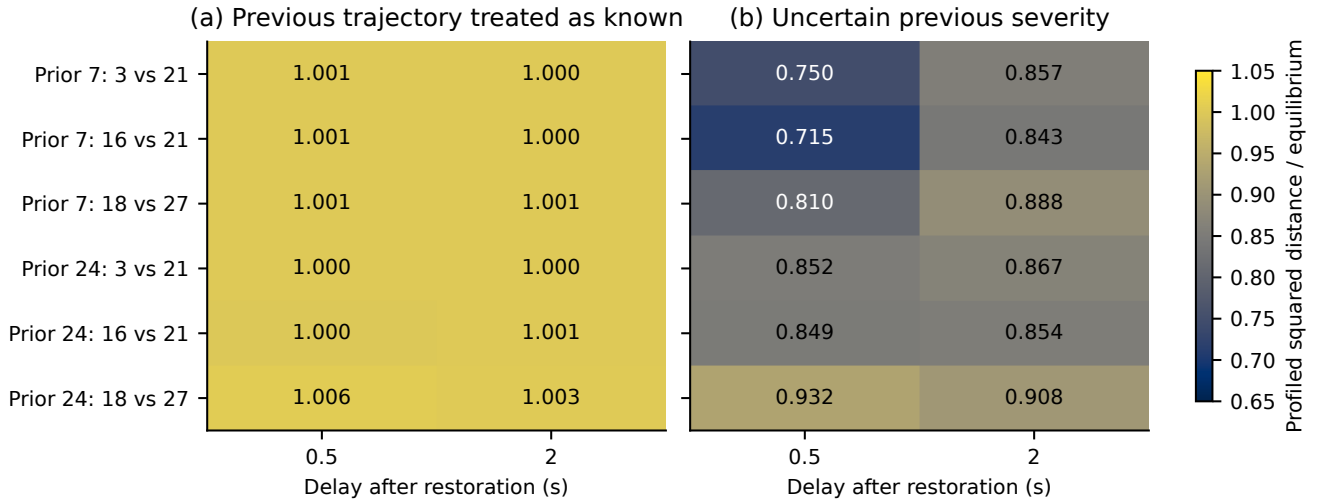


FIGURE 8. **Full-F01 physical control, not an R11 performance comparison. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

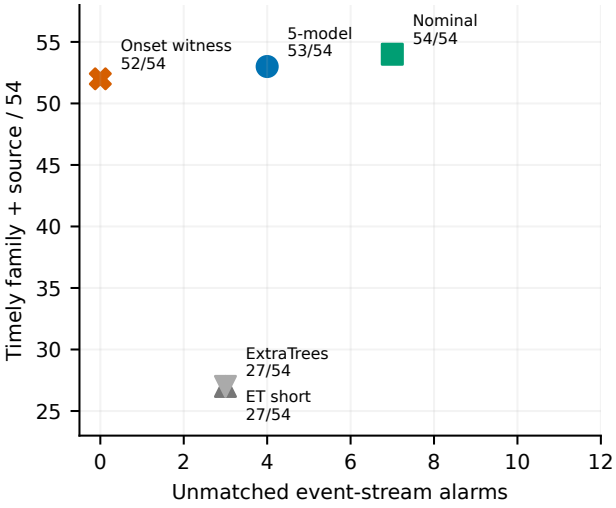


FIGURE 9. **The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.**

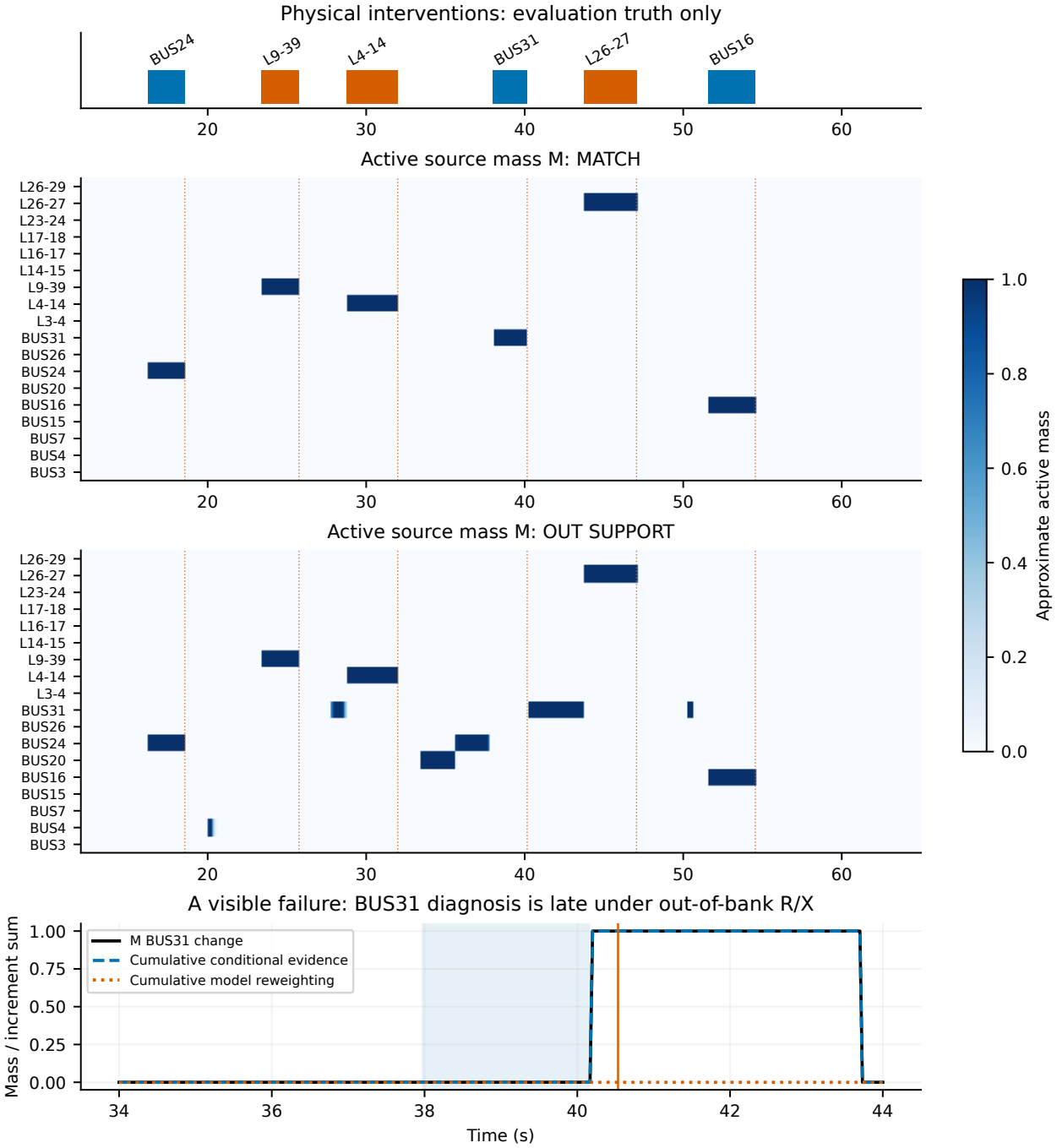


FIGURE 10. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in development parent H002. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

APPENDIX N RESEARCH DECISIONS BEHIND THE MAIN NARRATIVE

The article's main text now follows one question: which physical intervention explains a sparse PMU history, and when is that explanation distinguishable? The supporting studies below retain different estimators and denominators. They are evidence for particular steps in that argument, not a pooled test of a completed joint estimator.

A. THE CAUSAL IMPLEMENTATION'S LOCAL AMPLITUDE INTEGRAL

This calculation describes the retained local Gaussian history model used by R11, not a replacement for the batch estimator's GH31 quadrature. For one load-history interval, write its new-report residual as

$$e = sa + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, S), \qquad a \sim \mathrm{TN}_{[l,u]}(\mu, v).$$

The physical transition/observation operators supply s; conditional state and colored-noise propagation supply S. Here μ, v parameterize the underlying untruncated normal density, not its bounded mean and variance. Define

$$g = s^\top S^{-1} s, \quad q = s^\top S^{-1} e, \quad v' = \frac{v}{1+vg}, \quad \mu' = \mu + v'(q - \mu g),$$

and $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Completing the square gives

$$\begin{aligned} \log p(e \mid h) = & -\tfrac{1}{2}(e^\top S^{-1} e + \log|S| + n_y \log(2\pi)) \\ & + \tfrac{1}{2}\left[\log(v'/v) + \mu'^2/v' - \mu^2/v\right] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \end{aligned}$$

Reported amplitude moments use the corresponding truncated-normal moments. The integral is evaluated over the retained amplitude intervals; it is not replaced by the best-fitting amplitude alone. Fixed line-mode hypotheses use their switched operator without a load-amplitude integral.

The square completion is standard. Its role here is to show precisely how a physical sensitivity and an uncertain intensity enter the executable likelihood. It is exact for this conditional Gaussian model; linearization, covariance sharing, nuisance substitution and history pruning prevent an exact unrestricted nonlinear-posterior claim. The frozen implementation is `integrated_loglike` in the R11 `observer/static_rx.py`; the one-report model weights call it before current-report screening. Rolling proposals do not create a second copy of the same observations.

B. STATE RECOVERY AND MODEL ADEQUACY

The matched B2 mean hidden-voltage error is 0.009634%, compared with 0.3402% for the equilibrium reference. Under the tested operating shifts, frozen errors of 0.19885%, 0.38834% and 0.57795% become 0.00525%, 0.00749% and 0.00974% after regularized physical recentering. The nuisance information rank is 25 of 43. This supports a useful point target while leaving parameter non-uniqueness and joint covariance failure visible.

C. BATCH SOURCE RESOLUTION

The historical implementation has an effective cardinality prior different from its originally declared prior because of a single-source normalization factor. The archived outcomes below are preserved, not corrected-prior results or calibrated physical limits. The detailed qualification follows in Section O of this supplement. The implemented batch estimator retains GH31 amplitude integration, W2 covariance, a fixed operating point and known onset/family. It evaluates the complete zero-, one- and two-source catalog; it does not supply the causal R11 decisions.

D. WHAT THE REFERENCE ESTIMATOR STILL DOES NOT CLAIM

The historical supervised bank spans 690 ANDES trajectories, with detection F1 0.989 and physical Top-1 0.592 on its own ordinary split. Its held-source stress test gives 3/303 exact decisions. These are different populations from H04/C01, so their scores cannot be subtracted to claim an improvement of the hackathon leaderboard. The unrestricted joint state/event/integrity estimator and learned discrepancy layer remain proposals. The pre-revision manuscript and its complete original main-text derivations are preserved in `archive/narrative_before_20261030/`.

TABLE 24. Exploration changes the next inference question, not the test labels.

Question explored	Observation	Decision retained	Boundary
Can eight PMUs constrain hidden voltage?	Accurate matched reconstruction; shift degrades it	Keep the physical state and operating context explicit	State recovery is not source attribution
Can an offset repair all mismatch?	Hidden-state recovery needs AC-feasible recentering	Separate physical recentering from measured residual centering	R11 intercept is not a new hidden-state solution
Does a source label transfer?	Legacy held-source test fails	Generate physical hypotheses for candidate assets	Causal unseen-source transfer remains untested
Does detecting multiplicity locate both sources?	Weak exact supports overlap	Profile severity/nuisance and keep alternatives	Batch onset/family known
Does every plausible multi-family proposal become a correct final decision?	The exploratory pilot does not establish that	Keep E_J as proposed, not a validated replacement for E_B	Dossier-only results remain secondary
Can a frozen physical bank follow successive events?	H04 separates timely recognition from eventual attribution	Retain dynamic recovery and compare nominal/mixture	Development, known catalog
Can stricter episode admission remove extras for free?	It rejects a real BUS16 intervention	Keep the failed ablation and freeze co-references	Zero extras is not uniform improvement
Does the result persist under an independent draw?	C01 is prespecified	No further method selection on that examination	Results enter only after complete evaluation

TABLE 25. Fresh prospective 137-hypothesis confirmation. The bank contains H_0, 16 single-source hypotheses, and all 120 two-source supports. Repeated noise realizations are decisions, not independent physical trajectories. Top-k is exact support recovery and is undefined for H_0 and single-source rows in this exported support table.

Regime	Physical trajectories	Decisions	Cardinality (%)	Top-1 (%)	Top-3 (%)	Mean $p(K = 2)$
No event	—	100	100.0	—	—	0.000
Single source	128	384	93.8	—	—	0.051
Weak–weak pair	1920	5760	37.4	24.4	59.8	0.427
Weak–strong pair	480	1440	77.6	66.2	95.8	0.808
Moderate pair	480	1440	99.7	98.8	100.0	0.996
Finite pair	480	1440	100.0	100.0	100.0	1.000

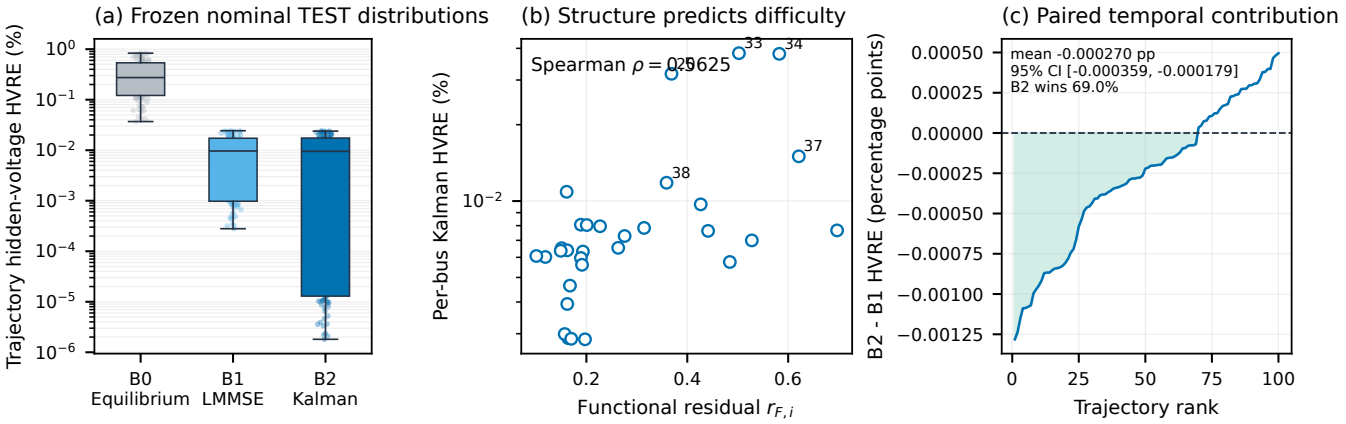


FIGURE 11. **Expanded matched state evidence, retained from the original frozen study. It evaluates hidden-voltage reconstruction and does not certify source identity.**

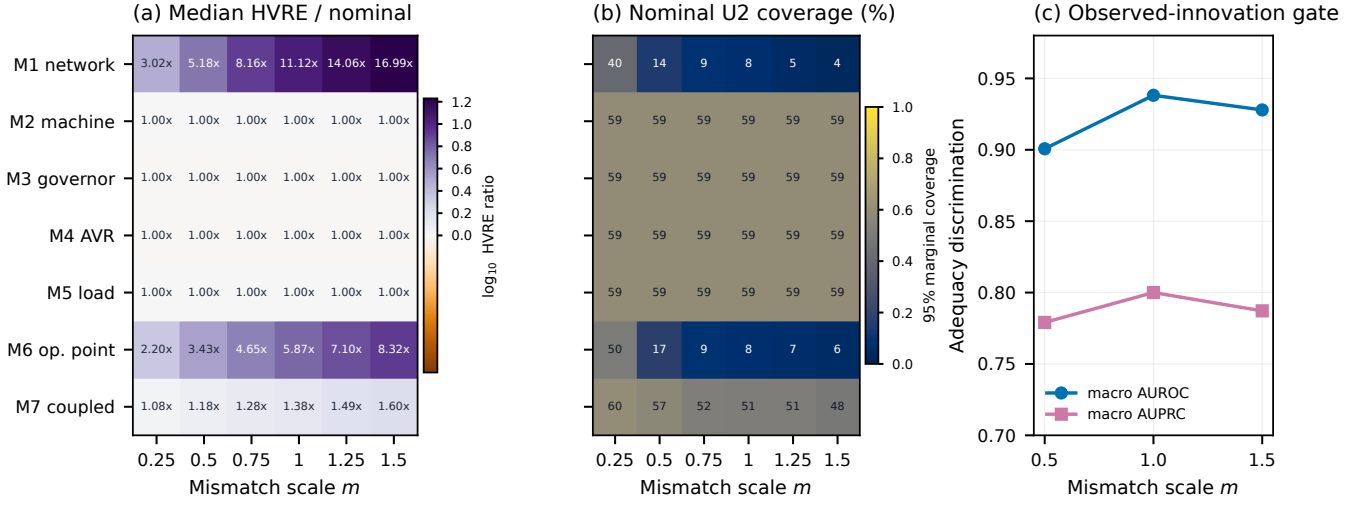


FIGURE 12. **Expanded model-mismatch evidence. Physical and operating shifts are distinct from instrumental covariance corrections.**

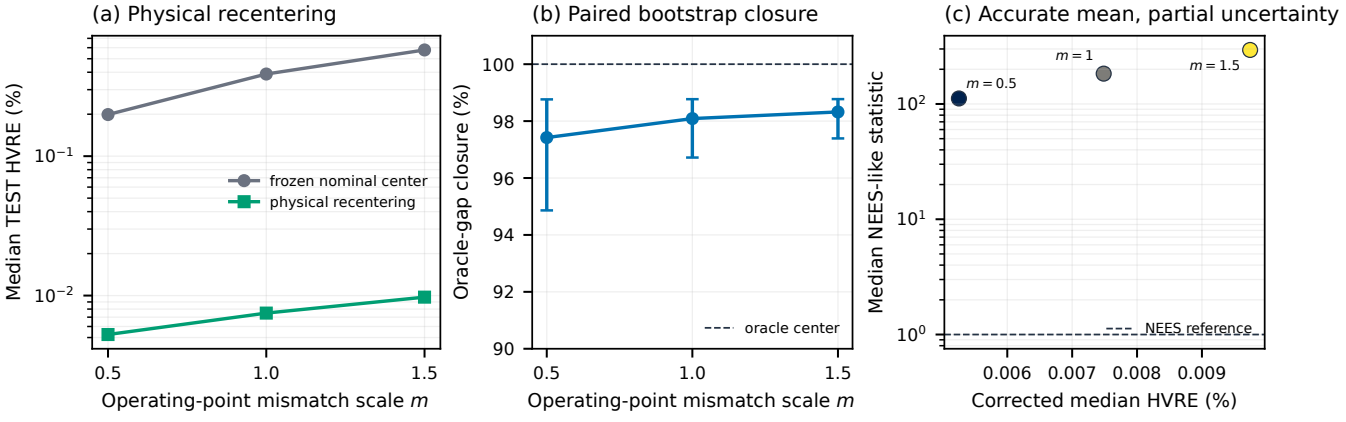


FIGURE 13. **Physical recentering repairs much of the tested point-estimate error while the joint uncertainty diagnostic remains inadequate.**

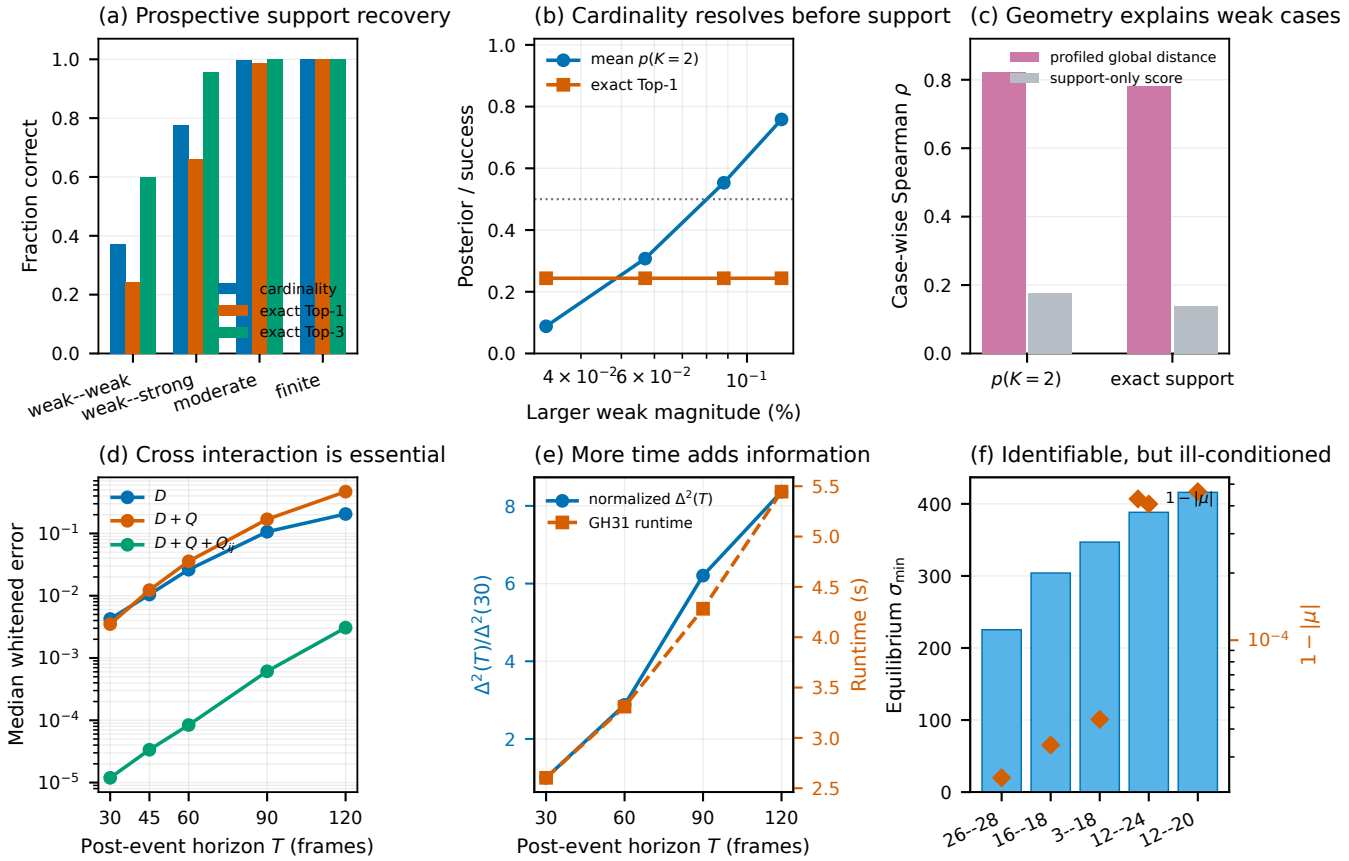


FIGURE 14. **Expanded source-resolution study. Prospective recovery, retrospective profiled geometry and the longer-horizon diagnostic have distinct roles; the latter panels do not replace a new prospective test.**

APPENDIX O WHY VISIBLE EVENTS CAN HAVE AMBIGUOUS SOURCES

The continuous results raise a separate physical question: can these measurements distinguish the candidate sources at all? A good state estimate does not answer it. Different interventions can produce similar histories after fitting their severity and initial state [29], [30].

For a response s_h at a specified severity and context, its distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\eta_j \in \mathcal{A}_j} \|s_h - s_j(\eta_j)\|_{\Sigma^{-1}}^2. \tag{182}$$

The set $\mathcal{A}_j$ specifies allowed severity, sign and nuisance coordinates. If zero severity is admitted, different support families can meet at the no-event boundary. Unrestricted amplitude minimization would then give a misleading source-separation interpretation. Under a common Gaussian law, coincident means cannot be distinguished at that intersection. Small distances indicate a noise-sensitive problem, not a universal error formula or an online uncertainty certificate.

A separate archived load-only experiment illustrates observed ambiguity. It uses 32 rectangular voltage/current coordinates, known onset and known operating context. Its original amplitude-integrated batch score compares 137 zero-, one- and two-source supports on 3488 fresh nonlinear trajectories. It is a different estimator and evidence contract from the causal bank. Its scores require the normalization qualification below before being interpreted under the originally declared priors.

Exact-support Top-1 increases from 24.4% for weak pairs to 100.0% for finite pairs (Fig. 15). A visible intervention therefore need not yield a reliable location in this implementation. Top-3 in the archive ranks only pair supports, whereas Top-1 also requires the full bank to select two sources. They must not be compared as though only the number of returned labels changed. These observed errors do not separate insufficient information from model, prior or numerical error; that separation requires an exact-response control.

The archived implementation also requires a prior qualification. Its single-source integral carries an extra factor $1/(2\pi)$ relative to the null and pair integrals, giving an effective cardinality prior proportional to $(0.20, 0.50/(2\pi), 0.30)$ instead of the originally declared $(0.20, 0.50, 0.30)$. A zero-response control on the historical code exposed this difference. An isolated successor restores the declared prior and passes independent linear-Gaussian checks for all 136 event supports; its physical performance has not been recomputed. Figure 15 retains the archived implementation's outcomes. This issue does not alter the local R11 amplitude integral, checked separately by numerical quadrature.

Longer history and different PMUs may help discrimination, but these curves do not certify an online uncertainty or abstention rule. Exact replay of the auxiliary study requires restoring source arrays absent from this artifact copy; no newly fitted dictionary replaces them. Detailed state-reconstruction studies remain supporting evidence rather than extra event-diagnosis scores.

APPENDIX P COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The frozen test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated

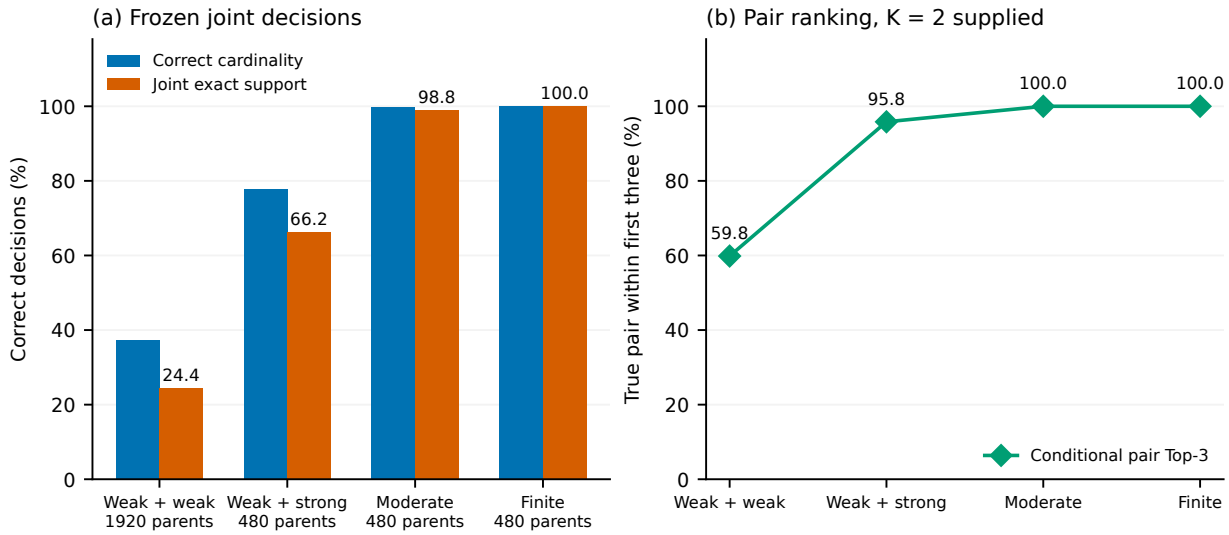


FIGURE 15. Archived load-support outcomes, not causal R11 accuracy. Left: cardinality and joint exact-source decisions of the frozen 137-support implementation. Right: Top-3 rank restricted to the 120 pairs, conditional on knowing that two sources occurred; it is a different decision contract. Labels give independent physical trajectories, each replayed with three noise seeds. The evidence-normalization audit below qualifies the cardinality prior; these curves are not corrected-prior results or proof of an information-theoretic limit.

directly from the sealed summary, records, event matches and paired intervals.

A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 16 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 33 describes those already recorded outputs. It does not change the frozen first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective RAW0001 replay under its startup noise-scale contract.

TABLE 26. Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.

Network	Noise	Physical bank	Nominal bank	ExtraTrees	ET short
Matched	1	108/108; 1	108/108; 1	56/108; 6	60/108; 5
Matched	2	108/108; 0	108/108; 0	54/108; 4	58/108; 0
In-bank	1	108/108; 2	108/108; 3	56/108; 5	60/108; 5
In-bank	2	108/108; 1	108/108; 2	54/108; 4	57/108; 0
Out-of-bank	1	106/108; 38	105/108; 47	56/108; 5	60/108; 6
Out-of-bank	2	106/108; 30	106/108; 39	54/108; 5	57/108; 0

TABLE 27. Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, false restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.

Method	Unavailable	Normal incidence	Normal alarms	Exposure (h)	False OFF	Open at end
Physical 5-model	0	0/72	0	11.760	58	0
Physical nominal	0	0/72	0	11.760	68	0
ExtraTrees	0	0/72	0	11.760	29	0
ET short	0	0/72	0	11.760	16	0

TABLE 28. Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.

R/X condition	Noise	Physical 5-model	Physical nominal	ExtraTrees	ET short
Matched	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
Matched	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
In-bank	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
In-bank	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
Out-of-bank	1	0.33/0.18	0.33/0.18	0.23/0.18	0.19/0.18
Out-of-bank	2	0.26/0.18	0.33/0.18	0.30/0.18	0.20/0.18

TABLE 29. Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the frozen evaluator.

R/X condition	Noise	Physical 5-model	Physical nominal	Chen2016
Matched	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Matched	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
In-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.04
In-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.05
Out-of-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Out-of-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.59

TABLE 30. Persistent-line coverage and stopping outcomes across six repeated strata, including both frozen ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen2016 normal exposure ends at its first stop; the other methods monitor the full available record.

Method	Unavailable	Correct first	Pre-change	Censored	Normal incidence	Exposure (h)
Physical 5-model	0	162/162	0	0	0/72	11.760
Physical nominal	0	162/162	0	0	0/72	11.760
ExtraTrees	0	162/162	0	0	0/72	11.760
ET short	0	162/162	0	0	0/72	11.760
Chen2016	0	74/162	66	0	72/72	0.701

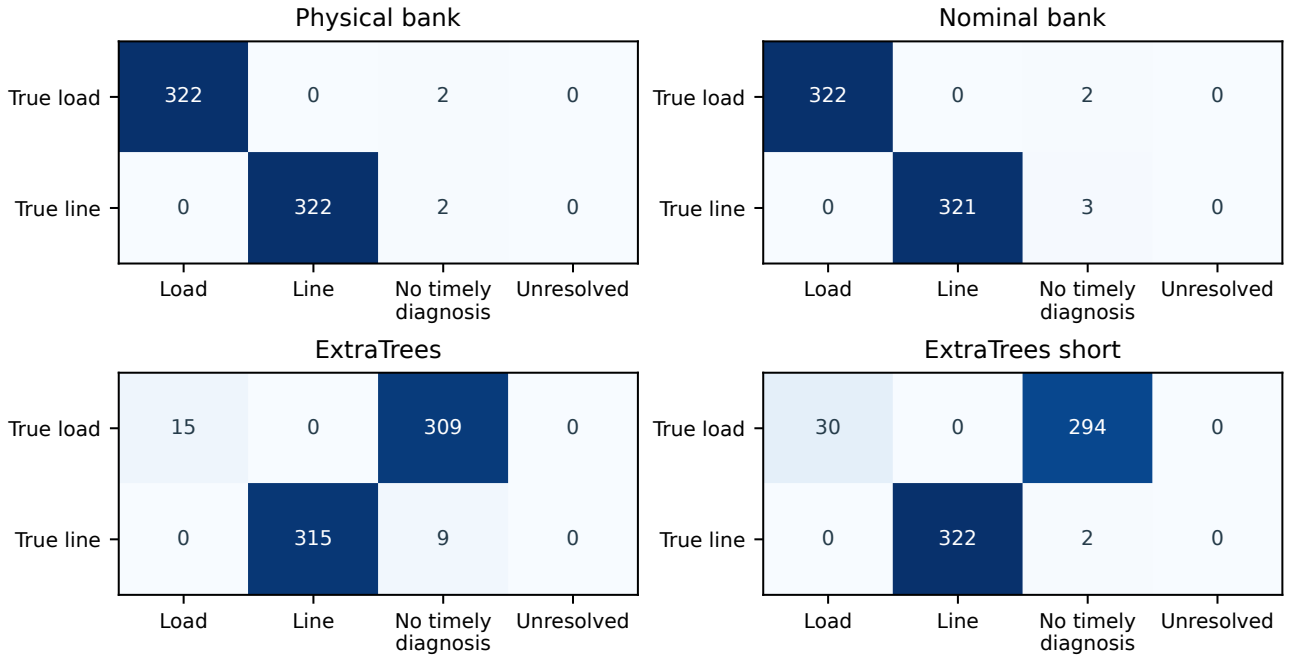


FIGURE 16. **Posthoc confusion audit of the frozen continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the prediction seal. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

TABLE 31. **Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.**

Family / asset	Physical bank	Nominal bank	ExtraTrees	ET short
Line / LINE14-15	36/36	36/36	27/36	34/36
Line / LINE16-17	36/36	36/36	36/36	36/36
Line / LINE17-18	48/48	48/48	48/48	48/48
Line / LINE23-24	30/30	30/30	30/30	30/30
Line / LINE26-27	24/24	24/24	24/24	24/24
Line / LINE26-29	24/24	23/24	24/24	24/24
Line / LINE3-4	42/42	40/42	42/42	42/42
Line / LINE4-14	46/48	48/48	48/48	48/48
Line / LINE9-39	36/36	36/36	36/36	36/36
Load / BUS15	42/42	42/42	6/42	6/42
Load / BUS16	42/42	42/42	0/42	3/42
Load / BUS20	12/12	12/12	6/12	12/12
Load / BUS24	42/42	42/42	0/42	0/42
Load / BUS26	18/18	18/18	0/18	0/18
Load / BUS3	42/42	42/42	3/42	3/42
Load / BUS31	40/42	40/42	0/42	0/42
Load / BUS4	36/36	36/36	0/36	6/36
Load / BUS7	48/48	48/48	0/48	0/48

TABLE 32. Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per monitored event-record hour, excluding 12 s of startup from each 120-s record. The registered rate differences and 95-percent parent-bootstrap bounds are exactly rescaled by 120/108; original counts, matches and intervals remain archived.

Network	Noise	Method	Reference	Timely difference	Extra-rate difference
Matched	1	Physical bank	ExtraTrees short	44.44 [40.48, 47.92]	-7.41 [-16.67, 0.00]
Matched	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	1	Nominal bank	ExtraTrees short	44.44 [40.74, 48.04]	-7.41 [-14.81, 0.00]
Matched	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	2	Physical bank	ExtraTrees short	46.30 [42.75, 49.12]	0.00 [0.00, 0.00]
Matched	2	Physical bank	ExtraTrees	50.00 [46.30, 53.70]	-7.41 [-14.81, -1.85]
Matched	2	Nominal bank	ExtraTrees short	46.30 [42.86, 49.12]	0.00 [0.00, 0.00]
Matched	2	Nominal bank	ExtraTrees	50.00 [46.38, 53.70]	-7.41 [-14.81, -1.85]
In-bank	1	Physical bank	ExtraTrees short	44.44 [40.62, 47.92]	-5.56 [-14.81, 5.56]
In-bank	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-5.56 [-14.81, 5.56]
In-bank	1	Nominal bank	ExtraTrees short	44.44 [40.62, 47.92]	-3.70 [-12.96, 7.41]
In-bank	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-3.70 [-12.96, 5.56]
In-bank	2	Physical bank	ExtraTrees short	47.22 [43.18, 50.98]	1.85 [0.00, 5.56]
In-bank	2	Physical bank	ExtraTrees	50.00 [46.49, 53.70]	-5.56 [-11.11, 0.00]
In-bank	2	Nominal bank	ExtraTrees short	47.22 [43.33, 51.04]	3.70 [0.00, 9.26]
In-bank	2	Nominal bank	ExtraTrees	50.00 [46.30, 53.70]	-3.70 [-11.11, 3.70]
Out-of-bank	1	Physical bank	ExtraTrees short	42.59 [38.60, 46.43]	59.26 [40.74, 79.63]
Out-of-bank	1	Physical bank	ExtraTrees	46.30 [42.11, 50.00]	61.11 [40.74, 81.48]
Out-of-bank	1	Nominal bank	ExtraTrees short	41.67 [36.11, 46.67]	75.93 [51.85, 100.00]
Out-of-bank	1	Nominal bank	ExtraTrees	45.37 [41.11, 50.00]	77.78 [50.00, 105.56]
Out-of-bank	2	Physical bank	ExtraTrees short	45.37 [41.11, 50.00]	55.56 [38.89, 72.22]
Out-of-bank	2	Physical bank	ExtraTrees	48.15 [43.59, 52.50]	46.30 [27.78, 66.67]
Out-of-bank	2	Nominal bank	ExtraTrees short	45.37 [41.11, 50.00]	72.22 [46.30, 98.15]
Out-of-bank	2	Nominal bank	ExtraTrees	48.15 [43.75, 52.56]	62.96 [35.19, 88.89]

TABLE 33. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.

Method	Openings	Extras	Unmatched OFF	Open at end	FN reports
Physical bank	162	0	0	162	810
Nominal bank	162	0	0	162	810
ExtraTrees	162	164	164	16	335374
ExtraTrees short	162	179	178	61	182600

TABLE 34. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.

Component	Positive-window hits	FP	FN
Missing	14/14	0	0
Corrupt	1/2	2	1

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